

WEB-BASED EMPLOYEE PERFORMANCE TRACKING SYSTEM

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by

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ABSTRACT

Job assignment and performance monitoring at Pakonek Solutions Inc., an internet service provider with a largely field-deployed workforce, are handled through verbal instructions, messaging applications, and manual job order forms, a practice prone to data loss that keeps job records and evaluation data separate and produces inconsistent employee assessment. This study asked whether a centralized web-based platform could improve the accuracy, consistency, and timeliness of workload assignment and performance evaluation. The researchers designed, developed, and evaluated a Web-Based Employee Performance Tracking System for administrators, managers, field technicians, and office staff. The study followed the developmental method supported by a qualitative approach, with requirements established through interviews with technicians, office staff, and managers and analysis of existing job orders and performance records. The system was built using ASP.NET Core with HTML, CSS, and JavaScript and a Microsoft SQL Server database under the Rapid Application Development model, refined through repeated cycles of prototyping, testing, and user feedback. It provides role-based access control, employee and user account management, job ticket assignment for technicians, task assignment with activity submission and approval for office staff, workload monitoring, criteria-based evaluation with configurable weights, and performance reports generated from existing records. The completed system was evaluated by ten respondents through User Acceptance Testing, and the data were analyzed using the System Usability Scale (SUS). The system obtained an overall SUS score of 78.25, above the 68 benchmark and rated "Good," with respondents reporting strong willingness to use it frequently (4.70), high confidence in operating it (4.30), and low dependence on technical assistance (1.70). The findings indicate that the system is suitable for its intended users and contributes a practical model for digitizing performance monitoring in field-service organizations.

Keywords: *employee performance tracking, job-based evaluation, workload monitoring, Rapid Application Development, System Usability Scale*

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Chapter 1

INTRODUCTION

A. Rationale of the Study

Employee performance is a key element that determines the success or failure of an organization [1], particularly of Internet Service Providers (ISPs) such as Pakonek Solution Inc., where a significant portion of the workforce consists of field-based and remotely deployed employees. At present, job assignments at Pakonek Solutions Inc. are managed through verbal instructions, messaging applications, and manual job order forms — coordination methods that are largely informal and prone to data loss or inconsistency. Monitoring and evaluating performance in such an environment is challenging due to limited physical supervision and reliance on these traditional methods. The ability to track and evaluate performance accurately is crucial for organizations to obtain timely, consistent, and accurate data on employee productivity, work quality, and task completion.

Despite the use of basic tools for task coordination, significant gaps remain in how employee performance, particularly for field-deployed technicians, is systematically evaluated. Current methods often rely on subjective managerial judgment and periodic reviews, leading to inconsistent evaluations and poor data quality. Traditional performance evaluation methods, such as manual documentation and subjective assessments, do not accurately reflect actual workloads, job completion efficiency, or work outcomes. Additionally, job assignments and performance data are managed separately, making it difficult to obtain a comprehensive view of an employee's performance. These fragmented

approaches contribute to delays in feedback, leading to missed opportunities for performance improvement and reducing overall employee satisfaction.

This study addresses these challenges by proposing a web-based employee performance tracking system that uses a job-based evaluation approach. Instead of relying primarily on subjective ratings, the system measures performance using objective indicators such as task completion, timeliness, and job outcomes. The system aims to centralize task management and performance records into a single platform, enabling real-time monitoring and more accurate evaluation. By automating data collection and consolidating performance information, the proposed system supports consistent assessments and more informed workforce management decisions.

The proposed system benefits both management and employees by promoting transparency, accountability, and data-driven evaluation. Managers can use the system to monitor workloads, identify performance issues, and allocate tasks more effectively, ensuring optimal resource distribution. At the same time, employees gain access to clear performance records and timely feedback, which will directly contribute to professional development. As employee development indicates that the organization cares about its employees and wants them to grow [1], the system ensures that employees have the tools they need to develop their skills. Furthermore, the system highlights the role of digital innovation in improving workforce management, showing how a unified platform can integrate task tracking, workload monitoring, and performance evaluation into one cohesive solution. This study may also serve as a reference model for future systems that aim to improve employee evaluation practices, foster accountability, and enhance organizational efficiency.

B. Objectives of the System

To develop a web-based employee performance tracking system that streamlines evaluations, enhances accountability, and improves workforce management for field-based and office employees of Pakonek Solutions Inc. Specifically, the system aims to:

- centralize employee performance-related data by storing employee profiles, job assignments, task records, and evaluation results in a single database, enabling a comprehensive and unified view of performance;
- implement a job-based performance evaluation mechanism for field technicians using task completion, timeliness, and job outcomes, and a standardized evaluation mechanism for office staff based on structured criteria and manager scores;
- support standardized performance evaluation and scoring by providing managers with structured rating forms and documented justifications derived from system-generated data, ensuring consistency and objectivity across all evaluations;
- monitor employee workload distribution by tracking the number of assigned and completed tasks per employee and presenting the data through visual dashboards to help identify overworked or underutilized personnel;
- generate performance reports and historical records by producing monthly and quarterly summaries, including job completion, workload, and evaluation results, to provide management with actionable insights for decision-making; and
- provide employees with secure access to their performance records and feedback through the system, enhancing transparency and accountability.

C. Significance of the Study

This study on developing a web-based employee performance tracking system is significant because it addresses challenges in monitoring and evaluating field-based or remote employees, particularly at Pakonek Solutions Inc. By centralizing data, automating evaluations, and providing actionable insights, the system enhances workforce management, ensures timely feedback, and reduces manual errors.

Management and Company Owners – Management will benefit from a structured and centralized view of employee performance, workload, and productivity. The system enables data-driven decision-making by providing accurate performance records, reducing reliance on subjective evaluations, and improving workforce planning for both field technicians and office staff.

Employees – Employees, including both office staff and field-based technicians, will benefit from increased transparency and accountability in their performance evaluations. By accessing their records, feedback, and progress, employees can monitor their development, understand performance expectations, and work toward continuous improvement. The system's job-based tracking of task completion and timeliness ensures fair recognition of work contributions and helps employees manage their workload more effectively.

Researchers and Developers – Future researchers and developers may use the system's design, documentation, and findings as a reference for similar performance tracking systems in service-oriented and field-based industries.

D. Scope and Limitations of the System

The proposed Web-Based Employee Performance Tracking System is designed to support the workforce management operations of Pakonek Solutions Inc. The system focuses on managing employee performance, job ticket assignments, task tracking, and evaluation records through a centralized web-based platform. The system includes the following functionalities:

User Account Management – allows the Administrator to create, update, deactivate, and manage user accounts with role-based access for Administrators, Managers, Field Technicians, and Office Staff.

Employee Profile Management – enables the Administrator and Manager to store, update, and retrieve employee information such as role, department, and deployment status.

Job Ticket Management – allows Managers to create, assign, monitor, and close job tickets for Field Technicians, including installation, repair, and maintenance tasks.

Task Assignment and Tracking (Office Staff Only) – allows Managers to assign tasks, set deadlines, and track completion for Office Staff.

Workload Monitoring – displays the number of active and completed job tickets per Field Technician to help Managers balance task distribution.

Job-Based Performance Evaluation – allows Managers to evaluate Field Technicians based on completed job tickets, timeliness, and manager validation rather than subjective ratings alone.

Performance Evaluation for Office Employees – enables Managers to assess Office Staff using standardized evaluation criteria aligned with their job roles.

Criteria Management – allows the Administrator to define, update, and manage performance evaluation criteria, scoring standards, and rating weights

Feedback and Appraisal Records – stores evaluation comments and appraisal history, accessible to Managers for review and to Employees for personal reference.

Performance Reports and Dashboards – generates summarized reports and visual dashboards to support performance analysis and decision-making by Managers and Administrators.

However, the system is not capable of the following functionalities:

- The system does not calculate salaries, bonuses, or promotions based on the performance results.
- The system requires manual data entry for job and task completion.
- The system does not include predictive analysis, and performance insights are based solely on recorded job tickets and task completion.
- The system supports documentation, alerts, and reporting, but does not enforce disciplinary actions or company rules.
- The system is not designed to replace customer service or CRM systems.

DEFINITION OF TERMS

The definition of terms serves as the glossary that provides a clear explanation of the terminology used throughout the research study.

Appraisal – a formal assessment process in which an employee's performance, behavior, and contributions are evaluated and documented by a manager or authorized personnel based on predefined criteria and system-generated records.

Employee Performance – the measurement of an employee's effectiveness in completing tasks, meeting deadlines, and fulfilling job responsibilities, evaluated using objective metrics such as task completion, timeliness, and manager validation.

Feedback and Appraisal Records – documentation of performance evaluations, comments, and appraisals stored within the system for managerial review and employee reference to promote transparency and continuous development.

Job-Based Evaluation – a performance assessment method that evaluates employees based on measurable task outcomes, such as task completion and timeliness, rather than subjective ratings.

Job Ticket – a record of a specific task or assignment for an employee, including task details, assigned personnel, deadlines, and completion status, used for task tracking and evaluation.

Workload Monitoring – the tracking and analysis of tasks assigned to employees to ensure balanced distribution and to identify overworked or underutilized personnel.

Chapter 2

THEORETICAL BACKGROUND

A. Review of Related Literature and Systems

The review of related systems will examine existing systems that have similar features or functions to the proposed system. This section will help the researchers understand how current systems operate and identify useful features, limitations, and improvements that can be considered in developing the proposed system.

1. Zoho People

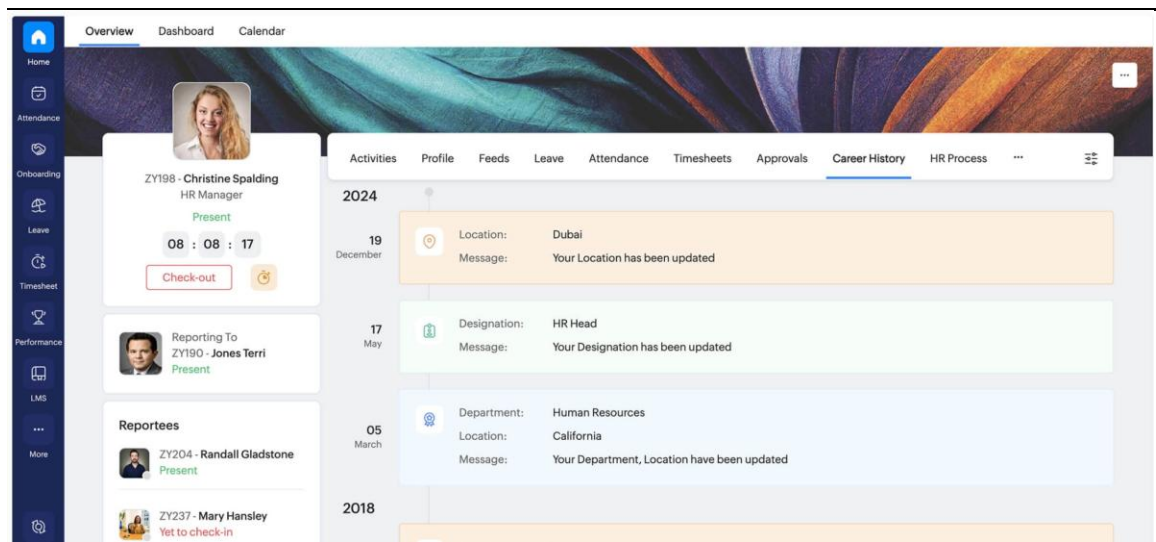


Figure 2.1: Zoho People Home Page

Zoho People is a cloud-based human resource management system designed to help organizations streamline HR tasks such as employee management, attendance tracking, and performance evaluations. The system automates HR workflows and centralizes employee data, enhancing productivity, reducing administrative burdens, and providing actionable

insights into workforce performance [2]. It is widely used by various organizations to efficiently manage and track employee-related processes.

The system includes features such as centralized employee profiles, role-based access control, automated attendance and leave management, customizable performance appraisals with 360-degree feedback and criteria management, analytics and reporting dashboards, recognition tools, continuous feedback tools, customizable review templates, and API integrations with other HR and business systems [2].

However, Zoho People relies primarily on subjective performance reviews and does not support job ticket management, job-based performance evaluation for field-based employees, or real-time workload monitoring per employee. The proposed system addresses this limitation by incorporating automated job ticket tracking, job-based performance evaluation, and workload monitoring specifically designed for field technicians.

2. BetterWorks

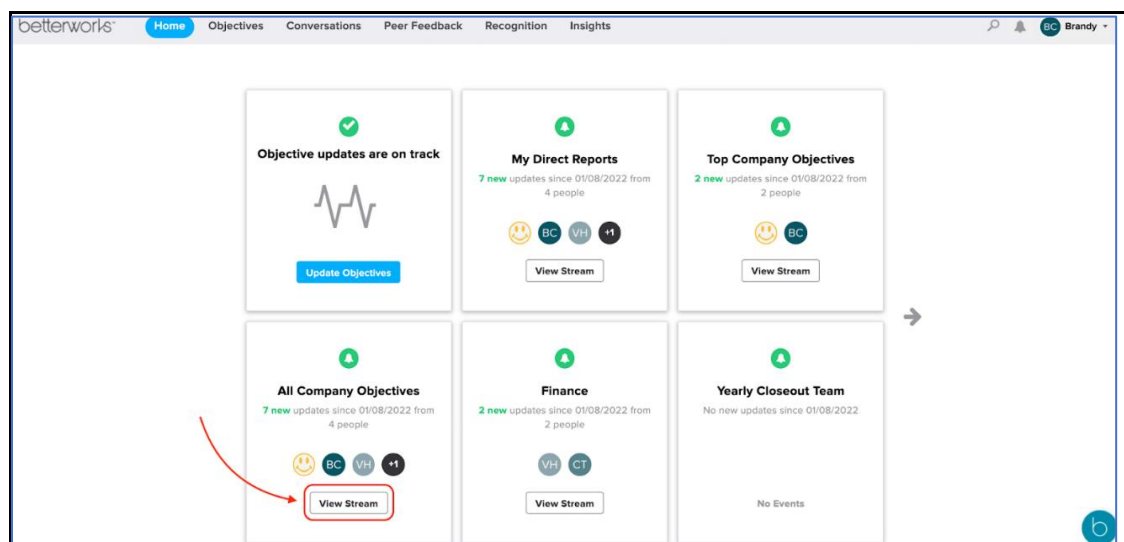


Figure 2.2: BetterWorks Home Page

BetterWorks is a performance management platform designed to align employee goals with company objectives and promote continuous performance improvement. Unlike traditional annual reviews, BetterWorks focuses on real-time goal setting, feedback, and performance conversations, helping organizations monitor and improve workforce performance across all levels [3].

Key features include a goal alignment framework using OKRs for setting and tracking organizational and individual goals, role-based access control, task assignment and tracking tools for office staff, real-time performance conversations and continuous feedback mechanisms, recognition tools, employee development tools, analytics dashboards that visualize employee progress and engagement, and API integrations with third-party HR and enterprise systems for centralized performance tracking and data sharing [3].

Despite its strong goal-setting and continuous feedback features, BetterWorks does not provide job ticket management, job-based performance evaluation, or workload monitoring for field-based employees. The proposed system addresses this gap by enabling job-specific monitoring, job ticket tracking, and workload distribution visualization for field technicians.

3. Leapsome

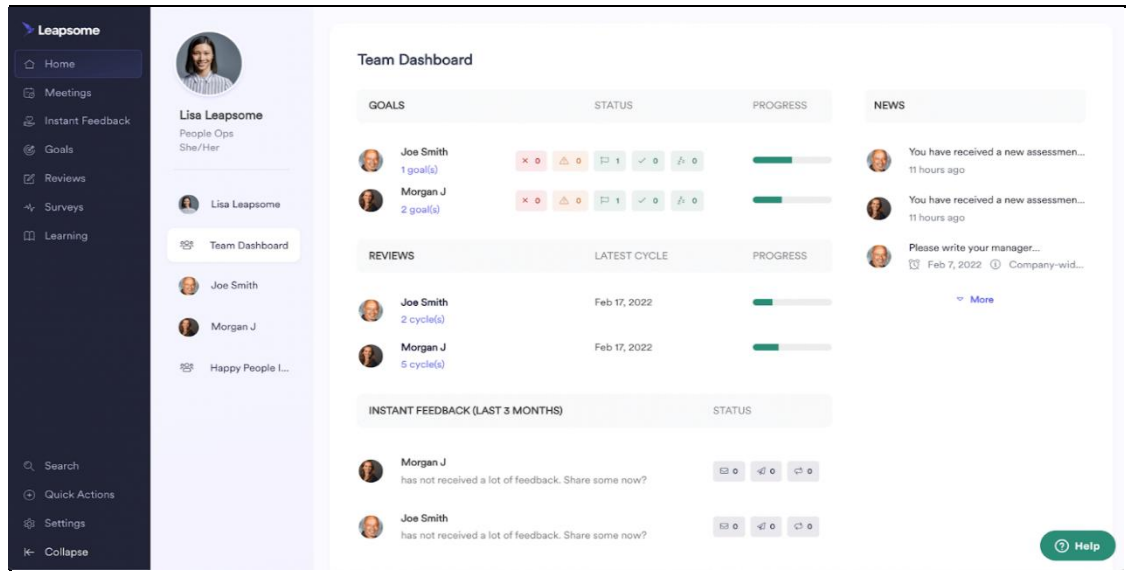


Figure 2.3: Leapsome Home Page

Leapsome is a cloud-based performance management and HR platform designed to help organizations manage employee reviews, set goals, and provide continuous feedback. It centralizes performance reviews and employee development tools in one place, improving how HR teams and managers evaluate and track employee progress [4].

Notable features of Leapsome include role-based access control, centralized employee profile management, task assignment and tracking for office staff, customizable performance reviews with flexible templates and criteria management, 360-degree feedback mechanisms from peers, managers, and self-assessments, employee development tools, goal-setting tools using OKRs, continuous feedback, attendance and leave management, real-time analytics and reporting dashboards, recognition tools, customizable review and feedback templates, and integrations with various HR tools [4].

Although Leapsome offers comprehensive performance management features, it does not provide job ticket management, job-based performance evaluation, or workload

monitoring tailored to field-based or operationally deployed employees. The proposed system addresses this by adding automated job ticket tracking, job-based evaluation, and workload monitoring specifically for field technicians.

4. 15five

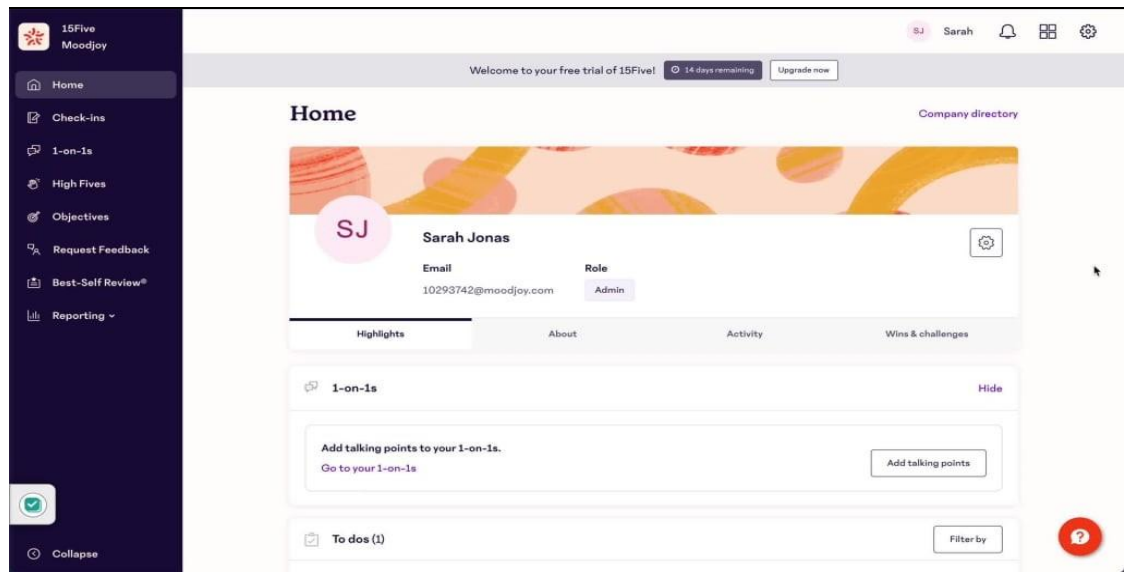


Figure 2.4: 15Five Dashboard

15Five is a cloud-based performance management platform designed to help organizations enhance employee engagement, performance, and retention. It allows HR teams and managers to collect feedback, conduct performance reviews, and support employee growth through regular check-ins and goal tracking. The system moves away from traditional annual reviews by making performance management more continuous and actionable [5].

Its features include role-based access control, employee profile management, task assignment and tracking for office staff, continuous feedback through regular and weekly check-ins, employee engagement surveys and tracking, employee development tools, OKRs for aligning individual and organizational goals, recognition tools, real-time performance conversations, analytics dashboards that provide insights into employee

performance and trends, attendance tracking, and API integrations with other HR and collaboration tools [5].

However, 15Five does not support job ticket management, job-based performance evaluation, or workload monitoring for field-based operational roles. The proposed system addresses this limitation by providing job-specific monitoring, automated job ticket tracking, and workload distribution management for field technicians.

5. BambooHR

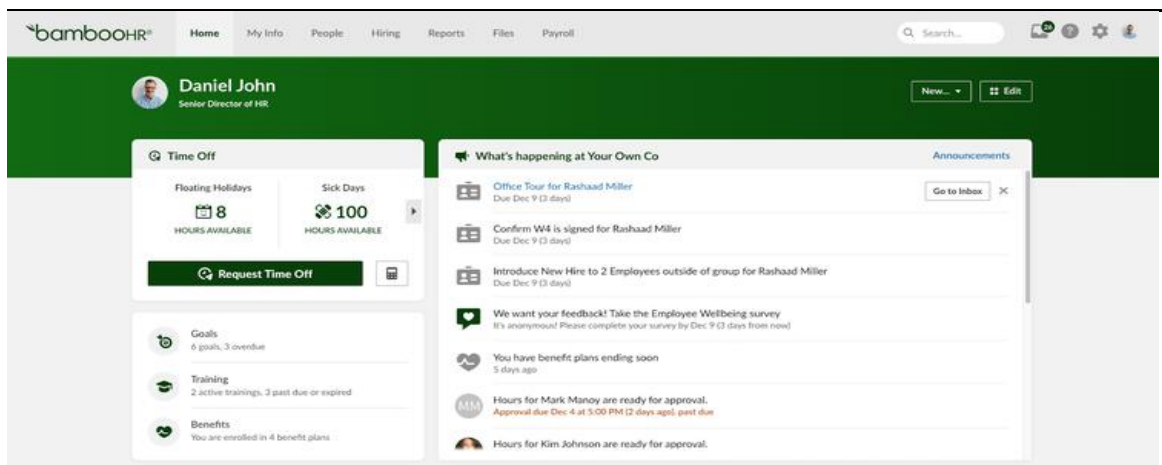


Figure 2.5: BambooHR Home Page

BambooHR is a cloud-based human resource management system used by organizations to manage a wide range of HR tasks on a single platform. It helps businesses simplify hiring, employee records, time-off tracking, and performance reviews, making HR processes more efficient and organized. The system is designed to reduce manual work and centralize key employee information for easier access and reporting [6].

BambooHR is a cloud-based human resource management system used by organizations to manage a wide range of HR tasks on a single platform. It helps businesses simplify hiring, employee records, time-off tracking, and performance reviews, making

HR processes more efficient and organized. The system is designed to reduce manual work and centralize key employee information for easier access and reporting [6].

However, BambooHR does not support job ticket management, job-based performance evaluation, or workload monitoring for field-based or operationally deployed employees. The proposed system addresses this by implementing automated job ticket tracking, job-based evaluation, and real-time workload monitoring for field technicians.

B. Matrix of Comparison

This section presents a comparison matrix between the proposed employee performance tracking system and existing systems with similar functionalities. The comparison highlights key features, capabilities, and limitations to show how the proposed system addresses current gaps and enhances performance management for field-based and remote employees. The matrix serves as a foundation for evaluating the system's effectiveness, uniqueness, and its ability to meet the needs of organizations and employees.

Table 2.1 Matrix of Comparison

System/ Function	Zoho People	Betterworks	Leapsome	15Five	BambooHR	Proposed System
User Account Managemen t	✓	✓	✓	✓	✓	✓
Employee Profile Managemen t	✓	✓	✓	✓	✓	✓

Job Ticket Management						✓
Task Assignment & Tracking (Office Staff Only)		✓	✓	✓	✓	✓
Workload Monitoring						✓
Job-Based Performance Evaluation						✓
Performance Evaluation for Office Employees	✓	✓	✓	✓	✓	✓
Criteria Management	✓	✓	✓	✓	✓	✓
Feedback and Appraisal Records	✓	✓	✓	✓	✓	✓
Performance Reports and Dashboards	✓	✓	✓	✓	✓	✓
360-Degree Feedback	✓	✓	✓	✓	✓	

Goal Setting (OKRs)	✓	✓	✓	✓	✓	
Employee Access to Performance Records	✓	✓	✓	✓	✓	✓
Attendance and Leave Management	✓		✓	✓	✓	
Employee Engagement Tracking	✓					
Continuous Feedback	✓	✓	✓	✓		
Real-Time Performance Conversation		✓		✓		
Recognition Tools	✓	✓	✓	✓	✓	
Customizable Reviews and Feedback	✓	✓	✓	✓	✓	
Employee Development Tools		✓	✓	✓		

Weekly Check-Ins				✓		
Employee Engagement Surveys				✓		
Time-Off Tracking					✓	
Onboarding and Offboarding Tools					✓	
Document Storage					✓	
Role-Based Access Control	✓	✓	✓	✓	✓	✓
API Integration	✓	✓	✓	✓	✓	

Chapter 3

METHODOLOGY

A. Research Environment

The study will be conducted at Pakonek Internet Service Provider, located at Phase 2, 2-78 Everlasting St, St Jude Acres Rd, Bulacao, Cebu City, focusing on its operational environment, particularly the monitoring of job completion and performance of field technicians and office staff. The research will gather data from selected technicians and administrative personnel who are directly involved in daily job assignments, reporting, and performance evaluation. These participants are chosen because their actual work experience provides accurate and relevant information about the current manual processes and the challenges encountered in monitoring employee performance. The study is limited to Pakonek personnel to ensure that the data reflects real operational conditions and supports the development of a system that will improve efficiency, accuracy, and transparency in employee performance monitoring.

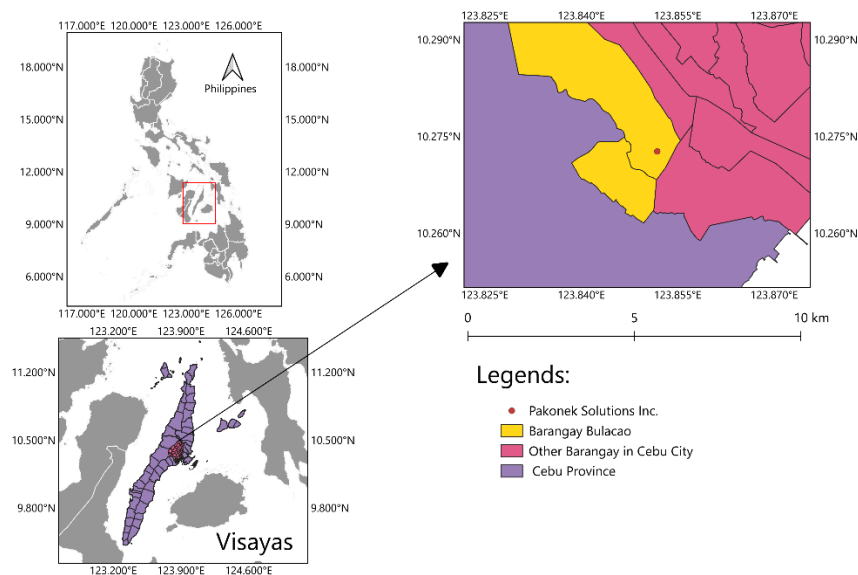


Figure 3.1: Research Locale

a. Methods and Techniques

The researchers will utilize the following methods and techniques in developing the system:

Developmental Method – This study used the developmental method to guide the design, development, and improvement of the Employee Performance Tracking System for Pakonek Solutions Inc. This iterative process involved planning, designing, coding, testing, and refining the system based on user feedback to ensure it met the needs of the organization.

Qualitative Method – The qualitative method was used to gather insights from employees at Pakonek Solutions Inc. to understand the challenges in tracking and evaluating employee performance. This method helped the researchers collect detailed feedback and identify gaps in the current system, ensuring the new system aligned with user needs.

Interview Technique – The researchers conducted interviews with field technicians, office staff, and managers to explore their challenges with the current performance monitoring system. These interviews provided valuable insights into the difficulties they faced and the features needed in the new system.

Document Analysis Technique – The document analysis technique involved reviewing existing records, such as job orders and performance evaluations, to identify inefficiencies in the current system. This ensured the new system addressed issues such as incomplete or inconsistent performance data.

User Acceptance Testing (UAT) – User Acceptance Testing (UAT) was conducted to ensure the system functioned correctly and met user expectations. Selected employees tested the system by performing tasks such as creating job tickets and evaluating performance. Feedback from UAT helped the researchers make necessary adjustments, and any issues identified during testing were addressed before full implementation.

b. Models

The researchers will use several system development and design models to illustrate how the system will function, how users will interact with it, and how data will be organized.

Rapid Application Development (RAD) Method – The RAD approach involves rapid prototyping and user feedback during the development of the system. The reason for using the RAD approach is to enable the creation of a system quickly, ensuring that user requirements are satisfied during the process. RAD allows for the evolutionary growth of the system based on user feedback.

Use Case Model – The Use Case Model defines the specific interactions between users, such as managers, field technicians, and office staff, and the system. By illustrating key actions like creating job tickets, submitting tasks, and viewing performance reports, this model ensures that the system addresses all necessary user functions and clearly outlines their roles. It helps ensure the system is designed to meet the practical needs of its users.

Class Model – The Class Model defines the system’s classes, attributes, and methods, and how they are connected. This model supports an object-oriented approach, helping to structure the system in a way that is scalable and maintainable. It ensures that the system’s components are organized logically, allowing for smoother development and easier future modifications.

Entity-Relationship Diagram – The Entity-Relationship Diagram illustrates the conceptual structure of the database by showing how entities from system’s entities, attributes, and relationships. It serves as the basis for the Data Schematic Diagram, where these entities and relationships are translated into tables, fields, primary keys, and foreign keys.

c. Tools

The following software tools will be used by the researchers in developing the proposed system:

ASP.NET Core – ASP.NET Core is a cross-platform, high-performance web framework developed by Microsoft. It served as the primary backend framework for handling business logic and data processing for the system.

Draw.io – Draw.io is an online diagramming tool. It was used to design Use Case Diagrams and ER Models to visually represent the system structure and data flow.

Figma – Figma is a collaborative interface design and prototyping tool. It was used to create wireframes and interactive prototypes to visualize the system.

Git and GitHub – Git is a version control system, and GitHub is a cloud-based repository platform. They were used to manage source code, track project versions, and enable efficient team collaboration.

HTML, CSS, and JavaScript – These are core web development technologies. They were used to structure content, design the interface, and add interactive functionality to the system.

Microsoft SQL Server – Microsoft SQL Server is a relational database management system. It will be used to store, manage, and retrieve system data securely and efficiently.

Microsoft Word – Microsoft Word is a word processing application. It was used to create, format, and document research papers and project reports professionally.

d. Development Process

The development of the proposed system follows the Rapid Application Development (RAD) model, which supports fast system development through iterative design and continuous user feedback. The RAD model consists of four major phases: Requirements Planning, User Design, Construction, and Cutover.

1. Requirements Planning Phase (January week 1 – February week 2) – During this phase, the researchers gathered system requirements through interviews and observations of the daily operations at Pakonek Solutions Inc. The researchers identified existing problems in monitoring employee performance, managing job ticket assignments, and tracking task completion. These findings

were used to determine the necessary system features and functionalities, ensuring that all stakeholders agreed on the project's objectives and functional boundaries before development began.

2. User Design Phase (February week 3 – April week 4) – In this phase, the researchers developed system prototypes and interface designs based on the identified requirements. Using Figma for interface prototypes and diagrams for data mapping, the prototypes were presented to the supervisor and staff for feedback. The design process involved continuous evaluation and improvement until the system design met the needs of the users.
3. Construction Phase (May week 1 – August week 2) – During the construction phase, the researchers developed the complete system by implementing the approved design. The researchers began with essential features such as user profiles and employee records before moving to more complex modules such as job ticket management and workload monitoring. Coding, database development, and system integration were performed during this stage. System testing was also conducted to ensure that the system functions correctly and meets the required specifications.
4. Cutover Phase (August week 3 – August week 4) – The cutover phase involves deploying the completed system for actual use through User Acceptance Testing (UAT). Employees performed real-world tasks to verify that the system was ready for operation. The system was then deployed to a live server and existing company data was migrated. Basic orientation and training were

conducted to ensure that users understand how to operate the system effectively, and ongoing monitoring was performed to ensure system stability.

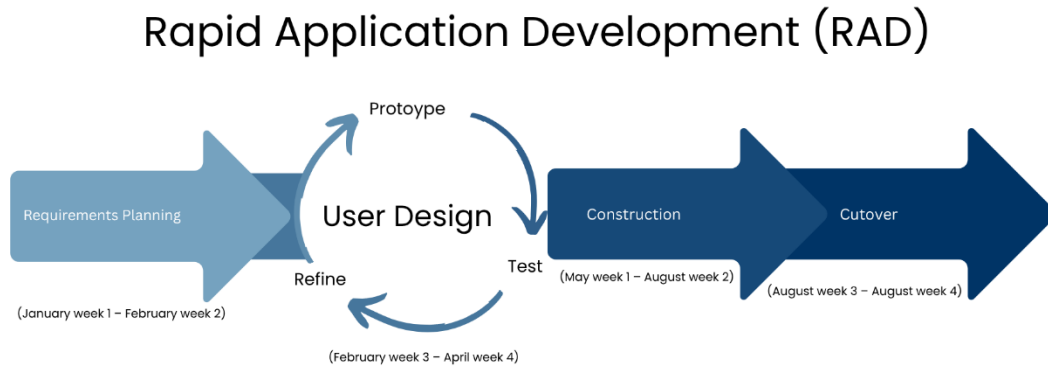


Diagram 3.1: Rapid Application Development Flow

B. Requirements Analysis

a. Functional Requirements

Functional requirements describe the specific tasks and functionalities that the Web-based Employee Performance Tracking System must perform to support operations and user interaction.

- User Account Management – The system shall allow administrators to create, update, deactivate, and manage user accounts with role-based access for managers, technicians, and office staff.
- Employee Profile Management – The system shall allow authorized users to store, update, and retrieve employee information such as name, employee ID, department, role, and deployment status.

- Job Ticket Management – The system shall allow managers to create, assign, monitor, and close job tickets for field technicians, including details such as job type, assigned technician, location, scheduled date, and status.
- Task Assignment and Tracking – The system shall allow managers to assign tasks to office staff, set deadlines, and track task completion.
- Workload Monitoring and Alerts – The system shall display the number of active and completed job tickets per technician to support workload management.
- Employee Performance Evaluation – The system shall allow managers to evaluate employee performance based on role. Field technicians shall be assessed using completed job tickets, timeliness, and manager validation, while office employees shall be evaluated using standardized criteria aligned with their job responsibilities.
- Criteria Management – The system shall allow administrators or managers to define and manage performance evaluation criteria, scoring standards, and rating weights.
- Feedback and Appraisal Records – The system shall store evaluation comments, scores, and appraisal history for each employee and allow employees to view their own feedback and performance records.
- Performance Reports and Dashboards – The system shall generate summarized reports and visual dashboards showing job completion, workload, and performance evaluation results

b. Non-Functional Requirements

Non-functional requirements describe how well the Web-based Employee Performance Tracking System should perform and what quality it must have.

- Performance – The system shall load pages and process user actions within an acceptable response time under normal internet conditions.
- Reliability and Availability – The system shall remain functional and accessible during the regular working hours of Pakonek Solutions Inc., except during scheduled maintenance.
- Security – The system shall require users to log in using valid credentials and protect stored data from unauthorized access.
- Usability – The system shall provide a simple and easy-to-navigate interface that allows users with basic computer skills to perform tasks without extensive training.
- Data Integrity and Backup – The system shall ensure that all stored data is accurate and consistent, and it shall perform regular data backups to allow recovery in case of system failure.

c. Usability Requirements

The system is designed to ensure ease of use for all identified user roles, including the Administrator, Manager, Field Technician, and Office Staff.

- Ease of Use – The system interface shall be simple and organized so that users can easily navigate different system modules without difficulty.

- Learnability – Users shall be able to understand and operate the system with minimal training. A brief walkthrough or orientation will be provided by the system administrator or the research team before first-time use.
- Secure Access – Each user must have a valid username and password provided by the system administrator. Users will log in using their assigned credentials before they can access any feature of the system.
- Role-Based Access – Each user role shall have access only to the features and data relevant to their responsibilities within the organization.
- Basic Computer Literacy – Users are expected to have basic computer literacy, including the ability to use a web browser, navigate web pages, fill out online forms, and perform simple actions such as clicking buttons and reading tables or dashboards.
- Device Compatibility – The system shall be accessible on both desktop and mobile devices, as long as the device is equipped with a modern and up-to-date web browser.
- Internet Connectivity – Since the system is web-based, users must have a stable internet connection to load pages, save data, and generate reports.

d. Use Case Model

This section presents the use case model of the Web-Based Employee Performance Tracking System, illustrating how the four actors—Admin, Manager, Field Technician, and Office Staff—interact with the system's functionalities. It shows key processes such as task assignment, workload monitoring, work

submission and review, performance evaluation, and report generation, providing a clear overview of user interactions within the system.

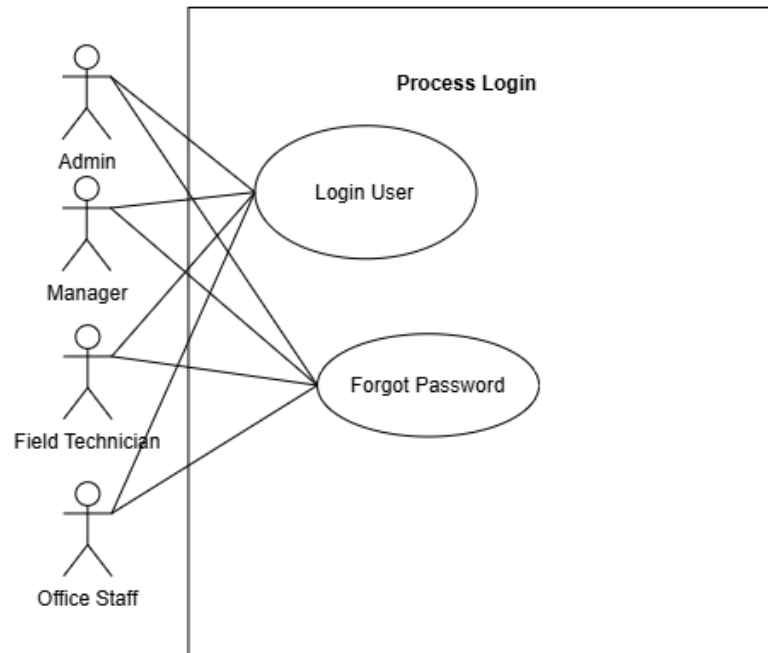


Diagram 3.2: Process Login

Written Use Case:

Use Case Name: Login User

Purpose: To let a registered employee sign in with their username and password, and be taken to the home page for their role.

Actors: Admin, Manager, Field Technician, Office Staff

Input Parameters

- Username
- Password

Output Parameters

- The user is signed in and taken to the home page for their role.

- Success or error message.

Pre-Condition

- The user has an account with a username and a password. Both the account and the employee record it belongs to are Active.

Post-Condition

- A session is written holding the user's employee id, full name, username and role. The role is what every page permission is checked against.
- The date and time of the sign-in are recorded on the account.

Successful Scenario

1. The user opens the Login page
2. The user types the username and the password.
3. The user clicks Log in.
4. The system finds the account that matches the username.
5. The system checks that the account is Active and that the employee record attached to it is Active.
6. The system checks the password against the hash stored for the account.
7. The system clears the session and writes the user's employee id, full name, username and role into it.
8. The system records the time of the sign-in on the account.
9. The system sends the user to their own home page: the Admin and the Manager to the Dashboard, the Field Technician to their job ticket list, and the Office Staff to their task list.

Exception Scenario

- If the username or the password is left empty, the system shows Please enter both a username and password.
- If the credentials are wrong or the account is inactive, the system shows Invalid username or password in every case, so the form cannot reveal which usernames exist..

Additional Remarks

- The password is never kept in a readable form. Only a hashed version and its salt are stored.
- Usernames are generated by the system when the account is created, so the user does not choose their own.

Written Use Case:

Use Case Name: Forgot Password

Purpose: To let a user who cannot sign in set a new password using a code sent to their email.

Actors: Admin, Manager, Field Technician, Office Staff

Secondary Actor: Email Service Provider

Input Parameters

- Email
- Verification code
- New password
- Confirm new password

Output Parameters

- The password is replaced and the user is returned to the Login page.
- Success or error message.

Pre-Condition

- The user has an active account, and the employee record it belongs to is active.

Post-Condition

- The new password replaces the old one, and the old one stops working.
- The user is returned to the Login page with a confirmation.

Successful Scenario

1. The user clicks Forgot Password on the Login page.
2. The user types the email on their employee record.
3. The system generates a four-digit code.
4. The system sends the code to that email through the email service provider.
5. The user types the code.
6. The system checks the code against the one it sent.
7. The user types the new password and confirms it.
8. The system saves the new password.
9. The system returns the user to the Login page with Your password was changed. Please log in with your new password.

Exception Scenario

- If no active account has that email, the system shows No active account was found with that email address.

- If the code is wrong, the system says how many attempts are left. After five wrong codes, or once the code has expired, the user must request a new one.
- If the new password is not 8 to 10 digits, or the two entries do not match, the password is not changed and a message appears.

Additional Remarks

- The code lasts ten minutes, can only be used once, and is never saved to the database.
- Setting a new password does not sign the user in. The user returns to the Login page and signs in with it.

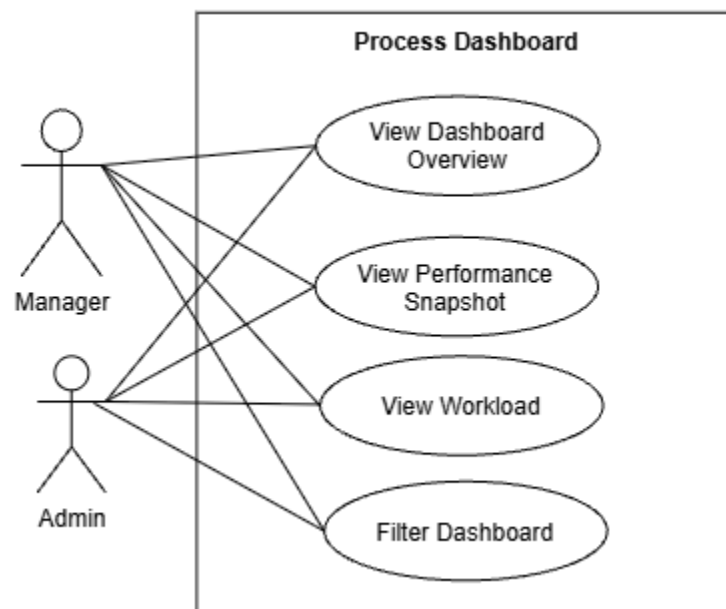


Diagram 3.3: Process Dashboard

Written Use Case:

Use Case Name: View Dashboard Overview

Purpose: To let the Manager and the Admin see what needs their attention and how much work the period holds.

Actors: Manager, Admin

Input Parameters

None

Output Parameters

- The items needing attention and the work counts for the period.

Pre-Condition

- The user is signed in.

Post-Condition

- The overview is shown for the current month, or for whichever period is selected.

Successful Scenario

- The user opens the Dashboard.
- The system shows the items needing attention with a count for each: pending reschedule requests, activities awaiting review, activities rejected and stalled, draft evaluations, employees not yet evaluated this period, and tickets completed this period.
- The system shows how many job tickets and office tasks fall in the period.
- The user may click an item to go straight to the records it counts, already filtered.
- The system shows the score trend across the last twelve months, or the last eight quarters when a quarter is selected.

Exception Scenario

- If nothing needs attention, the counts read zero and are shown as an all-clear.
- If no evaluation has been finalized yet, the trend has nothing to draw.

Additional Remarks

- The Manager and the Admin see the same dashboard. Only the sidebar differs.
- Completed tickets this period is shown as information rather than as something to act on.
- The trend covers every period on record, not just the one selected, since it is there to show the direction over time.

Written Use Case:

Use Case Name: View Performance Snapshot

Purpose: To let the Manager and the Admin see how the workforce scored for the period and how much of it has been evaluated.

Actors: Manager, Admin

Input Parameters

- None

Output Parameters

- The average score, the evaluation coverage, and the highest and lowest performers.

Pre-Condition

- The user is signed in and the Dashboard is open.

Post-Condition

- The snapshot is shown for the selected period.

Successful Scenario

1. The system counts the employees who can be evaluated and the ones evaluated in the period.
2. The system shows the average score for the period, along with how many of the eligible employees it covers.
3. The system shows the highest performer and the lowest performer with their score and rating.

Exception Scenario

- If nobody has been evaluated in the period, there is no average and no performers to name.
- If only one employee has been evaluated, only the highest performer is shown, since the lowest would be the same person.

Additional Remarks

- Only finalized evaluations count. A draft is still work in progress.
- Only Field Technicians and Office Staff are counted, since Admins and Managers are never evaluated.
- The coverage is shown alongside the average so the average is read against how many people it is drawn from.
- Over a quarter an employee may hold more than one evaluation, so the highest and lowest name an employee by their average rather than by a single evaluation.

Written Use Case:

Use Case Name: View Workload

Purpose: To let the Manager and the Admin see how work is spread across the workforce and who is overloaded or idle.

Actors: Manager, Admin

Input Parameters

- None

Output Parameters

- The workload of every employee, heaviest first.

Pre-Condition

- The user is signed in and the Dashboard is open.

Post-Condition

- The workload distribution is shown.

Successful Scenario

1. The system works out each employee's workload from their active and overdue work.
2. The system lists every employee with their role and workload, heaviest first, each marked Idle, Normal or Heavy.

Exception Scenario

- If there are no active employees to show, nothing is listed.

Additional Remarks

- Office staff and field technicians are scored the same way as on the Workload Monitoring page, so a band means the same thing on both.
- Idle is flagged rather than shown as good news, since someone with too little work is as much worth knowing about as someone with too much.

Written Use Case:

Use Case Name: Filter Dashboard

Purpose: To let the Manager and the Admin change the period the dashboard covers.

Actors: Manager, Admin

Input Parameters

- Period type
- Month or quarter
- Year

Output Parameters

- The dashboard rebuilt for the chosen period.

Pre-Condition

- The user is signed in and the Dashboard is open.

Post-Condition

- The attention items, the work counts, the performance snapshot and the score trend are all rebuilt for the chosen period.

Successful Scenario

1. The user chooses either a month or a quarter, and the year.
2. The system rebuilds the dashboard for that period.

Exception Scenario

- If nothing falls in the chosen period, the counts read zero.

Additional Remarks

- The dashboard opens on the current month.
- The years offered are the last five, along with any year already selected.
- Choosing a quarter also changes the trend from twelve months to eight quarters.
- The workload distribution is not tied to the period. It always shows the work outstanding right now.

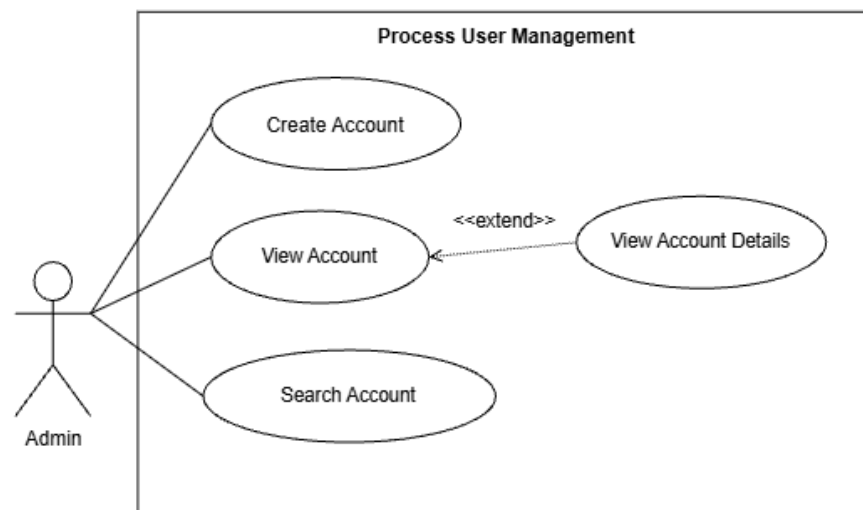


Diagram 3.4: Process User Account Management

Written Use Case:

Use Case Name: Create Account

Purpose: To let the Admin create a user account for an existing employee.

Actor: Admin

Input Parameters

- Selected employee
- Password
- Confirm password

Output Parameters

- The account is created and appears in the account list.

Pre-Condition

- The Admin is signed in.
- The employee record already exists, created through Employee Management.

Post-Condition

- The account is saved with the status Active and a username generated by the system.

Successful Scenario

1. The Admin opens the User Management page.
2. The Admin chooses the employee the account is for.
3. The Admin types the password and confirms it.
4. The Admin clicks Create Account.
5. The system generates the username from the employee's role and id.
6. The system saves the account and returns to the account list.

Exception Scenario

- If the password is shorter than 8 digits, the browser blocks the submission and shows, Please match the requested format.

- If the two entries do not match, the form shows, Passwords do not match. and the account is not created.
- If the Admin clicks Cancel, nothing is saved.

Additional Remarks

- The password field accepts numbers only and stops at 10 digits, so anything else cannot be typed in the first place.
- The Admin does not choose the username. The system builds it from the role and the employee id, for example Pakonek-A001.

Written Use Case:

Use Case Name: View Account

Purpose: To let the Admin see the user accounts and manage the access settings of any one of them.

Actor: Admin

Input Parameters

- Selected account
- New password
- Confirm new password

Output Parameters

- The list of accounts, and the details of the one selected.
- Success or error message.

Pre-Condition

- The Admin is signed in.

Post-Condition

- If the password was reset, the new one replaces the old and the old stops working.
- If the account was disabled, it can no longer sign in

Successful Scenario

1. The Admin opens the User Management page.
2. The system lists the user accounts.
3. The Admin may open an account to view its details.
4. The system shows the account overview: the username, the date the account was created, the date the employee was hired, and the last sign-in.
5. The system shows the employee's information: full name, email, role, department, account status and employee status.
6. The Admin may optionally reset the password by typing a new one and confirming it. The system shows *Password reset successfully*.
7. The Admin may optionally disable the account, or open the employee's record to edit it.

Exception Scenario

- If no accounts exist yet, the list comes back empty.
- If the new password is shorter than 8 digits, the form is not submitted and a message appears on that field.
- If the two entries do not match, the system shows Passwords do not match. and the password is not changed

Additional Remarks

- Disabling takes effect immediately and is not confirmed first. A disabled account cannot sign in, and the login page shows Invalid username or password rather than saying the account is disabled.
- Editing the employee's information is done in Employee Management, not here.

Written Use Case:

Use Case Name: Search Account

Purpose: To let the Admin find a user account.

Actor: Admin

Input Parameters

- Search keyword

Output Parameters

- Matching accounts.

Pre-Condition

- The Admin is signed in.

Post-Condition

- Matching accounts are listed.

Successful Scenario

1. The Admin types a keyword in the search box.
2. The system compares the keyword with the username and the employee's name.
3. The system lists the matches.

Exception Scenario

- If nothing matches, the list comes back empty.
- If the search box is cleared, all accounts are listed.

Additional Remarks

- The list narrows as the Admin types. There is no button to press.

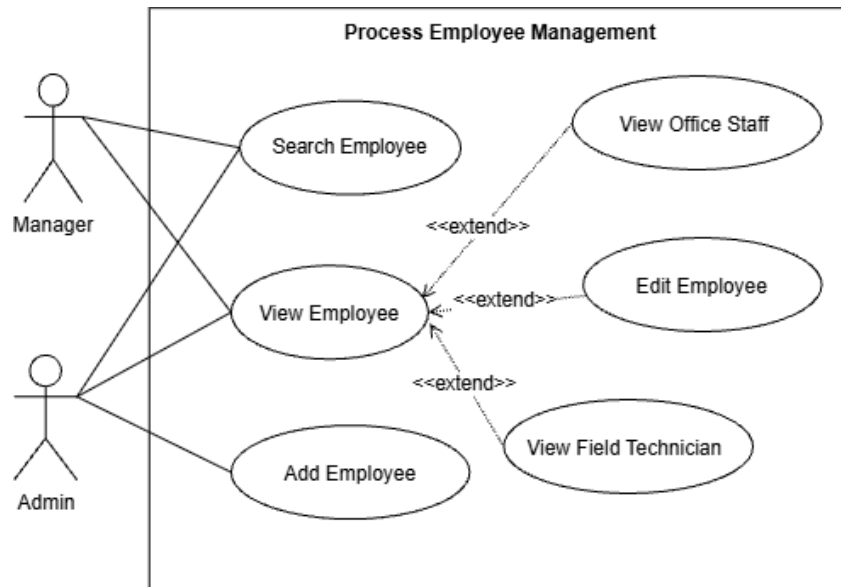


Diagram 3.5: Process Employee Management

Written Use Case:

Use Case Name: Add Employee

Purpose: To let the Admin create an employee record.

Actor: Admin

Input Parameters

- Full name
- Email
- Role type

- Department

Output Parameters

- The employee is created and appears in the employee list.

Pre-Condition

- The Admin is signed in.
- There is at least one active department for the role type being chosen.

Post-Condition

- The employee is saved with the status Active.

Successful Scenario

1. The Admin opens the Employees page and clicks New Employee.
2. The system shows the form with the active departments to choose from.
3. The Admin types the full name and the email, and chooses the role type and the department.
4. The Admin clicks Save.
5. The system saves the employee and returns to the employee list.

Exception Scenario

- If a required field is left empty, the form is not submitted and a message appears on that field.
- If the full name contains anything other than letters, spaces, apostrophes or hyphens, the system shows Full name should only contain letters, spaces, apostrophes, or hyphens.

- If the full name is already used by another employee, the system shows This full name is already taken.
- If the email is not written correctly, the system shows Please enter a valid email address.
- If the email is not from an allowed provider, the system shows that the domain is not allowed.
- If the email is already used by another employee, the system shows This email is already taken.
- If the department does not belong to the chosen role type, the system shows Select a department that matches the chosen role type.

Additional Remarks

- The email must be from one of the allowed providers: gmail.com, yahoo.com, outlook.com, hotmail.com, icloud.com, live.com, msn.com or mail.com.
- Creating the employee does not create their account. That is done separately in User Management.

Written Use Case:

Use Case Name: View Employee

Purpose: To let the Admin and the Manager see the employee records, and to let the Admin update any one of them.

Actors: Admin, Manager

Input Parameters

- Selected employee
- Full name
- Email
- Role type
- Department
- Active

Output Parameters

- The list of employees, and the changes saved to any one of them.
- Success or error message.

Pre-Condition

- The user is signed in.

Post-Condition

- The employees are listed in order of name.
- If an employee was edited, the updated details are saved.

Successful Scenario

1. The user opens the Employees page.
2. The system lists every employee in order of name, with their email, department, role type, date hired and whether they are active.
3. The user may narrow the list to the office staff or the field technicians.
4. The Admin may open an employee to edit their details.
5. The system fills the form with the employee's current details.
6. The Admin changes what needs changing and clicks Save.
7. The system saves the changes and returns to the employee list.

Exception Scenario

- If no employees exist yet, the list comes back empty.
- If a required field is left empty, the form is not submitted and a message appears on that field.
- If the full name contains anything other than letters, spaces, apostrophes or hyphens, the system shows Remove the Symbols, Numbers, or Characters (1!@#\$\$%...)
- If the email now belongs to another employee, the system shows This email is already taken.
- If the email is not written correctly or is not from an allowed provider, the system asks for a valid address.
- If the department does not belong to the chosen role type, the system shows Select a department that matches the chosen role type.

Additional Remarks

- Only the Admin can add or change employees. The Manager sees them only.
- Setting an employee to inactive stops them signing in, since the login checks the employee record as well as the account.

Written Use Case:

Use Case Name: Search Employee

Purpose: To let the Admin and the Manager find an employee.

Actors: Admin, Manager

Input Parameters

- Search keyword

Output Parameters

- Matching employees.

Pre-Condition

- The user is signed in.

Post-Condition

- Matching employees are listed.

Successful Scenario

1. The user types a keyword in the search box.
2. The system compares the keyword with the full name and the email.
3. The system lists the matches in order of name.

Exception Scenario

- If nothing matches, the list comes back empty.
- If the search box is left empty, all employees are listed.

Additional Remarks

- The search looks at the full name and the email only, not the department.

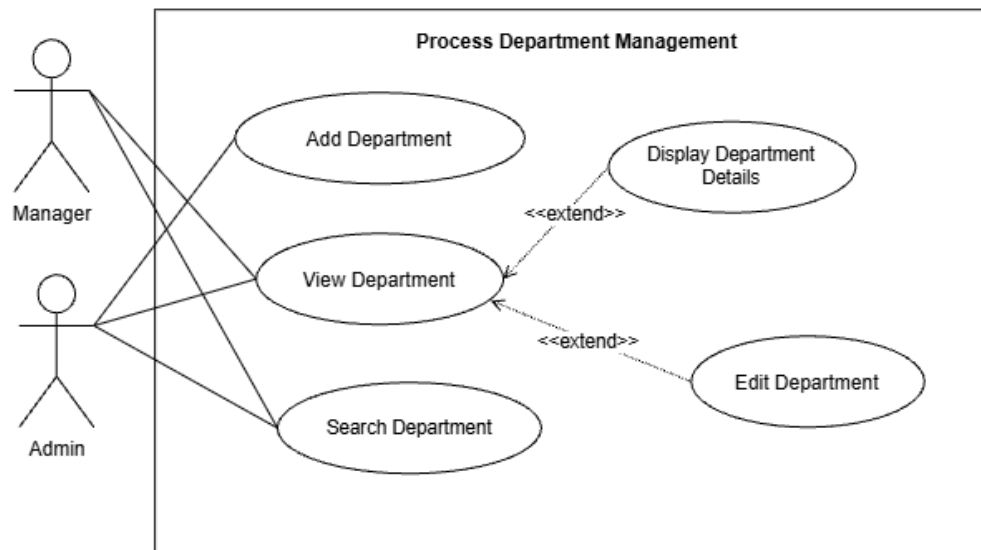


Diagram 3.6: Process Department Management

Written Use case:

Use Case Name: Add Department

Purpose: To let the Admin create a department.

Actor: Admin

Input Parameters

- Department name
- Description
- Role type
- Active

Output Parameters

- The department is created and appears in the department list.

Pre-Condition

- The Admin is signed in.

Post-Condition

- The department is saved and can be chosen when creating an employee of its role type.

Successful Scenario

1. The Admin opens the Departments page and clicks New Department.
2. The Admin types the department name and the description, and chooses the role type.
3. The Admin clicks Save.
4. The system saves the department and returns to the department list.

Exception Scenario

- If a required field is empty, the system shows a message beside that field.

Additional Remarks

- A department belongs to one role type. An employee can only be placed in a department that matches their own role type.
- Only active departments are offered when adding an employee.

Written Use Case:

Use Case Name: View Department

Purpose: To let the Admin and the Manager see the departments and the employees in them, and to let the Admin update any one of them.

Actor: Admin, Manager

Input Parameters

- Selected department
- Department name
- Description
- Role type
- Active

Output Parameters

- The list of departments, the employees in the one selected, and the changes saved to it.
- Success or error message

Pre-Condition

- The user is signed in.

Post-Condition

- The departments are listed in order of name.
- If a department was edited, the updated details are saved.

Successful Scenario

1. The user opens the Departments page.
2. The system shows every department as a card with its name, description, role type, whether it is active, and how many employees are in it.
3. The user may view a department's details, and the system shows its name, the number of employees in it, and each employee's name, email, role and status.

4. The Admin may open a department to edit it.
5. The system fills the form with the department's current details.
6. The Admin changes what needs changing and clicks Save.
7. The system saves the changes and returns to the department list.

Exception Scenario

- If no departments exist yet, no cards are shown.
- If the department has no employees, none are listed in its details.
- If a required field is left empty, the form is not submitted and a message appears on that field.

Additional Remarks

- Only the Admin can add or change departments. The Manager sees them only.
- Setting a department to inactive keeps it off the list offered when adding a new employee, but the employees already in it stay where they are.

Written Use Case:

Use Case Name: Search Department

Purpose: To let the Admin and the Manager find a department by name.

Actors: Admin, Manager

Input Parameters

- Search keyword

Output Parameters

- Matching departments.

Pre-Condition

- The user is signed in.

Post-Condition

- Matching departments are listed.

Successful Scenario

1. The user types a keyword in the search box and submits it.
2. The system compares the keyword with the department name.
3. The system lists the matches in order of name.

Exception Scenario

- If nothing matches, no cards are shown.
- If the search box is left empty, all departments are listed.

Additional Remarks

- The search looks at the department name only, not the description or the role type.

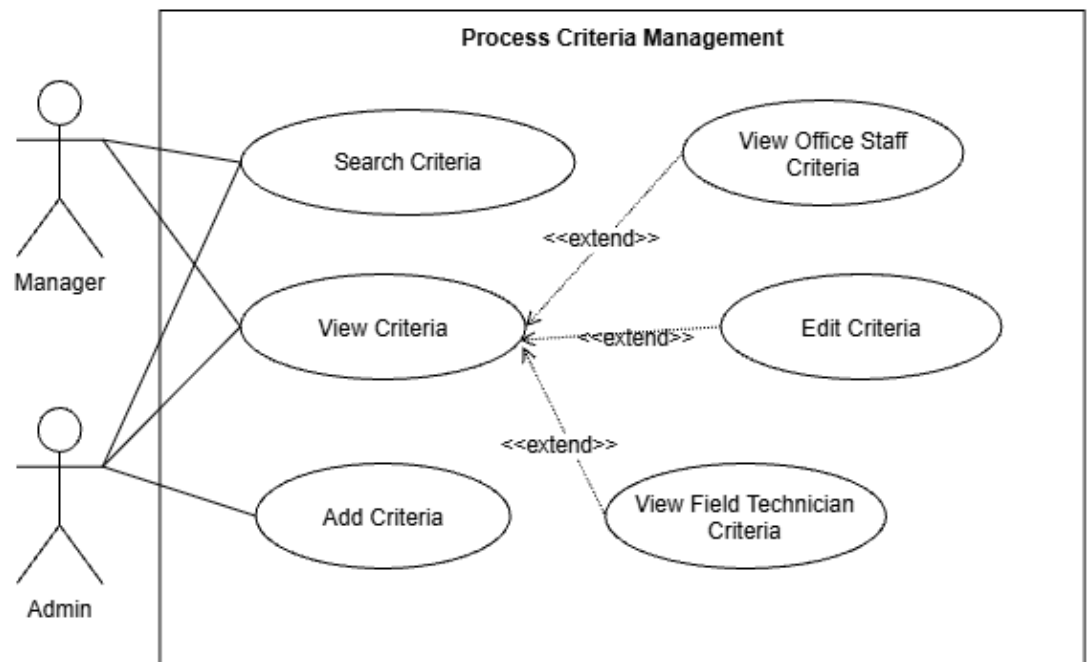


Diagram 3.7: Process Criteria Management

Written Use Case:

Use Case Name: Add Criteria

Purpose: To let the Admin create a performance criterion for a role type.

Actor: Admin

Input Parameters

- Criteria name
- Description
- Weight
- Role type
- Active

Output Parameters

- The criterion is created and appears in the criteria list.

Pre-Condition

- The Admin is signed in.

Post-Condition

- The criterion is saved and counts toward the weight of its role type.

Successful Scenario

1. The Admin opens the Criteria page and clicks Add.
2. The system shows the form together with how much weight is already used by the active office staff criteria and the active field technician criteria.
3. The Admin types the criteria name, sets the weight, and chooses the role type.
4. The Admin clicks Save.
5. The system saves the criterion and returns to the criteria list.

Exception Scenario

- If a required field is left empty, the form is not submitted and a message appears on that field.
- If the name is already used by another criterion, the system says that a criterion of that name already exists and asks for a different one.
- If the weight is zero or less, the system shows Please set a weight greater than 0 - a criterion can't be saved without one.
- If the weight would push that role type's active criteria past 100%, the system says what the total would come to and asks the Admin to lower the weight or deactivate another criterion first.

Additional Remarks

- The active criteria of a role type can never total more than 100%, because the weights are what the overall score is worked out from.
- A criterion that is not active does not count toward the 100%.

Written Use Case:

Use Case Name: View Criteria

Purpose: To let the Admin and the Manager see the performance criteria, and to let the Admin update any one of them.

Actors: Admin, Manager

Input Parameters

- Selected criterion
- Criteria name
- Description
- Weight
- Role type
- Active

Output Parameters

- The list of criteria, and the changes saved to any one of them.
- Success or error message.

Pre-Condition

- The user is signed in.

Post-Condition

- The criteria are listed in order of name.

- If a criterion was edited, the updated details are saved.

Successful Scenario

1. The user opens the Criteria page.
2. The system lists every criterion in order of name, with its description, role type, weight and whether it is active.
3. The user may narrow the list to the office staff criteria or the field technician criteria.
4. The Admin may open a criterion to edit it.
5. The system fills the form with its current details and shows how much weight the other active criteria of each role type already use.
6. The Admin changes what needs changing and clicks Save.
7. The system saves the changes and returns to the criteria list.

Exception Scenario

- If no criteria exist yet, the list comes back empty.
- If a required field is left empty, the form is not submitted and a message appears on that field.
- If the name is already used by another criterion, the system says so and asks for a different one.
- If the weight is zero or less, the system shows Please set a weight greater than 0 - a criterion can't be saved without one.
- If the weight would push that role type's active criteria past 100%, the system says what the total would come to and asks the Admin to lower the weight or deactivate another criterion first.

Additional Remarks

- Only the Admin can add or change criteria. The Manager sees them only.
- Setting a criterion to inactive frees up its weight for the rest of its role type. The criterion being edited is not counted against itself, so keeping its own name or weight is never treated as a clash.

Written Use Case:

Use Case Name: Search Criteria

Purpose: To let the Admin and the Manager find a criterion by name.

Actors: Admin, Manager

Input Parameters

- Search keyword

Output Parameters

- Matching criteria.

Pre-Condition

- The user is signed in.

Post-Condition

- Matching criteria are listed.

Successful Scenario

1. The user types a keyword in the search box and submits it.
2. The system compares the keyword with the criteria name.
3. The system lists the matches in order of name.

Exception Scenario

- If nothing matches, the list comes back empty.
- If the search box is left empty, all criteria are listed.

Additional Remarks

- The search looks at the criteria name only, not the weight or the role type.

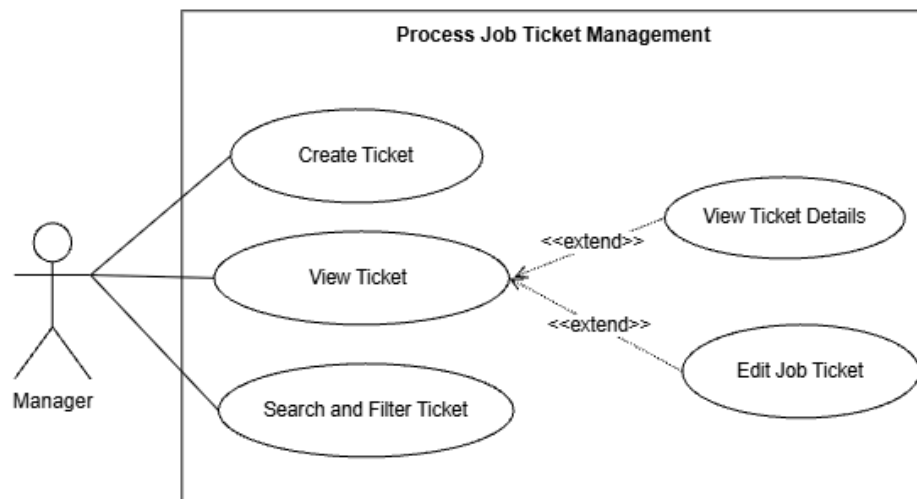


Diagram 3.8: Process Job Ticket Management

Written Use Case:

Use Case Name: Create Ticket

Purpose: To let the Manager create a job ticket and assign it to field technicians.

Actor: Manager

Input Parameters

- Job type
- Client full name
- Primary number

- Secondary number
- Fiber plan
- Description
- Service date
- Due date
- Location address
- Nearest landmark
- Assigned field technicians
- Team leader

Output Parameters

- The ticket is created and appears in the ticket list.

Pre-Condition

- The Manager is signed in.
- There is at least one active field technician who has a user account.

Post-Condition

- The ticket is saved with the status Pending and a ticket number generated by the system.
- The Manager who created it is recorded on the ticket.
- Each assigned field technician receives a notification about the new job.

Successful Scenario

1. The Manager opens the Job Tickets page and clicks Create.
2. The system shows the form with the next ticket number, the service date set to today, and the due date set two days ahead.

3. The Manager selects the job type.
4. The Manager fills in the client and location details, and either the fiber plan or the description depending on the job type.
5. The Manager sets the service date and the due date.
6. The Manager adds the field technicians and chooses which one will lead the job.
7. The Manager clicks Save Job Ticket.
8. The system generates the ticket number and saves the ticket as Pending.
9. The system records the assignments and the leader, and notifies each assigned technician.
10. The system returns to the ticket list

Exception Scenario

- If a required field is left empty, or the service date is earlier than today, the form is not submitted and a message appears on that field.
- If the service date is earlier than today, the form is not submitted and a message appears on that field.
- If the due date is earlier than the service date, the system shows Date of completion cannot be before the service date.
- If a contact number is not an 11-digit Philippine mobile number, the system shows Enter a valid 11-digit Philippine mobile number (e.g. 09171234567).
- If no field technician is assigned, the system shows Please assign at least two field technician.

Additional Remarks

- The ticket number is generated by the system in the form JOB-YYYY-0001. The Manager only sees it as a preview and cannot change it.
- Only active technicians who already have a user account appear in the list of assignees, since a technician without an account has no way to see the job.
- Maintenance jobs do not collect client contact details.
- An Installation records a fiber plan and no description. Every other job type records a description and no fiber plan. Maintenance jobs do not collect client contact details

Written Use Case:

Use Case Name: View Ticket

Purpose: To let the Manager see the job tickets, open any one of them, and update or reschedule it.

Actor: Manager

Input Parameters

- Selected ticket
- Job type
- Client full name
- Primary number
- Secondary number
- Fiber plan

- Description
- Service date
- Due date
- Location address
- Nearest landmark
- Team leader

Output Parameters

- The list of tickets, the details of the one selected, and the changes saved to it.
- Success or error message.

Pre-Condition

- The Manager is signed in.

Post-Condition

- The tickets are listed, newest first.
- If a ticket was edited, the updated details are saved.
- If the service date was changed, the ticket goes back to Pending, its remarks are cleared, and whatever was already submitted is kept in the reschedule history and the submission history.

Successful Scenario

1. The Manager opens the Job Tickets page.
2. The system lists the tickets, newest first.
3. The Manager may open a ticket to view its details.

4. The system shows the ticket details, the assigned technicians and their leader, and the Manager who created it.
5. The system shows the history of reschedules and the history of submissions, newest first, along with the remarks and files kept with each one.
6. The Manager may optionally download a file a technician submitted.
7. The Manager may open the ticket for editing, change what needs changing, and choose a different leader from among the technicians already assigned.
8. The Manager clicks Save Job Ticket, and the system saves the changes and returns to the ticket list.

Exception Scenario

- If no tickets exist yet, the list comes back empty.
- If the ticket is already Completed or Cancelled, the Manager is sent to the details page instead and cannot edit it.
- If the ticket has a pending reschedule request and the Manager saves without changing the service date, the system shows This job order has a pending reschedule request. Please set a new service date to resolve it.
- If the service date is being changed to a day earlier than today, the system shows Date cannot be rescheduled to a previous day.
- If the due date is earlier than the service date, the system shows Date of completion cannot be before the service date.

Additional Remarks

- The technicians assigned to a ticket cannot be changed here. Only who leads them can change, and only while the ticket is Pending, Overdue or In Progress.
- Changing the service date is how a reschedule happens, and it is also how the Manager answers a technician's reschedule request. Both are the same action and the Manager does not have to give a reason.
- The status is never taken from the form. It only changes through a reschedule, through the field technician's own work, or when the Manager closes the ticket.

Written Use Case:

Use Case Name: Search and Filter Ticket

Purpose: To let the Manager narrow the ticket list down to the ones they are looking for.

Actor: Manager

Input Parameters

- Search keyword
- Status

Output Parameters

- Matching tickets.

Pre-Condition

- The Manager is signed in and the ticket list is open.

Post-Condition

- Matching tickets are listed.

Successful Scenario

1. The Manager types a keyword in the search box, chooses a status, or both.
2. The system narrows the list to the tickets that match.

Exception Scenario

- If nothing matches, the list comes back empty.
- If the keyword is cleared and the status is set back to all, the full list returns.

Additional Remarks

- The list narrows as the Manager types. There is no button to press.

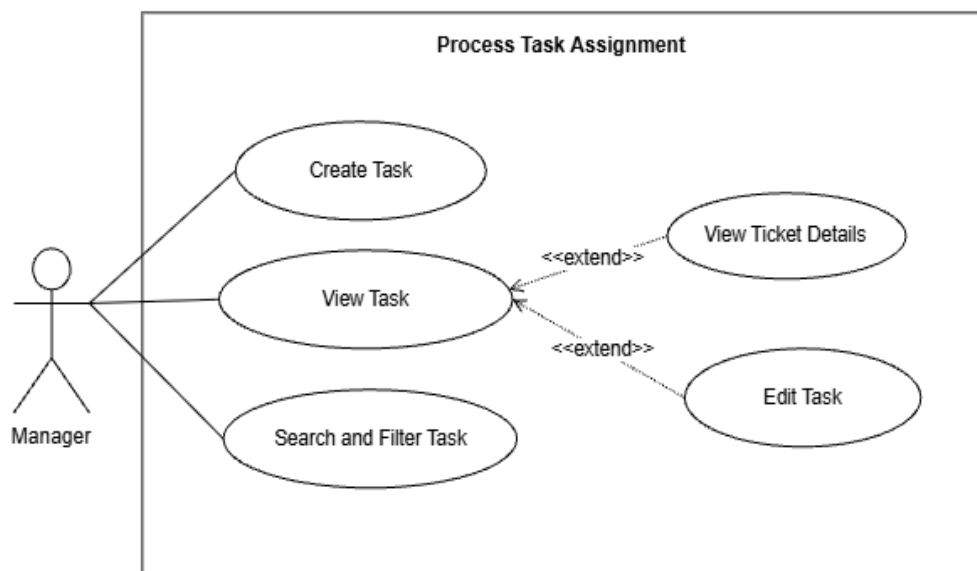


Diagram 3.9: Process Task Assignment

Written Use Case:

Use Case Name: Create Task

Purpose: To let the Manager create an office task, break it into activities, and assign it to office staff.

Actor: Manager

Input Parameters

- Task name
- Description
- Due date
- Assigned employees
- Activities

Output Parameters

- The task is created and appears in the task list.

Pre-Condition

- The Manager is signed in.
- There is at least one active office staff who has a user account.

Post-Condition

- The task is saved with the status Pending, no progress, and a task number generated by the system.
- Each activity is saved as Pending.
- Each assigned employee receives a notification about the new task.

Successful Scenario

1. The Manager opens the Office Task page and clicks Create.
2. The system shows the form with the next task number and the due date set two days ahead.

3. The Manager fills in the task name, the description and the due date.
4. The Manager adds the employees who will handle the task.
5. The Manager adds the activities that make up the task.
6. The Manager clicks Save.
7. The system generates the task number, saves the task as Pending, records the assignments and the activities, and notifies each assigned employee.
8. The system returns to the task list.

Exception Scenario

- If a required field is left empty, or the due date is earlier than today, the form is not submitted and a message appears on that field.
- If no employee is assigned, the system shows Please assign at least one employee.
- If no activity is added, the system shows Please add at least one activity.

Additional Remarks

- The task number is generated by the system in the form TASK-YYYY-0001. The Manager only sees it as a preview and cannot change it.
- Only active office staff who already have a user account can be assigned, since a staff member without an account has no way to see the task.
- If only one employee is assigned, every activity belongs to them. If more than one is assigned, the activities stay unclaimed until a staff member uploads a file for one.

Written Use Case:

Use Case Name: View Task

Purpose: To let the Manager see the office tasks, review the work submitted for any one of them, and update its details.

Actor: Manager

Input Parameters

- Selected task
- Approve or reject for each activity
- Feedback
- Task name
- Description
- Due date

Output Parameters

- The list of office tasks, the details of the one selected, and the changes saved to it.

Pre-Condition

- The Manager is signed in.

Post-Condition

- The tasks are listed, newest first.
- The reviewed activities are saved with their status and feedback, together with who reviewed them and when.
- The task's progress and status are recalculated from its activities.

Successful Scenario

1. The Manager opens the Office Task page.
2. The system lists the tasks, newest first.
3. The Manager may open a task to view its details.
4. The system shows the task details, the assigned employees, and every activity with its status and the file submitted for it.
5. The Manager may view the history of submissions for an activity, or download a submitted file.
6. The Manager approves or rejects each activity, and types feedback where needed.
7. The Manager clicks Save, and the system saves the reviews, recalculates the task, and returns to the task list.
8. The Manager may instead open the task for editing, change the name, description or due date, and save.

Exception Scenario

- If no tasks exist yet, the list comes back empty.
- If the Manager approves or rejects an activity that has no file, the system shows that it cannot be approved or rejected until the employee attaches a file.
- If the Manager rejects an activity without typing feedback, the system shows that feedback is required.
- If the task is already Completed, the Manager is sent to the details page instead and cannot edit it

Additional Remarks

- Nothing is saved until the Manager clicks Save. Cancel discards every approval and rejection made on the page. Once an activity is approved it is locked and cannot be changed again.
- Progress is the percentage of the task's activities that have been approved. A task becomes Completed when every activity is approved, In Progress once at least one is approved or submitted, and Overdue if it is unfinished past its due date.
- The assigned employees and the activities cannot be changed when editing. Only the name, the description and the due date can.

Written Use Case:

Use Case Name: Search and Filter Task

Purpose: To let the Manager narrow the task list down to the ones they are looking for.

Actor: Manager

Input Parameters

- Search keyword
- Status

Output Parameters

- Matching tasks.

Pre-Condition

- The Manager is signed in and the task list is open.

Post-Condition

- Matching tasks are listed.

Successful Scenario

1. The Manager types a keyword in the search box, chooses a status, or both.
2. The system narrows the list to the tasks that match.

Exception Scenario

- If nothing matches, the list comes back empty.
- If the keyword is cleared and the status is set back to all, the full list returns.

Additional Remarks

- The list narrows as the Manager types. There is no button to press and the page does not reload.

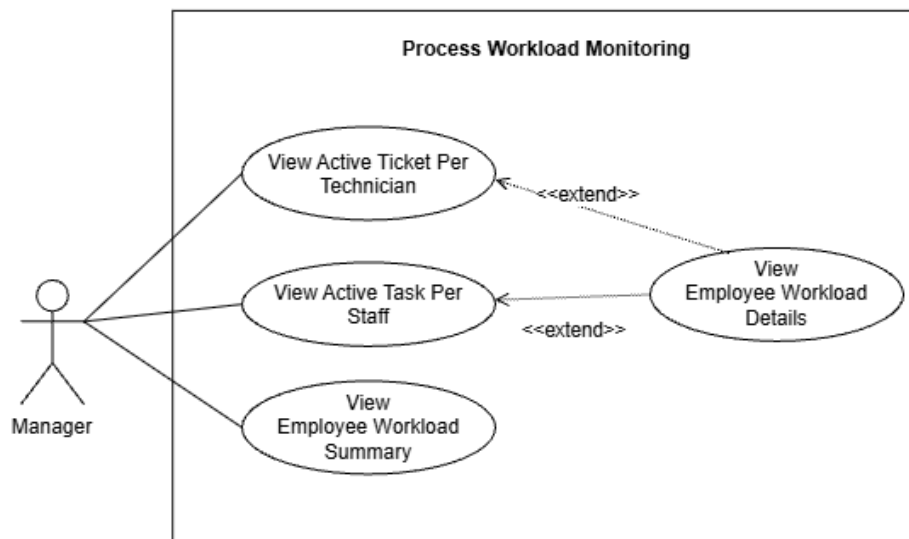


Diagram 3.10: Process Workload Monitoring

Written Use Case:

Use Case Name: View Active Ticket Per Technician

Purpose: To let the Manager see how many active job tickets each field technician is carrying.

Actor: Manager

Input Parameters

- Selected technician

Output Parameters

- The list of field technicians with their active ticket counts, and the active tickets of the one selected.

Pre-Condition

- The Manager is signed in.

Post-Condition

- The technicians are listed, busiest first.

Successful Scenario

1. The Manager opens the Workload Monitoring page.
2. The system lists every active field technician with their department and the number of active tickets, busiest first.
3. The Manager may click View on a technician.
4. The system shows the technician's name, department and number of active tickets, along with the job id, type of job, client, status, date of job and due date of each one.

Exception Scenario

- If a technician has no active tickets, the count reads zero and the view shows No active tickets.

Additional Remarks

- Active means Pending or In Progress. Completed and cancelled tickets are not counted.
- The details open over the list. The page does not reload and nothing is saved

Written Use Case:

Use Case Name: View Active Task Per Staff

Purpose: To let the Manager see how many active office tasks each staff member is carrying.

Actor: Manager

Input Parameters

- Selected staff member

Output Parameters

- The list of office staff with their active task counts, and the active tasks of the one selected.

Pre-Condition

- The Manager is signed in.

Post-Condition

- The staff are listed, busiest first.

Successful Scenario

1. The Manager opens the Active Tasks Per Staff tab.
2. The system lists every active office staff with their department and the number of active tasks, busiest first.
3. The Manager may click View on a staff member.

4. The system shows the staff member's name, department and number of active tasks, along with the task number, task name, status, date created and due date of each one.

Exception Scenario

- If a staff member has no active tasks, the count reads zero and none are listed.

Additional Remarks

- Active means Pending, In Progress or Overdue.
- The details open over the list. The page does not reload and nothing is saved.

Written Use Case:

Use Case Name: View Employee Workload Summary

Purpose: To let the Manager compare workload across the whole workforce and spot who is overloaded and who is idle.

Actor: Manager

Input Parameters

- None

Output Parameters

- The office staff workload table and the field technician workload table.

Pre-Condition

- The Manager is signed in.

Post-Condition

- Both tables are listed, heaviest workload first.

Successful Scenario

1. The Manager opens the Employee Workload Summary tab.
2. The system shows the office staff table with each staff member's active tasks, overdue tasks, completed tasks, pending activities, average score and workload level.
3. The system shows the field technician table with each technician's active tickets, overdue tickets, completed tickets and workload level.
4. Both tables are ordered with the heaviest workload first.

Exception Scenario

- If there are no active employees to show, nothing is listed.

Additional Remarks

- The workload level is worked out from the employee's active and overdue work, and is shown as Light, Moderate or Heavy.
- Office staff and field technicians are scored the same way, so a level means the same thing in both tables.

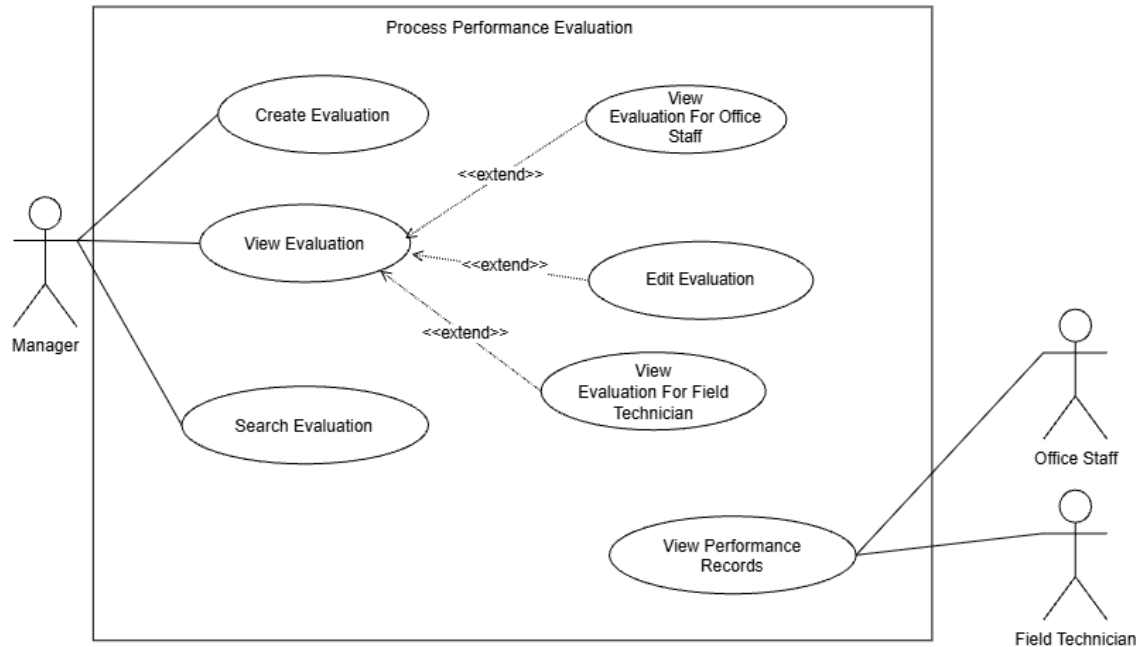


Diagram 3.11: Process Performance Evaluation

Written Use Case:

Use Case Name: Create Evaluation

Purpose: To let the Manager rate an employee against the criteria for their role and record the result for a period.

Actor: Manager

Input Parameters

- Evaluation period month and year
- Employee role
- Employee
- Evaluation date
- Star rating for each criterion
- Feedback for each criterion
- General feedback

Output Parameters

- The evaluation is saved as a draft or finalized.
- Success or error message.

Pre-Condition

- The Manager is signed in.
- There are active criteria for the employee's role type.
- The employee has no evaluation yet for that period.

Post-Condition

- The evaluation is saved with the overall score worked out from the stars and the weights, and the overall rating that goes with it.
- The Manager who saved it is recorded as the evaluator.
- The employee is notified.

Successful Scenario

1. The Manager opens the Performance Evaluation page and clicks Create.
2. The Manager chooses the period, the employee role and the employee.
3. The system shows that employee's performance for the period, and the criteria for their role type with their weights.
4. The Manager rates each criterion in stars and types feedback.
5. The Manager may type general feedback.
6. The Manager clicks Save as Draft or Finalize.
7. The system works out the score for each criterion and adds them into the overall score.
8. The system saves the evaluation and notifies the employee.

9. The system returns to the Create page with a confirmation, keeping the period and role so the next employee can be evaluated straight away.

Exception Scenario

- If no employee is chosen, the form is not submitted and a message appears on that field.
- If the Manager finalizes without rating every criterion, the system says which ones are still unrated.

Additional Remarks

- Only Field Technicians and Office Staff can be evaluated. An employee who already has an evaluation for the chosen period is hidden from the picker, so a duplicate cannot be created.
- A criterion's score is its stars measured against its weight, so a heavier criterion moves the overall score more. The evaluator is always the Manager who is signed in.

Written Use Case:

Use Case Name: View Evaluation

Purpose: To let the Manager see the evaluations that have been recorded, and to finish or correct a draft.

Actor: Manager

Input Parameters

- Selected evaluation
- Evaluation date

- Star rating for each criterion
- Feedback for each criterion
- General feedback

Output Parameters

- The list of evaluations, the details of the one selected, and the changes saved to it.
- Success or error message.

Pre-Condition

- The Manager is signed in.

Post-Condition

- The evaluations are listed, newest first.
- If a draft was edited, the ratings are replaced and the overall score and rating are worked out again.
- If it was finalized, the employee is notified.

Successful Scenario

1. The Manager opens the Performance Evaluation page.
2. The system lists every evaluation, newest first, with the employee, the period, the evaluator, the overall score and rating, and whether it is a draft or finalized.
3. The Manager may narrow the list to the office staff evaluations or the field technician evaluations.
4. The Manager may open an evaluation to see the full rating for every criterion, along with the employee's performance for that period.

5. The Manager may open a draft for editing and change the ratings or the feedback.
6. The Manager clicks Save as Draft or Finalize.
7. The system works out the overall score again, saves the evaluation, and shows its details.

Exception Scenario

- If no evaluations exist yet, the list comes back empty.
- If the evaluation is already finalized, the Manager is sent to the details page instead and cannot edit it.
- If the Manager finalizes without rating every criterion, the system says which ones are still unrated.

Additional Remarks

- The employee and the period cannot be changed when editing. Only the ratings, the feedback and the date can.
- A criterion that has since been deactivated still appears if this evaluation already scored it, so an old score is never quietly dropped.

Written Use Case:

Use Case Name: Search Evaluation

Purpose: To let the Manager find an evaluation.

Actor: Manager

Input Parameters

- Search keyword
- Status

Output Parameters

- Matching evaluations.

Pre-Condition

- The Manager is signed in.

Post-Condition

- Matching evaluations are listed.

Successful Scenario

1. The Manager types a keyword in the search box, chooses a status, or both.
2. The system compares the keyword with the employee's name and the evaluator's name.
3. The system lists the matches, newest first.

Exception Scenario

- If nothing matches, the list comes back empty.
- If the search box is left empty and no status is chosen, all evaluations are listed.

Additional Remarks

- The status choices are Draft and Finalized.

Written Use Case:

Use Case Name: View Performance Records

Purpose: To let an employee see their own evaluation history and feedback.

Actors: Office Staff, Field Technician

Input Parameters

- Evaluation period
- Year

Output Parameters

- The employee's own evaluations and feedback.

Pre-Condition

- The employee is signed in.

Post-Condition

- The employee's records are shown, newest first.

Successful Scenario

1. The employee opens their performance records.
2. The system shows their overall rating across all their finalized evaluations, together with their completed work, on-time rate, and how many were rescheduled or cancelled.
3. The system shows their last evaluation and their evaluation history, newest first, along with any appraisal feedback recorded for them.
4. The employee may narrow the history to one period or year, or open an evaluation to see the rating for every criterion.

a. Exception Scenario

5. If they have not been evaluated yet, the system shows that no rating has been given.
6. If no feedback has been recorded, the system shows that none has been recorded yet.

Additional Remarks

- Only finalized evaluations are shown. A draft is still the Manager's work in progress and is not visible to the employee.
- The page is read-only. An employee only ever sees their own records and can never change them.

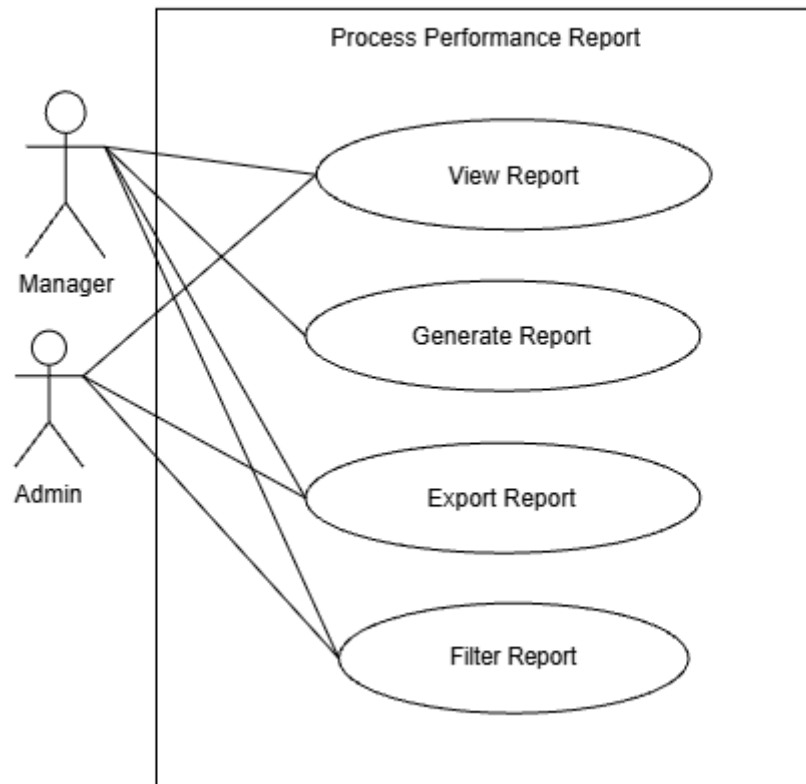


Diagram 3.12: Process Performance Report

Written Use Case:

Use Case Name: View Report

Purpose: To let the Manager and the Admin see how the workforce performed over a period.

Actors: Manager, Admin

Input Parameters

- None

Output Parameters

- The performance report.

Pre-Condition

- The user is signed in.

Post-Condition

- The report is shown for the selected period and department.

Successful Scenario

1. The user opens the Performance Report page.
2. The system shows the number of employees, the appraisals in the period, the average score and the completed work items, along with how many of the listed employees have an evaluation.
3. The system shows a bar chart of the average score of each employee and a pie chart of how many fall into each rating band.
4. The system lists every employee with their role, department, average and rating, best first.

Exception Scenario

- If no employee in the selection has been evaluated, the charts have nothing to draw.

Additional Remarks

- Only finalized evaluations count, and only Field Technicians and Office Staff, since Admins and Managers are never evaluated.

- An employee who has not been evaluated still appears, shown as not yet evaluated, so the report never understates how many people there are. The average is worked out only from those who have an evaluation.

Written Use Case:

Use Case Name: Filter Report

Purpose: To let the Manager and the Admin narrow the report to one period or one department.

Actors: Manager, Admin

Input Parameters

- Period
- Department

Output Parameters

- The report for the chosen period and department.

Pre-Condition

- The user is signed in and the report is open.

Post-Condition

- The report, the charts and the totals are all rebuilt for the chosen filters.

Successful Scenario

1. The user chooses a period, a department, or both.
2. The system rebuilds the report for that selection.

Exception Scenario

- If nobody in the chosen department was evaluated in the chosen period, the employees are still listed but shown as not yet evaluated.

Additional Remarks

- The periods offered are only the ones that actually have a finalized evaluation on record. Leaving the period unset covers every period.
- Only the departments of field technicians and office staff are offered, since no one else appears on the reports.

Written Use Case:

Use Case Name: Generate Report

Purpose: To let the Manager rebuild the report for the filters currently chosen.

Actor: Manager

Input Parameters

- Period
- Department

Output Parameters

- The report for the chosen filters.

Pre-Condition

- The Manager is signed in and the report is open.

Post-Condition

- The report is shown for the chosen filters.

Successful Scenario

1. The Manager chooses the period and the department.

2. The Manager clicks Generate Report.
3. The system builds the report and shows it.

Exception Scenario

- If nobody in the selection has been evaluated, the charts have nothing to draw.

Additional Remarks

- The report is already built when the page opens, and it rebuilds whenever a filter changes, so this is a way of asking for it again rather than the only way of getting it.
- The Admin does not have this button. The Admin's access to this page is to view the report and export it.

Written Use Case:

Use Case Name: Export Report

Purpose: To let the Manager and the Admin save the report as a file.

Actors: Manager, Admin

Input Parameters

- None

Output Parameters

- A CSV file of the report.

Pre-Condition

- The user is signed in and the report is open.

Post-Condition

- The file is downloaded.

Successful Scenario

1. The user clicks Export CSV.
2. The system builds the file from the report currently on screen, carrying the same period and department.
3. The file is downloaded, named with the date and time it was made.

Exception Scenario

- None

Additional Remarks

- The file holds the period, the department, when it was made, the four totals, and then a row for each employee with their role, department, average and rating.
- An employee with no evaluation has an empty average in the file rather than a zero, so they are not mistaken for someone who scored nothing.

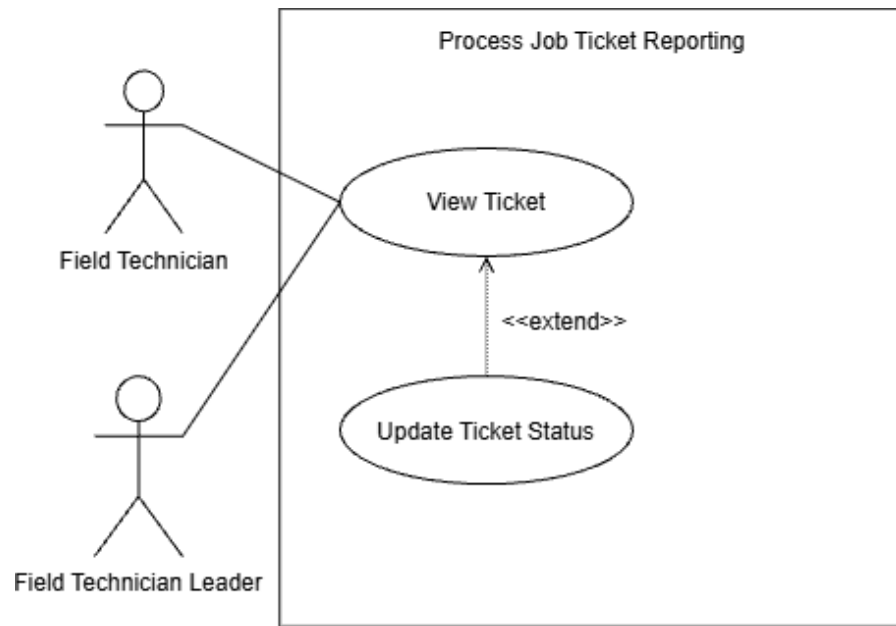


Diagram 3.13: Process Job Ticket Reporting

Written Use Case:

Use Case Name: View Ticket

Purpose: To let an assigned field technician see the job orders given to them, and to let the team leader report progress on one.

Actors: Field Technician, Field Technician Leader

Input Parameters

- Selected job order
- Photo or file
- Status
- Remarks

Output Parameters

- The job order's details.
- The status and remarks are updated.

- Error message.

Pre-Condition

- The technician is signed in and is assigned to the job order.
- To update it, the technician is the assigned leader, the job order is not Completed and has no pending reschedule request, and the service date has arrived.

Post-Condition

- The status and remarks are saved on the job order.
- The update is written to the ticket history together with the files uploaded for it.
- The manager who created the job order is notified.
- If the status is Completed, the date of completion is recorded.
- If a reschedule was requested, the job order is set to Reschedule Request and locked until the manager acts on it.

Successful Scenario

1. The technician opens their job order list.
2. The system lists only the job orders they are assigned to, newest first.
3. The technician selects a job order.
4. The system shows the status, the dates, the description, the location, the assigned team and who leads it, and the remarks.
5. The system shows the ticket history, newest first, with the status, the actor, the remarks and the files attached to each update.
6. The technician may view or download a submitted file.

7. The leader may upload a photo or file as proof, select the status and type the remarks.
8. The leader clicks Save.
9. The system saves the status and the remarks, files the update into the ticket history along with the uploaded files, and notifies the manager.
10. The system returns the leader to their job order list.

Exception Scenario

- If the leader chooses Completed or In Progress without uploading a file, the system says a photo or file must be uploaded first.
- If the leader chooses In Progress, Cancelled, or requests a reschedule without remarks, the system says a reason must be given in the remarks.
- If the leader requests a reschedule without uploading a file, the system says proof must be uploaded first.
- If the service date has not arrived, the system shows that the job order is not accessible yet and names the date it is scheduled for.
- If the job order is already Completed, or a reschedule request is still awaiting the manager, the system says it can no longer be updated.

Additional Remarks

- Only the leader can update the job order. A technician who is not the leader sees the details and the history but has no update controls on the page.
- Uploading a file on its own changes nothing. The status and the remarks are only saved when the leader presses Save. The leader may remove a file, or

cancel the edit, which discards the files uploaded this round after confirming.

- The status choices are Pending, In Progress, Completed, Cancelled and Reschedule Request. Rescheduled is not offered, since only the manager sets it

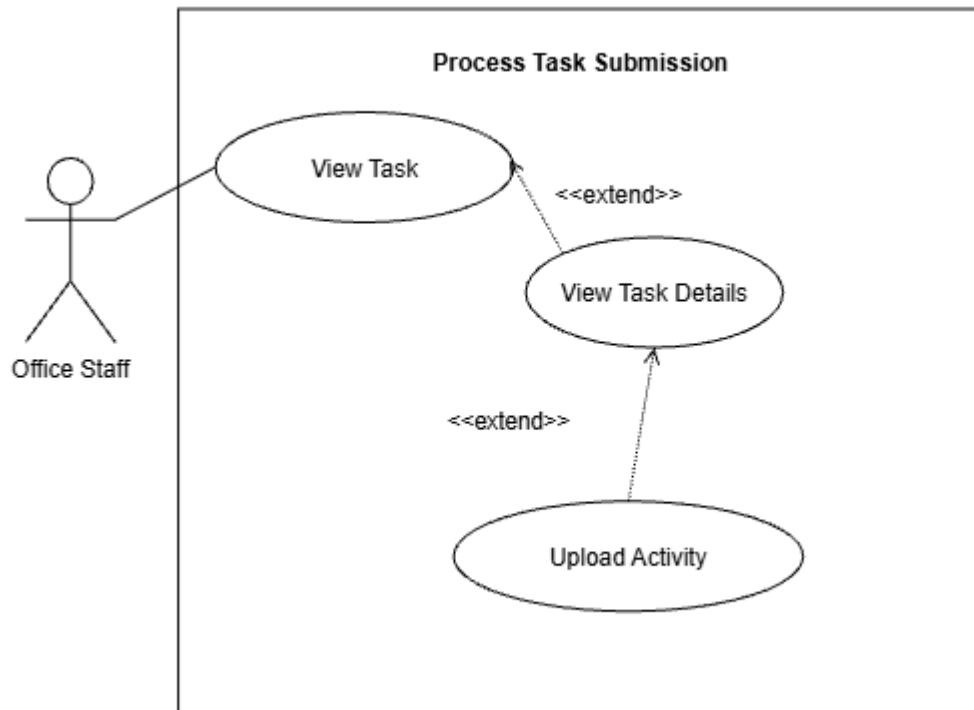


Diagram 3.14: Process Task Submission

Written Use Case:

Use Case Name: View Task

Purpose: To let an office staff member see the tasks assigned to them and submit files as proof of work for their activities.

Actor: Office Staff

Input Parameters

- Selected task

- Selected activity
- File

Output Parameters

- The list of tasks and the details of the one selected.
- The file is submitted and the activity is marked Submitted.
- Error message.

Pre-Condition

- The staff member is signed in and is assigned to the task.

Post-Condition

- The file is saved as a new submission awaiting review.
- The activity is claimed by the staff member who uploaded, and its status becomes Submitted.
- The task's progress and status are recalculated.
- The manager who assigned the task is notified.

Successful Scenario

1. The staff member opens their task list.
2. The system lists only the tasks they are assigned to, newest first, together with the activities of each.
3. The staff member selects a task.
4. The system shows the task details, the employees assigned to it, and the manager who assigned it.
5. The system shows every activity with its status, the file most recently submitted for it, and the manager's feedback.

6. The staff member may view the history of submissions for an activity, or download a submitted file.
7. The staff member chooses a file for an activity and uploads it.
8. The system saves the file as a new submission awaiting review and marks the activity Submitted.
9. The system records the staff member as the one handling that activity.
10. The system recalculates the task and notifies the manager.

Exception Scenario

- If the staff member is not signed in, they are returned to the Login page.
- If no tasks are assigned to them, the list comes back empty.
- If no file is chosen, the system shows *Please choose a file to upload*.

Additional Remarks

- If the task has only one assigned staff member, every activity is theirs. If it has more than one, the first person to upload for an activity claims it, and only they can upload for it afterwards. The upload control does not appear on activities claimed by someone else, or on activities already approved.
- Every upload is kept as its own attempt, so the history shows each file submitted along with the outcome the manager recorded against it. The staff member may remove a file they uploaded by mistake, but only while it is still awaiting review.
- Uploading a new file for a rejected activity clears the old feedback, since that feedback applied to the earlier file.

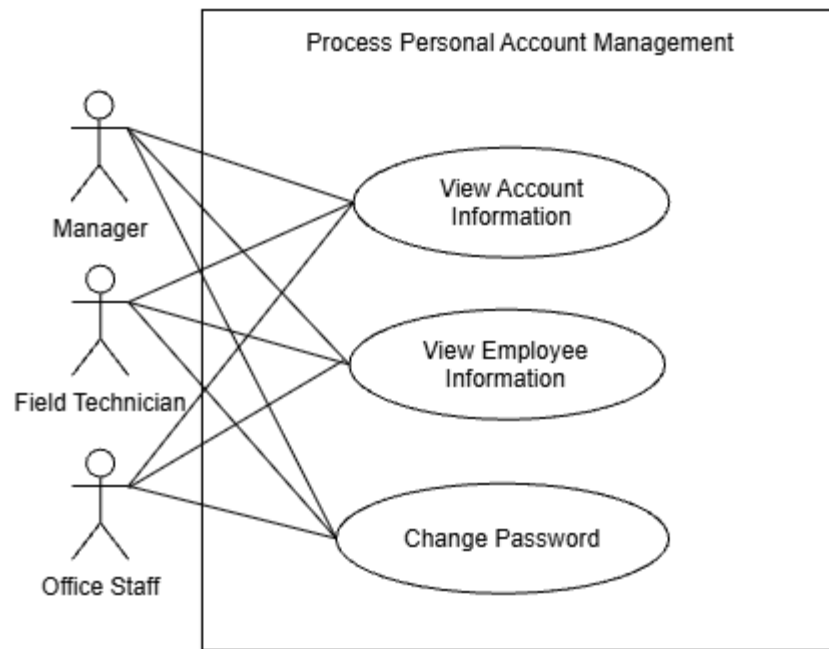


Diagram 3.15: Process Personal Account Management

Written Use Case:

Use Case Name: View Account Information

Purpose: To let a signed-in user see their own account details.

Actors: Manager, Field Technician, Office Staff

Input Parameters

- None

Output Parameters

- The user's own account details.

Pre-Condition

- The user is signed in.

Post-Condition

- The account details are shown.

Successful Scenario

1. The user opens My Account.
2. The system finds the account belonging to the signed-in user.
3. The system shows the account details.

Exception Scenario

- If the session has ended, or the account can no longer be found, the user is returned to the Login page.

Additional Remarks

- A user only ever sees their own account. The page shows the sidebar for whichever role the user holds, so one page serves all three.
- The Admin does not have this page. The Admin's own account is managed through User Management like every other account.

Written Use Case:

Use Case Name: View Employee Information

Purpose: To let a signed-in user see their own employee record and how much work they are carrying.

Actors: Manager, Field Technician, Office Staff

Input Parameters

- None

Output Parameters

- The user's own employee details and workload summary.

Pre-Condition

- The user is signed in.

Post-Condition

- The employee details and the workload summary are shown.

Successful Scenario

1. The user opens My Account.
2. The system shows their employee record together with their department.
3. For a Field Technician, the system shows their active, overdue and completed job tickets, and their workload level.
4. For an Office Staff member, the system shows their active, overdue and completed tasks, their pending activities, their average score, and their workload level.

Exception Scenario

- If the session has ended, the user is returned to the Login page.

Additional Remarks

- The workload is worked out the same way as on the Workload Monitoring page, so a level means the same thing in both places. A Manager has no personal workload here, since job tickets and tasks are assigned to field technicians and office staff.
- The page is read-only. A user cannot change their own employee record. That is done by the Admin in Employee Management.

Written Use Case:

Use Case Name: Change Password

Purpose: To let a signed-in user replace their own password.

Actors: Manager, Field Technician, Office Staff

Input Parameters

- Current password
- New password
- Confirm new password

Output Parameters

- Success or error message.

Pre-Condition

- The user is signed in and knows their current password.

Post-Condition

- The new password replaces the old one, and the old one stops working.

Successful Scenario

1. The user opens My Account.
2. The user types their current password.
3. The user types the new password and confirms it.
4. The user clicks Change Password.
5. The system checks the current password.
6. The system saves the new password and shows Password changed successfully.

Exception Scenario

- If the current password is wrong, the system shows Current password is incorrect.
- If the new password is not 8 to 10 digits, the system shows New password must be 8-10 digits (numbers only).

- If the two entries do not match, the system shows New passwords do not match.
- If the new password is the same as the current one, the system shows New password must be different from your current password.

Additional Remarks

- The current password is asked for so that a password cannot be changed by someone who walks up to an unattended computer.
- This is not the same as Forgot Password, which is for a user who cannot sign in at all and goes through their email instead.

e. Class Model

This section presents the class model of the Web-Based Employee Performance Tracking System, illustrating the structure of the system through its classes, attributes, methods, and relationships. The class model provides a visual representation of how the system's key components, such as the Administrator, Manager, Field Technician and Office Staff are organized and how they interact with one another to support the system's functionalities.

f. Entity-Relationship Model

This section presents the Entity-Relationship (ER) model, which illustrates the system's data structure by showing the relationships between different entities in the database. It includes entities, attributes, and relationships, providing a clear visual representation of how data is organized and connected within the system.

g. Database Schema

This section outlines the database schema, which defines the structure of the database through tables, fields, relationships, and constraints. It provides a detailed view of how data is organized, stored, and accessed within the system, ensuring that data management is efficient and consistent.

h. Implementation Strategies

The implementation strategies toward modernizing the employee performance tracking process at Pakonek Solutions Inc. follow a structured and organized approach to ensure a smooth transition from development to operational use. The process starts from a detailed project timeline. It outlines each phase of development, testing, and deployment so that all milestones are met on schedule, beginning with requirements planning from the first week of January to the second week of February, followed by user design from the third week of February to the fourth week of April, construction from the first week of May to the second week of August, and cutover from the third to the fourth week of August.

Resource allocation plays a major role in this phase because the [number] researchers are assigned to various tasks such as backend development, front-end implementation, database administration, testing, and documentation. This allocation ensures that personnel are adequately distributed and that the tools required at every critical step of the project, including Visual Studio 2022, Microsoft SQL Server, Figma, and Git and GitHub, are available.

During the deployment phase, proactive actions address the risk management topic. As contingency plans to reduce these risks are developed, early identification of possible dangers such as software defects, system downtime, unauthorized access to employee records, loss of stored data, and user resistance occurs. This includes maintaining regular database backups, retaining every uploaded submission as its own attempt instead of overwriting the previous one, protecting accounts through SHA256 password hashing with a unique salt for each account, role-based access control that limits each user to the pages permitted to their role, establishing troubleshooting protocols, and maintaining open communication with all stakeholders so that issues are resolved promptly as they arise.

Quality assurance was also included throughout the development process. Continuous feedback was gathered from the research adviser to identify areas that required improvement. During the user design phase, the research adviser reviewed the prototype and provided comments and suggestions, which were incorporated into succeeding iterations. After the necessary improvements were made, the completed system underwent User Acceptance Testing involving managers, field

technicians, and office staff to determine whether the system met the expected user requirements and was suitable for its intended users.

After the quality assurance process, the researchers implemented User Acceptance Testing to evaluate the completed system with managers, administrator, field technicians, and office staff. The respondents performed tasks based on their assigned roles to assess whether the system met the expected requirements and was suitable for its intended use. Feedback and results gathered during the testing were used to evaluate the system's usability and user acceptance. The completed Web-Based Employee Performance Tracking System was then prepared for operational use in supporting work assignment, performance evaluation, feedback, workload monitoring, and reporting processes.

i. System User Interface

This section presents the user interface of the Employee Performance Tracking System through screenshots of its key pages and screens. Each screenshot is accompanied by a brief description explaining its function and purpose, providing a clear understanding of how users navigate and utilize the system's features.

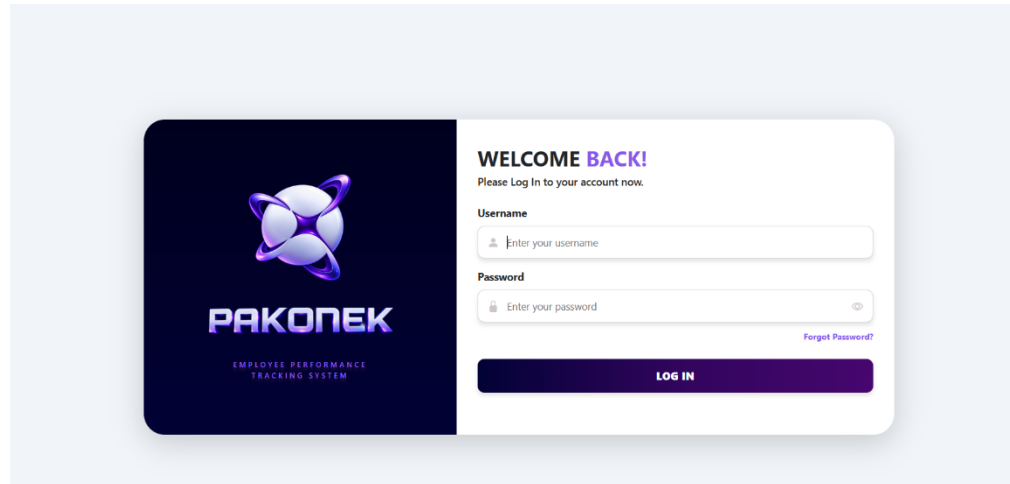


Figure 3.5: Log In

Description: This interface allows managers, administrators, technicians, and office staff to log in to the system.

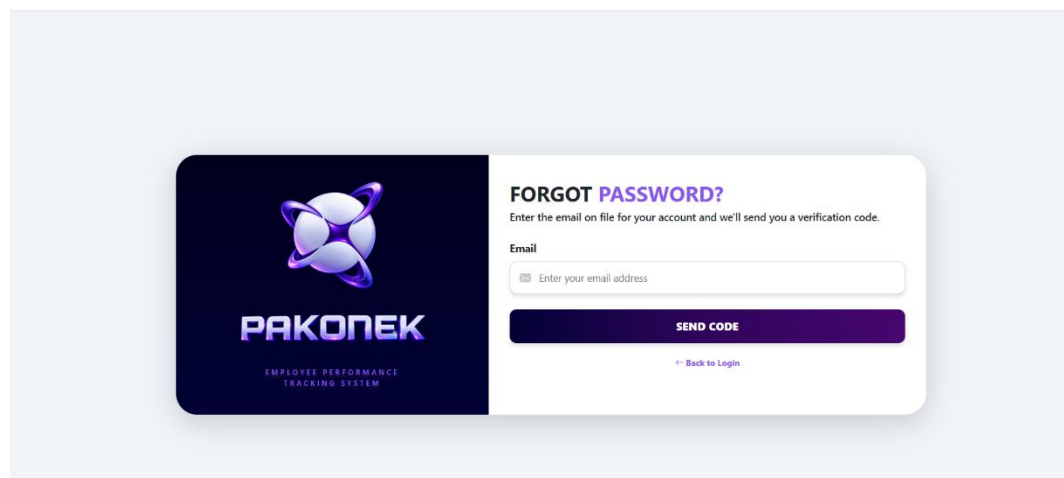


Figure 3.6: Forgot Password

Description: This interface allows users who have forgotten their passwords to request a verification code by email.

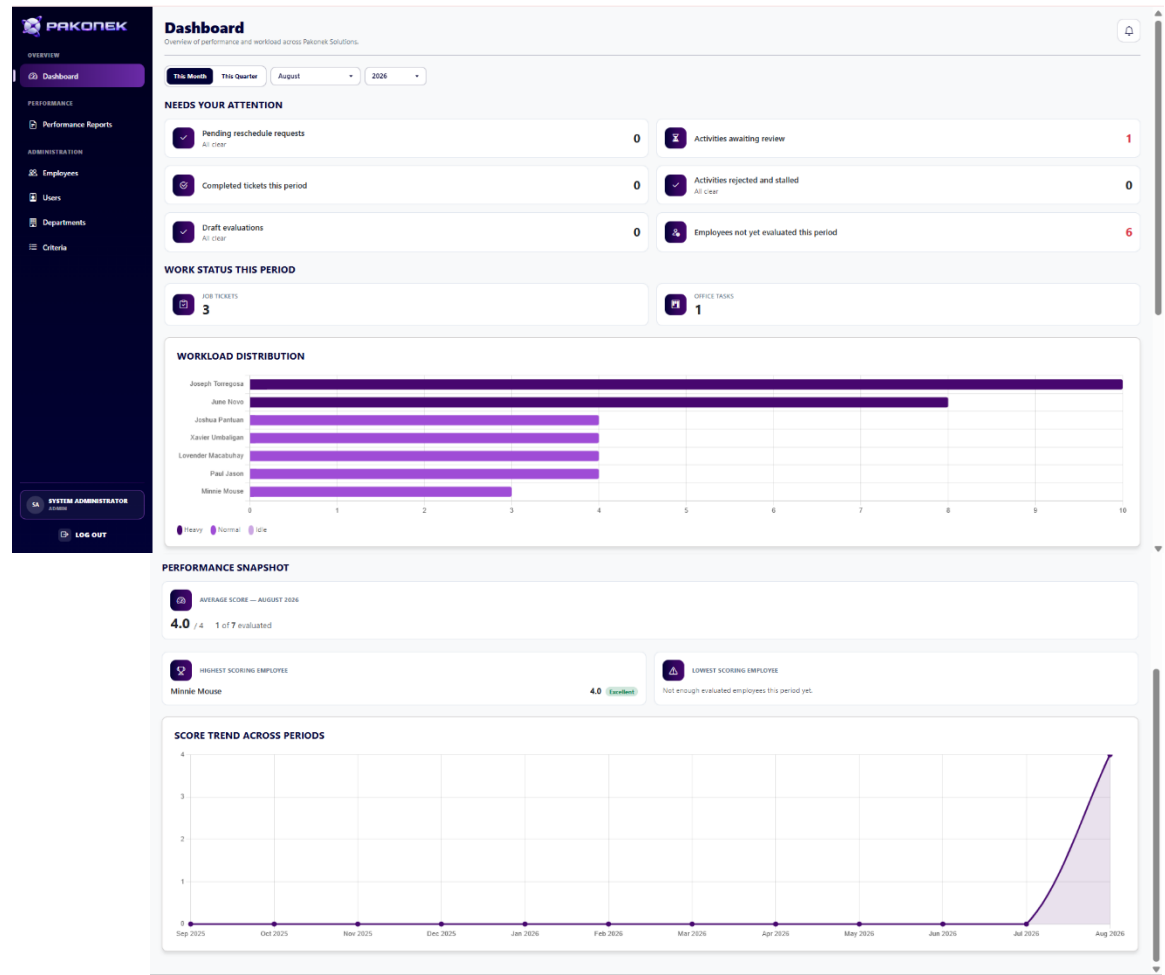


Figure 3.7: Administrator and Manager Dashboard

Description: This interface provides an overview of employee performance, workload distribution, evaluations, and system activities.

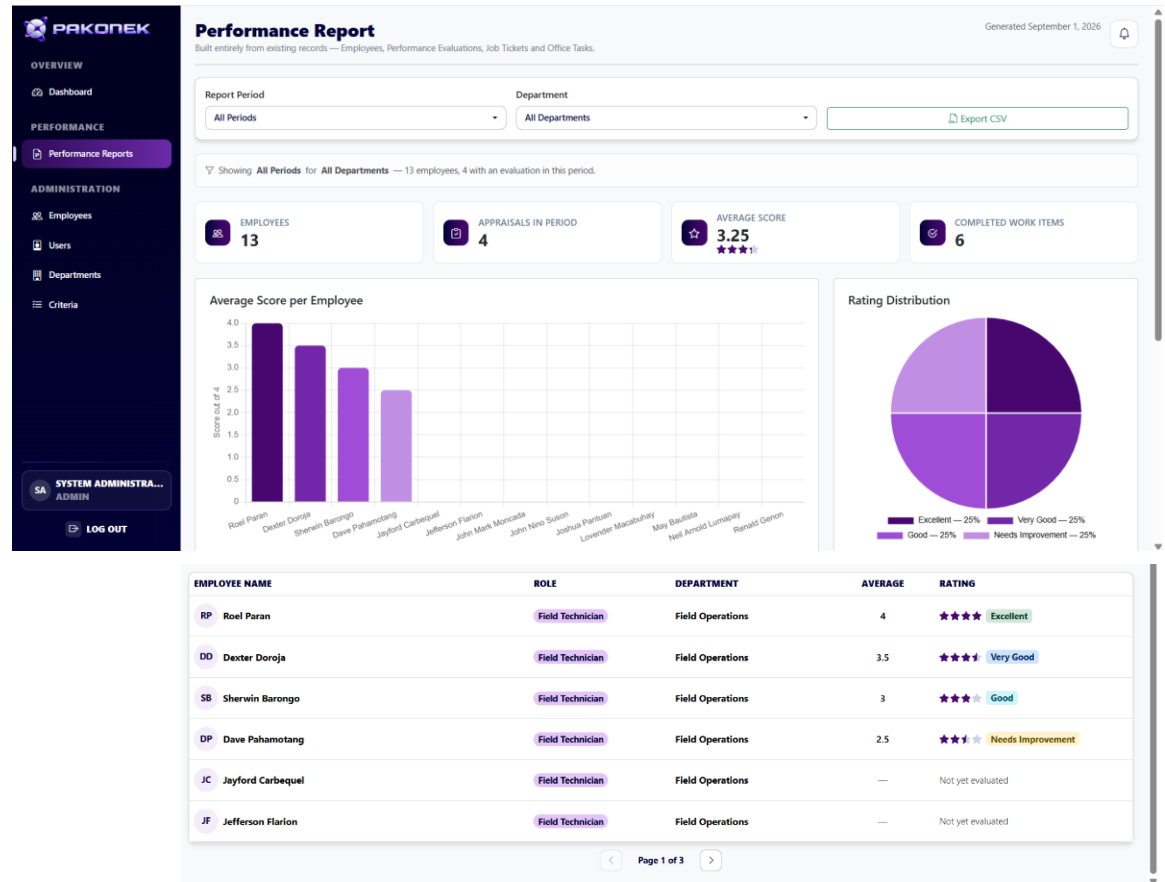


Figure 3.8: Administrator and Manager Performance Reports

Description: This interface displays the Performance Reports screen, providing an overview of employee evaluation results, average performance scores, rating distributions, and performance summaries.

Employees
Workforce profiles and deployment status.

Search by name or email... **All Roles** **Office Staff** **Field Technician** **+ New Employee**

NAME	DEPARTMENT	ROLE	DATE HIRED	STATUS	ACTIONS
CA Clifford Aguilar CliffordAguilar02@gmail.com	Administration	Field Technician	08/31/2026	ACTIVE	
DP Dave Pahamotang Davepahamotang@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE	
DD Dexter Doroja Dexterdorja@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE	
DJ Dexter John Baoy Dexteribaoy04@gmail.com	Administration	Field Technician	08/31/2026	ACTIVE	
JC Jayford Carbequel JayfordCarbequel@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE	
JF Jefferson Flarion Jeffersonflarion@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE	

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SA SYSTEM ADMINISTRATOR ADMIN LOG OUT

Figure 3.9: Employees

Description: This interface allows administrators to view, edit, search, and manage employee records, roles, departments, and employment status.

New Employee

Full Name *
e.g. Juan Dela Cruz ☐ Active

Email *
e.g. delacruz@gmail.com

Role Type *
Admin

Department
Administration

Only departments matching the selected role type are shown.

Cancel Save Employee

SA SYSTEM ADMINISTRATOR ADMIN LOG OUT

Figure 3.10: New Employee

Description: This interface allows administrators to add new employee records by entering employee information, role, email, department, and employment status.

PAKONEK

OVERVIEW

- Dashboard

PERFORMANCE

- Performance Reports

ADMINISTRATION

- Employees**
- Users
- Departments
- Criteria

SA SYSTEM ADMINISTRATOR ADMIN

LOG OUT

Edit Employee

Full Name *
Clifford Aguilar ☒ Active

Email *
CliffordAguilar02@gmail.com

Role Type *
Admin

Department
Administration

Only departments matching the selected role type are shown.

Cancel Save Changes

Figure 3.11: Edit Employee

Description: This interface allows administrators to update employee information, including personal details, role assignments, department, and employment status.

PAKONEK

User Management
Login accounts for every employee.

Search by username or employee...

All Roles Admin Manager Office Staff Field Technician

+ New User Account

USERNAME	EMPLOYEE	ROLE	CREATED	LAST LOGIN	STATUS	ACTIONS
Pakonek-A001	SA System Administrator	Admin	Aug 20, 2026	Sep 1, 2026 5:44 AM	ACTIVE	View
Pakonek-A003	DJ Dexter John Baoy	Admin	Aug 20, 2026	Aug 31, 2026 4:29 PM	ACTIVE	View
Pakonek-M012	XN Xavier Nathaniel Umbaligan	Manager	Aug 30, 2026	Sep 1, 2026 12:15 AM	ACTIVE	View
Pakonek-O011	NN Noel Novela	Manager	Aug 31, 2026	Never	ACTIVE	View
Pakonek-O019	LM Lovender Macabuhay	OfficeStaff	Aug 31, 2026	Aug 31, 2026 10:52 PM	ACTIVE	View
Pakonek-O020	MB May Bautista	OfficeStaff	Aug 31, 2026	Never	ACTIVE	View

Page 1 of 4

SYSTEM ADMINISTRATOR
ADMIN

LOG OUT

Figure 3.12: User Management

Description: This interface displays the User Management page, allowing administrators to create, view, search, and manage user accounts, roles, account status, and login activity.

PAKONEK

User Account Details
View and manage user account information and access settings.

Account Overview

Username Pakonek-A001	Hired At August 20, 2026
Account Created August 20, 2026	Last Login September 1, 2026 5:44 AM

Employee's Information

Full Name SYSTEM ADMINISTRATOR	Email admin@pakonek.local
Role ADMIN	Status ACTIVE
Department ADMINISTRATION	

Reset Password

New Password

8-10 digits, numbers only.

Confirm New Password

Change password

Disable Account **Edit Employee**

SA SYSTEM ADMINISTRA... ADMIN

LOG OUT

Figure 3.13: View User Account Details

Description: This interface allows administrators to view and monitor individual user accounts and reset account passwords.

PAKONEK

New User Account

Select Employee *

Select an employee

① Every active employee already has a user account.

Username

Auto generated username for employee

① Fixed automatically as Pakonek-(Role)(ID) (e.g. Pakonek-T001 for Field Technician #1) — this can't be changed later.

Password *

Enter a password

8-10 digits, numbers only.

Confirm Password *

Confirm your password

Cancel **Create Account**

SA SYSTEM ADMINISTRA... ADMIN

LOG OUT

Figure 3.14: New User Account

Description: This interface shows the New User Account page, which allows administrators to create new user accounts.

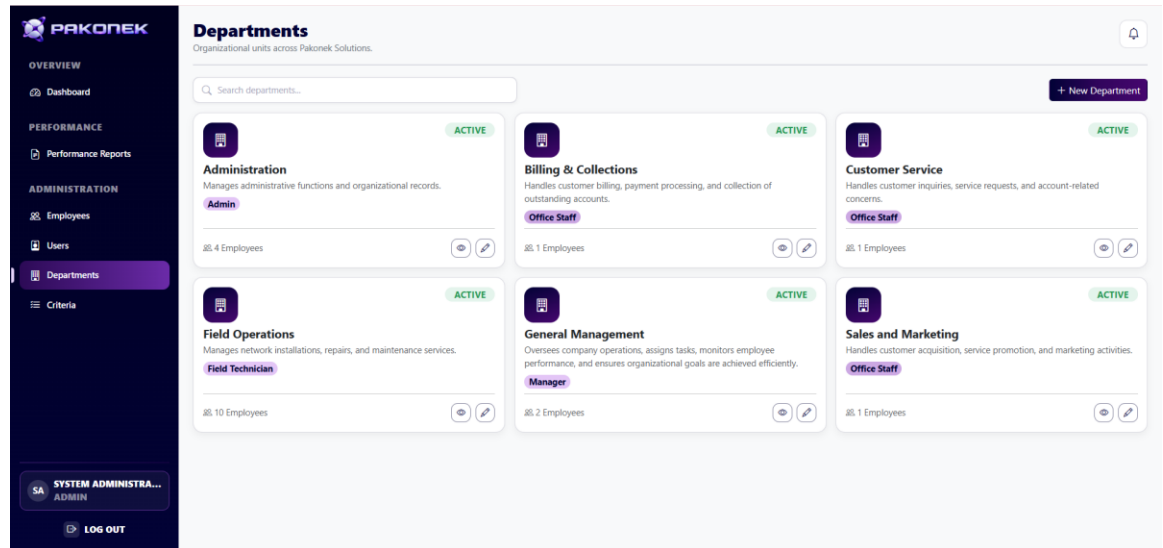


Figure 3.15: Departments

Description: This interface allows administrators to create and manage departments within the organization.

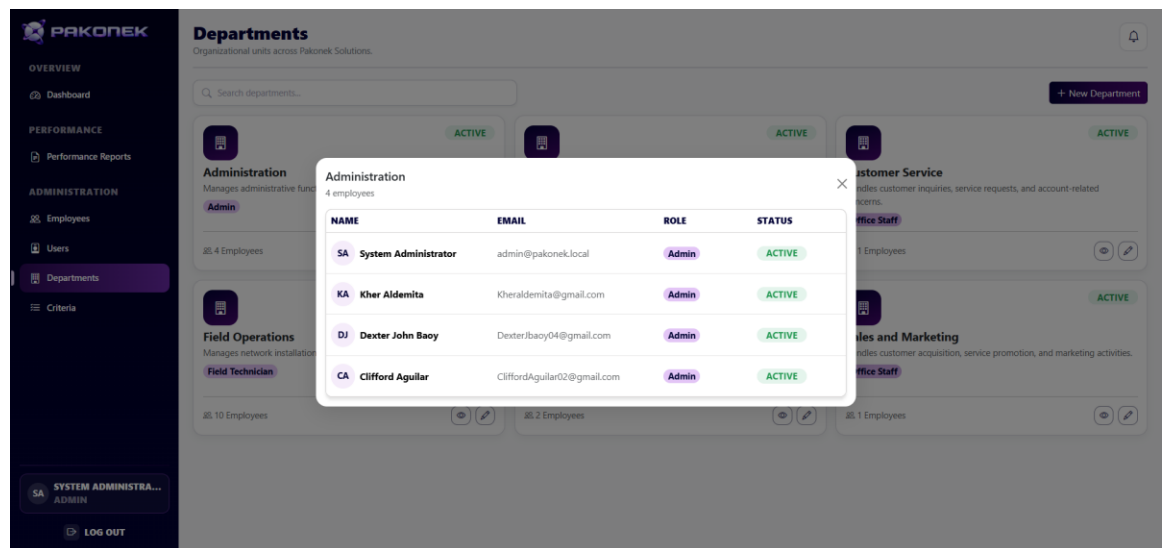


Figure 3.16: View Department

Description: This interface allows administrators to view and manage department records within the organization.

The screenshot shows the 'Edit Department' form in the PAKONEK system. The left sidebar contains the following menu items: OVERVIEW, Dashboard, PERFORMANCE, Performance Reports, ADMINISTRATION, Employees, Users, Departments (highlighted), and Criteria. At the bottom of the sidebar, there is a user profile for 'SA SYSTEM ADMINISTRATOR ADMIN' and a 'LOG OUT' button. The main form area is titled 'Edit Department' and contains the following fields: 'Name' with the value 'Administration' and an 'Active' checkbox checked, 'Description' with the text 'Manages administrative functions and organizational records.', and 'Role Type' with a dropdown menu showing 'Admin'. At the bottom right of the form are 'Cancel' and 'Save Changes' buttons.

Figure 3.17: Edit Department

Description: This interface allows administrators to view and manage department records within the organization.

The screenshot shows the 'New Department' form in the PAKONEK system. The left sidebar is identical to the previous figure, with 'Departments' highlighted. The main form area is titled 'New Department' and contains the following fields: 'Name' with the placeholder text 'e.g. Field Operations' and an 'Active' checkbox checked, 'Description' with the placeholder text 'Enter a description.', and 'Role Type' with a dropdown menu showing 'Admin'. At the bottom right of the form are 'Cancel' and 'Save Department' buttons.

Figure 3.18: New Department

Description: This interface allows administrators to create a new department by entering the department name, description, and role type.

CRITERIA	DESCRIPTION	ROLE TYPE	WEIGHT	STATUS	ACTIONS
Job Completion	Completes assigned job orders successfully.	Field Technician	30%	ACTIVE	
Quality of Service	Performs installations and repairs accurately.	Field Technician	20%	ACTIVE	
Quality of Work	Produces accurate, organized, and error-free work.	Office Staff	20%	ACTIVE	
Task Completion	Completes assigned tasks accurately and on time.	Office Staff	30%	ACTIVE	
Teamwork	Cooperates and works well with others.	Office Staff	20%	ACTIVE	
Time Management	Prioritizes and manages tasks efficiently.	Office Staff	30%	ACTIVE	

Figure 3.19: Administrator's Criteria Management

Description: This interface allows administrators to manage performance evaluation criteria and scoring standards.

Edit Criteria

Criteria Name *
Job Completion
☒ Active

Description
Completes assigned job orders successfully.

Role Type *
Field Technician

Weight (%)
30.00

How much this criterion counts toward the Overall Score in a Performance Evaluation. Weights for the same Role Type should add up to 100.

Other active Field Technician criteria are using 50% of 100% - 50% available (20% left if saved).

Cancel Save Changes

Figure 3.20: Edit Criteria

Description: This interface allows administrators and managers to modify existing evaluation criteria by updating the criteria name, description, weight, and status.

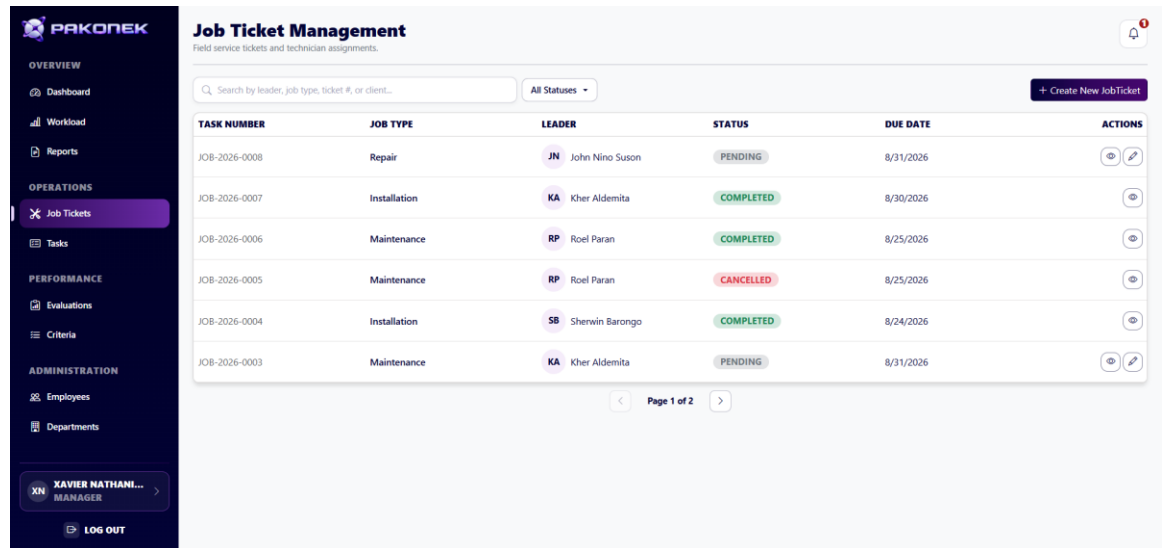
Figure 3.21: New Criteria

Description: This interface allows administrators to create new performance evaluation criteria, including the criterion name, description, role type, and weight.

TECHNICIANS	DEPARTMENTS	ACTIVE TICKETS	ACTION
SB Sherwin Barongo	Field Operations	2	View
JF Jefferson Flarion	Field Operations	1	View
JN John Nino Suson	Field Operations	1	View
NA Neil Arnold Lumapay	Field Operations	1	View
RG Renald Genon	Field Operations	1	View
DP Dave Pahamotang	Field Operations	0	View

Figure 3.22: Manager's Workload Monitoring

Description: This interface enables managers to monitor employee workloads and track task distribution across the workforce.



PAKONEK **Job Ticket Management**
Field service tickets and technician assignments.

Search by leader, job type, ticket #, or client... All Statuses [+ Create New JobTicket](#)

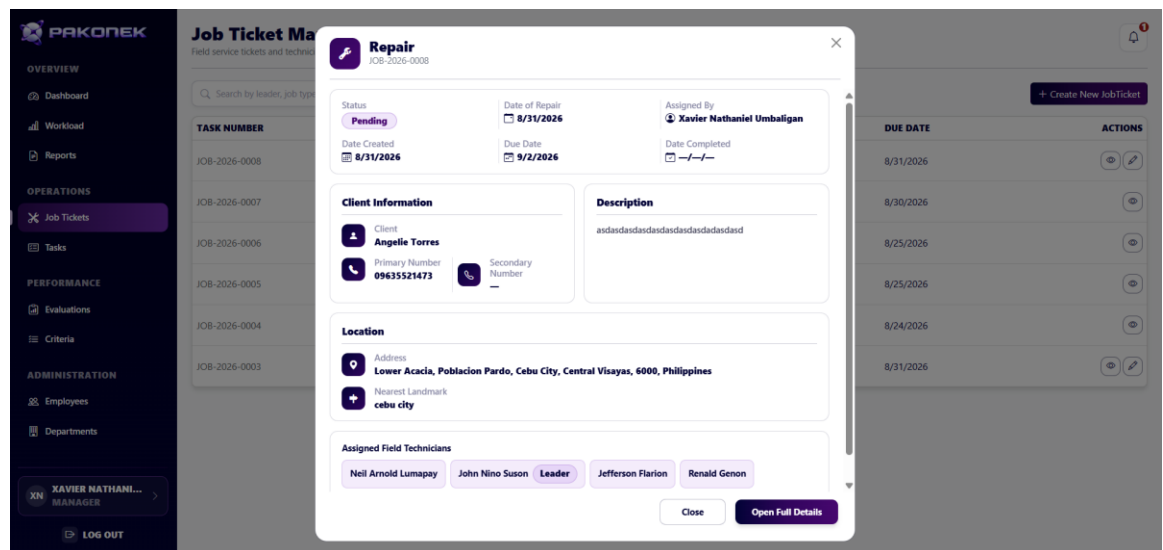
TASK NUMBER	JOB TYPE	LEADER	STATUS	DUE DATE	ACTIONS
JOB-2026-0008	Repair	JN John Nino Suson	PENDING	8/31/2026	
JOB-2026-0007	Installation	KA Kher Aldemita	COMPLETED	8/30/2026	
JOB-2026-0006	Maintenance	RP Roel Paran	COMPLETED	8/25/2026	
JOB-2026-0005	Maintenance	RP Roel Paran	CANCELLED	8/25/2026	
JOB-2026-0004	Installation	SB Sherwin Barongo	COMPLETED	8/24/2026	
JOB-2026-0003	Maintenance	KA Kher Aldemita	PENDING	8/31/2026	

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XAVIER NATHANI...
MANAGER [LOG OUT](#)

Figure 3.23: Job Tickets

Description: This interface allows managers to create, view, edit, and monitor job tickets for field technicians.



PAKONEK **Job Ticket Management**
Field service tickets and technician assignments.

Search by leader, job type, ticket #, or client... [+ Create New JobTicket](#)

Repair
JOB-2026-0008

Status: **Pending** Date of Repair: 8/31/2026 Assigned By: **Xavier Nathaniel Umbaligan**
 Date Created: 8/31/2026 Due Date: 9/2/2026 Date Completed:

Client Information

Client: **Angelle Torres**
 Primary Number: **09635521473** Secondary Number:

Description
 andasdasdasdasdasdasdasdasdasdas

Location

Address: **Lower Acacia, Poblacion Pardo, Cebu City, Central Visayas, 6000, Philippines**
 Nearest Landmark: **cebu city**

Assigned Field Technicians

Neil Arnold Lumapay **John Nino Suson** **Leader** **Jefferson Flarion** **Renald Genon**

[Close](#) [Open Full Details](#)

XAVIER NATHANI...
MANAGER [LOG OUT](#)

Figure 3.24: View Job Tickets

Description: This interface allows managers to view and monitor all job tickets within the system.

PAKONEK

Repair Job
JOB-2026-0008

Overview

- Dashboard
- Workload
- Reports
- Job Tickets**
- Tasks
- Evaluations
- Criteria
- Employees
- Departments

Client Information

Client: **Angelle Torres**

Primary Number: **09635521473**

Secondary Number: **—**

Location

Address: **Lower Acacia, Poblacion Pardo, Cebu City, Central Visayas, 6000, Philippines**

Nearest Landmark: **cebu city**

Assigned Field Technicians

Neil Arnold Lumapay John Nino Suson **Leader** Jefferson Flarion Renald Genon

The leader can be changed from the Edit page while this job order is Pending or In Progress.

Status: Pending

Date of Repair: 8/31/2026

Date Created: 8/31/2026

Due Date: 9/2/2026

Assigned By: Xavier Nathaniel Umbaligan

Date Completed: —/—/—

Description:
asdasdasdasdasdasdasdasdasdasd

XAVIER NATHANIEL...
MANAGER

LOG OUT

Back **Edit**

Figure 3.25: Open Full Details Job Ticket

Description: This interface allows managers to view the complete details of a selected job ticket, including customer information, assigned field technician, job description, status, dates, and work progress.

PAKONEK

Edit Job Ticket
Update the job details below. Assigned field technicians can't be added or removed here.

Ticket Number **JOB-2026-0008** | Date Created **08/31/2026** | Assigned by **Xavier Nathaniel Umbaligan**

1 Job Details

Job Type *
Repair

Full Name (Client) *
Angelle Torres

Primary Number *
09635521473

Secondary Number

Description *
asdasdasdasdasdasdasdasdasdas

Date of Repair *
08/31/2026

Due Date *
09/02/2026

① The deadline for this job to be finished. If this date passes and the job still isn't done, it's automatically marked Overdue.

2 Job Location *

Address
Lower Acacia, Poblacion Pardo, Cebu City, Central Visayas, 6000, Philippines

Nearest Landmark
cebu city

3 Assigned Field Technicians
Choose who leads this job order. The assigned team itself can't be added to or removed from here — only who leads them.

☐ Neil Arnold Lumapay

☒ John Nino Suson **Leader**

☐ Jefferson Flarion

☐ Renald Genon

Leader Remarks

① Remarks submitted by the field technician leader. Assignees are fixed at creation and can't be added or removed from this page.

Cancel Save Changes

Figure 3.26: Edit Job Ticket

Description: This interface allows managers to edit job ticket details, including assigned technicians, schedule and status.

PAKONEK

Job Ticket Form
Fill out the job details, enter the installation location, and assign field technicians.

Ticket Number: **JOB-2026-0009** | Date Created: **09/01/2026** | Assigned by: **Xavier Nathaniel Umbaligan**

1 Job Details

Job Type *
Select job type

Full Name (Client) *
e.g. Juan Dela Cruz

Primary Number *
09171234567

Secondary Number
Optional

Fiber Plan *
Select a plan

Date *
09/01/2026

Due Date *
09/03/2026

① The deadline for this job to be finished. If this date passes and the job still isn't done, it's automatically marked Overdue.

2 Job Location

Address *
Enter the job site address

Nearest Landmark
e.g. Near the barangay hall

3 Assign Field Technicians *
Only Field Technicians can be assigned to a job ticket. Assign at least 2, and pick one to be the leader.

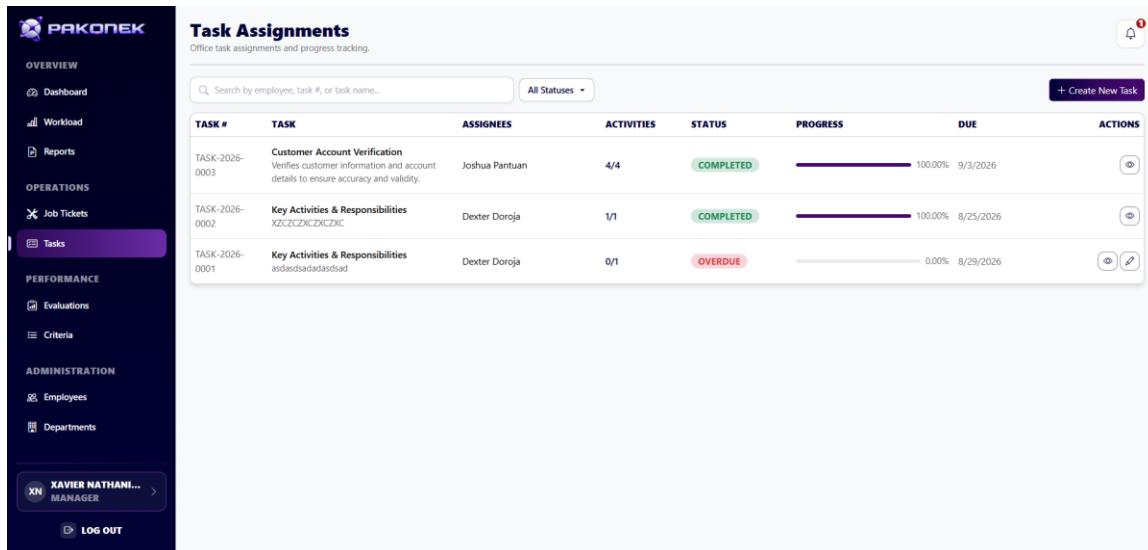
Select a field technician

No field technicians added yet.

Cancel | Save Job Ticket

Figure 3.27: Create New Job Ticket

Description: This interface allows managers to create and assign new job tickets to field technicians, including job details, location, and assigned Technicians



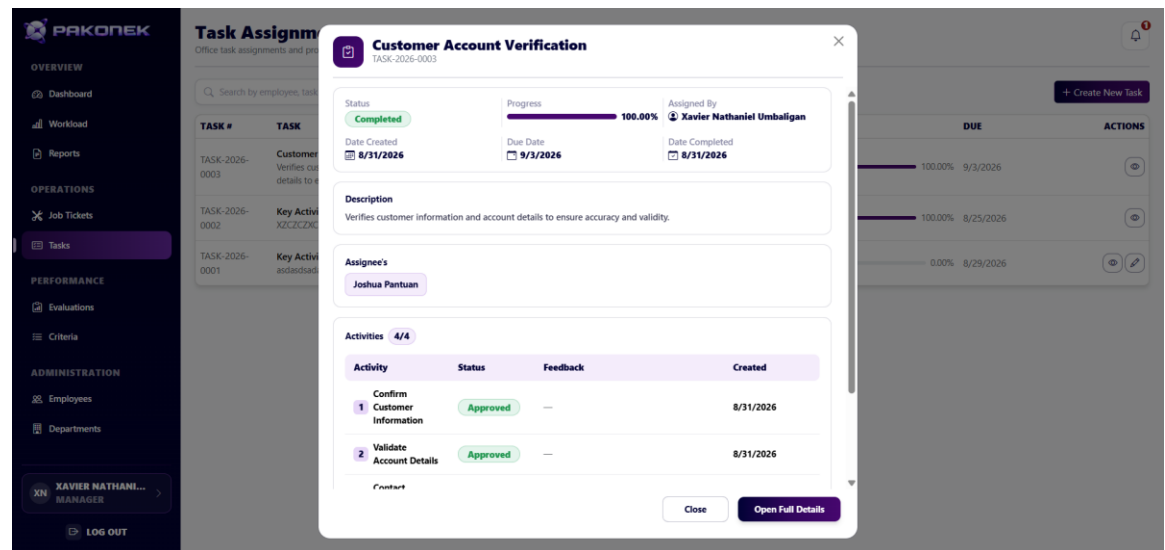
Task Assignments
Office task assignments and progress tracking.

Search by employee, task #, or task name... All Statuses + Create New Task

TASK #	TASK	ASSIGNEES	ACTIVITIES	STATUS	PROGRESS	DUE	ACTIONS
TASK-2026-0003	Customer Account Verification Verifies customer information and account details to ensure accuracy and validity.	Joshua Pantuan	4/4	COMPLETED	100.00%	9/3/2026	
TASK-2026-0002	Key Activities & Responsibilities XXXXXXXXXX	Dexter Doroja	1/1	COMPLETED	100.00%	8/25/2026	
TASK-2026-0001	Key Activities & Responsibilities asdasdasdasdad	Dexter Doroja	0/1	OVERDUE	0.00%	8/29/2026	

Figure 3.28: Task Assignments

Description: This interface allows managers to view assigned tasks, monitor progress, and track task completion for office staff.



Customer Account Verification
TASK-2026-0003

Status: **Completed** Progress: 100.00% Assigned By: **Xavier Nathaniel Umbaligan**

Date Created: **8/31/2026** Due Date: **9/3/2026** Date Completed: **8/31/2026**

Description
Verifies customer information and account details to ensure accuracy and validity.

Assignee's
Joshua Pantuan

Activities 4/4

Activity	Status	Feedback	Created
1 Confirm Customer Information	Approved	—	8/31/2026
2 Validate Account Details	Approved	—	8/31/2026

Close Open Full Details

Figure 3.29: View Task Assignments

Description: This interface allows managers to view assigned tasks, monitor task status, track progress, and review completion details.

PAKONEK

Customer Account Verification
TSA-00003

View task details and manage assignees and activities.

Status: **Completed** Progress: **100.00%** Approved: **4/4**
 Date Assigned: **8/31/2026** Due Date: **8/31/2026** Assigned By: **Xavier Nathaniel Umbaligan**

Description
Verifies customer information and account details to ensure accuracy and validity.

Assigned Employees
Joshua Panhuan

Activity	Status	Submission	Action	Feedback	History
Confirm Customer Information	Approved	Umbaligan, Xavi-	Locked	Add feedback...	
Validate Account Details	Approved	pakonek, use, cas-	Locked	Add feedback...	
Contact Customer for Clarification	Approved	BLANKS, MOODLE-	Locked	Add feedback...	
Submit Verification Report	Approved	Macabuhay, Lope-	Locked	Add feedback...	

Task status updates automatically from activity statuses: all Approved marks the task Completed; any Submitted or Approved sets it to In Progress.

XAVIER NATHANIL...
MANAGER

LOG OUT

Cancel Save

Figure 3.29: Open Full Details Task Assignment

Description: This interface allows managers to view complete information about a task assignment, including assigned employees, activities, submissions, deadlines, progress, status, and completion details

PAKONEK

Edit Task
Update the task details, reassign employees, or adjust its activities.

Task Number: **TASK-2026-0001** | Date Created: **08/24/2026** | Assigned by: —

1 Task Details

Task Name *
Key Activities & Responsibilities

Description
asdasdasdasdasd

Due Date *
08/29/2026

① The final deadline to complete this task. If it isn't finished by this date, the system will automatically mark it Overdue.

2 Assigned Employees

Assignees are fixed at creation and can't be added or removed from this page.

Dexter Doroja

3 Task Activities

dasdasdasdasd Rejected

Cancel Save Changes

Figure 3.30: Edit Task Assignment

Description: This interface allows managers to edit task assignment details, including the assigned employee, task information, due date and status.

PAKONEK

Task Assignment Form
Fill out the task details, assign employees, and break the task into activities.

Task Number: **TASK-2026-0004** | Date Created: **09/01/2026** | Assigned by: **Xavier Nathaniel Umbaligan**

1 Task Details

Task Name *
e.g. Quarterly inventory audit

Description
Instructions, objectives, or additional details...

Due Date *
09/09/2026

① The final deadline to complete this task. If it isn't finished by this date, the system will automatically mark it Overdue.

2 Assign Employees *
Only active Office Staff with a user account can be assigned. Add at least one.

Select an employee ▼ **Add Employees**

No employees added yet.

3 Task Activities *

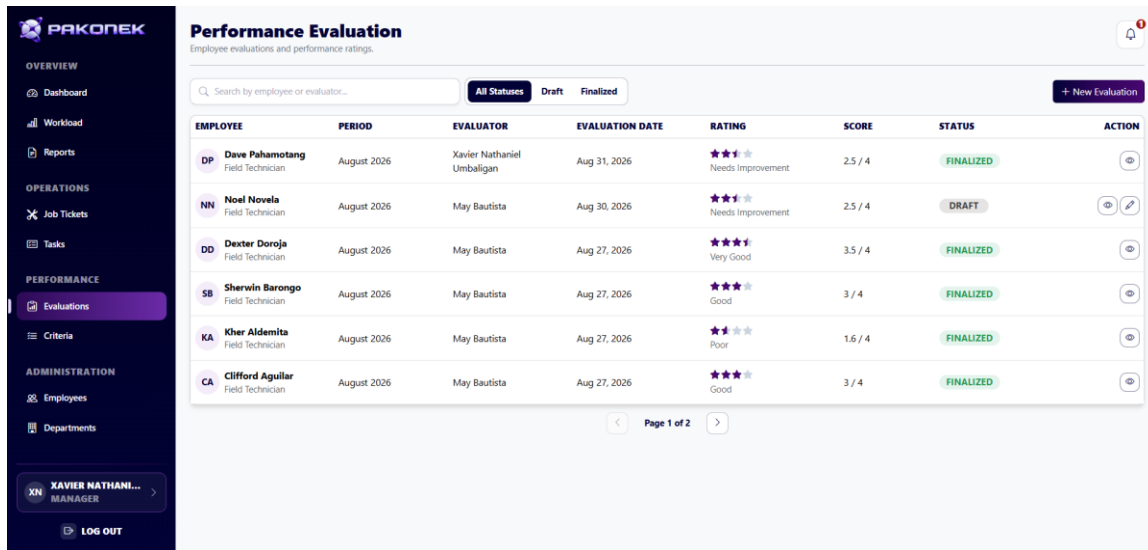
Enter an activity name ▼ **Add Activities**

No activities added yet.

Cancel **Save Task**

Figure 3.30: Create New Task Assignment

Description: This interface allows managers to create and assign tasks to office staff, specify activities, set deadlines, and define task requirements.



PAKONEK Performance Evaluation
Employee evaluations and performance ratings.

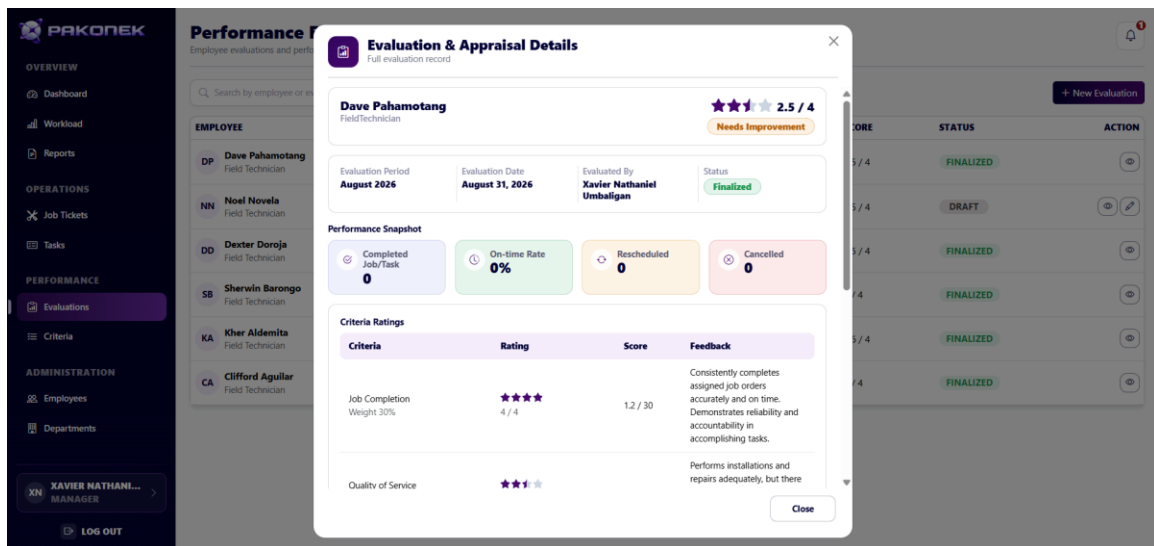
Search by employee or evaluator... All Statuses Draft Finalized + New Evaluation

EMPLOYEE	PERIOD	EVALUATOR	EVALUATION DATE	RATING	SCORE	STATUS	ACTION
DP Dave Pahamotang Field Technician	August 2026	Xavier Nathaniel Umbaligan	Aug 31, 2026	☆☆☆☆ Needs Improvement	2.5 / 4	FINALIZED	
NN Noel Novela Field Technician	August 2026	May Bautista	Aug 30, 2026	☆☆☆☆ Needs Improvement	2.5 / 4	DRAFT	
DD Dexter Doroja Field Technician	August 2026	May Bautista	Aug 27, 2026	☆☆☆☆ Very Good	3.5 / 4	FINALIZED	
SB Sherwin Barongo Field Technician	August 2026	May Bautista	Aug 27, 2026	☆☆☆☆ Good	3 / 4	FINALIZED	
KA Kher Aldemita Field Technician	August 2026	May Bautista	Aug 27, 2026	☆☆☆☆ Poor	1.5 / 4	FINALIZED	
CA Clifford Aguilar Field Technician	August 2026	May Bautista	Aug 27, 2026	☆☆☆☆ Good	3 / 4	FINALIZED	

Page 1 of 2

Figure 3.31: Performance Evaluation

Description: This This interface allows managers to evaluate employee performance using predefined criteria and provide feedback.



PAKONEK Performance Evaluation
Employee evaluations and performance ratings.

Search by employee or evaluator... All Statuses Draft Finalized + New Evaluation

Evaluation & Appraisal Details
Full evaluation record

Dave Pahamotang
Field Technician
☆☆☆☆ 2.5 / 4
Needs Improvement

Evaluation Period: August 2026
Evaluation Date: August 31, 2026
Evaluated By: Xavier Nathaniel Umbaligan
Status: Finalized

Performance Snapshot

Completed Job/Task 0	On-time Rate 0%	Rescheduled 0	Cancelled 0
-------------------------	--------------------	------------------	----------------

Criteria Ratings

Criteria	Rating	Score	Feedback
Job Completion Weight 30%	☆☆☆☆ 4 / 4	1.2 / 3.0	Consistently completes assigned job orders accurately and on time. Demonstrates reliability and accountability in accomplishing tasks.
Quality of Service	☆☆☆☆		Performs installations and repairs adequately, but there

Close

Figure 3.32: View Performance Evaluation

Description: This interface allows managers to view employee performance evaluation records, including ratings, feedback, evaluation criteria, and overall performance results.

PAKONEK

Edit Performance Evaluation
Update ratings, feedback, or finalize this evaluation

OVERVIEW

- Dashboard
- Workload
- Reports

OPERATIONS

- Job Tickets
- Tasks

PERFORMANCE

- Evaluations**
- Criteria

ADMINISTRATION

- Employees
- Departments

XAVIER NATHANI...
MANAGER

LOG OUT

Evaluation Details

Employee
Noel Novela (Manager)

Evaluator
May Bautista

The employee on an evaluation can't be changed — create a new evaluation instead.

Set automatically when this evaluation was created — not editable.

Evaluation Period
August 2026

Evaluation Date *
08/30/2026
Also used as the appraisal date.

Manager Criteria

Rate each criterion from 0.5 to 4 stars — click the left half of a star for a half rating. Click the same rating again to clear it.

No active criteria configured for this role type yet. Add some under Manager > Criteria.

Feedback

Summary of the employee's overall performance and areas for improvement.

Overall feedback for this evaluation period...

Covers both the evaluation and the appraisal decision.

Completed Job/Task
0

On-time Rate
0%

Rescheduled
0

Cancelled
0

Overall Score (preview)
0.0
out of 4
☆☆☆☆ **Poor**

Save as Draft

Finalize Evaluation

A draft can still be edited later; finalizing locks it.

Figure 3.33: Edit Performance Evaluation

Description: This interface allows managers to edit employee performance evaluation records, including ratings, feedback, and evaluation results.

PAKONEK

New Performance Evaluation
Evaluate and recognize employee performance

1 Select Evaluation Period

Period Month *
September

Period Year *
2026

2 Select Employee Role

Field Technician
Evaluate your field-based technicians

Office Staff
Evaluate your office-based staff

3 Evaluation Details

Employee *
Select an employee...

Evaluation Date *
09/01/2026

Evaluator
Xavier Nathaniel Umbaligan
Automatically set to your manager account.

4 Evaluation Criteria

Select an employee above to load their evaluation criteria.

5 Feedback

Summary of the employee's overall performance and areas for improvement.

Overall feedback for this evaluation period...

Covers both the evaluation and the appraisal decision.

Completed Jobs/Tasks

On-Time Rate

Rescheduled

Cancelled

Overall Score (preview)
0.0
out of 4
☆☆☆☆ **Poor**

Save as Draft

Finalize Evaluation

XN XAVIER NATHANI... MANAGER

LOG OUT

Figure 3.34: Create Evaluation

Description: This interface allows managers to create a new performance evaluation.

CRITERIA	DESCRIPTION	ROLE TYPE	WEIGHT	STATUS
Job Completion	Completes assigned job orders successfully.	Field Technician	30%	ACTIVE
Quality of Service	Performs installations and repairs accurately.	Field Technician	20%	ACTIVE
Quality of Work	Produces accurate, organized, and error-free work.	Office Staff	20%	ACTIVE
Task Completion	Completes assigned tasks accurately and on time.	Office Staff	30%	ACTIVE
Teamwork	Cooperates and works well with others.	Office Staff	20%	ACTIVE
Time Management	Prioritizes and manages tasks efficiently.	Office Staff	30%	ACTIVE

Figure 3.35: Manager Criteria Management

Description: This interface allows managers to view the performance evaluation criteria, role types, and weights used in evaluating employees.

NAME	DEPARTMENT	ROLE	DATE HIRED	STATUS
CA Clifford Aguilar CliffordAguilar02@gmail.com	Administration	Field Technician	08/31/2026	ACTIVE
DP Dave Pahamotang Davepahamotang@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE
DD Dexter Doraja Dexterdoraja@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE
DJ Dexter John Baoy Dexterjbaoy04@gmail.com	Administration	Field Technician	08/31/2026	ACTIVE
JC Jayford Carbequel Jayfordcarbequel@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE
JF Jefferson Flarion Jeffersonflarion@gmail.com	Field Operations	Field Technician	08/31/2026	ACTIVE

Figure 3.36: Manager Employee Management

Description: This interface provides managers with a view of employee records, roles, departments, and employment status.

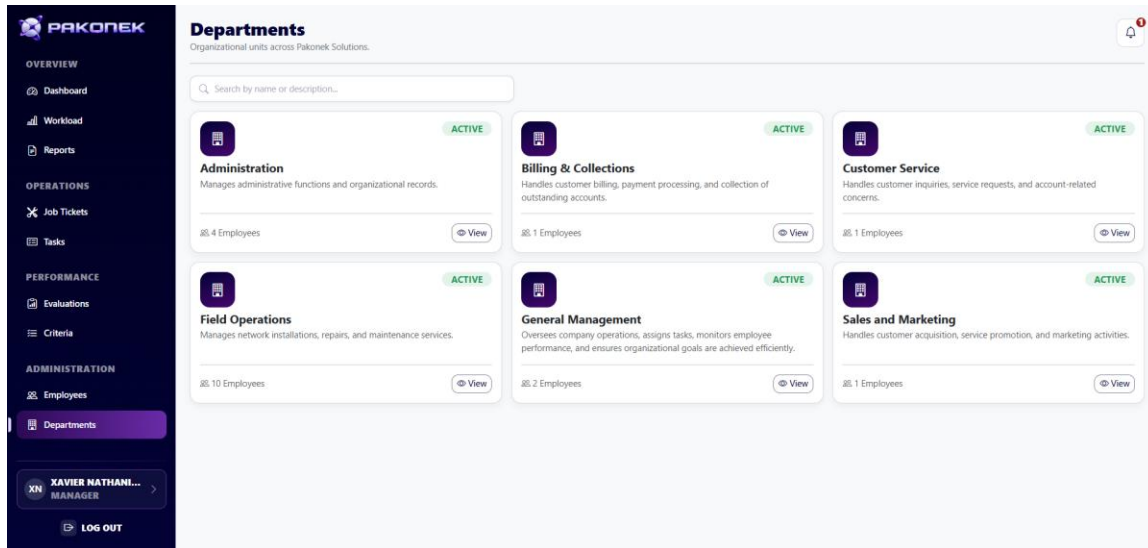


Figure 3.37: Manager's Departments

Description: This interface allows managers to view department information and monitor the departments within the organization.

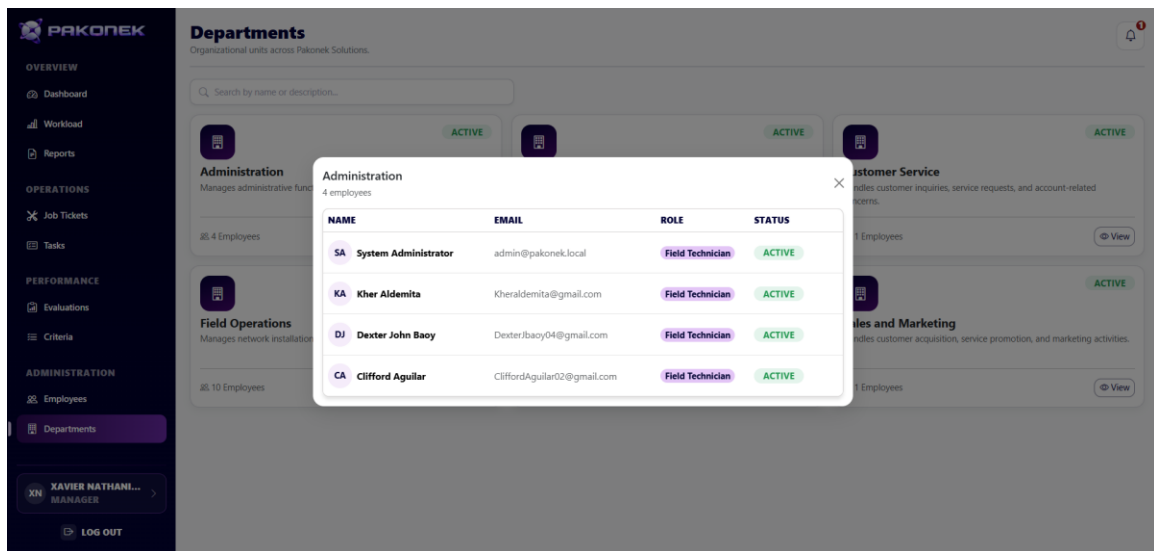


Figure 3.38: Manager's View Departments

Description: This interface allows managers to view department records and their corresponding employees.

My Account
Manage your account information and security settings.

Account Overview

Username Pakonek-M012	Hired At August 29, 2026
Account Created August 30, 2026	Last Login September 1, 2026 6:12 AM

Employee's Information

Full Name XAVIER NATHANIEL UMBALIGAN	Email xumbaligan@gmail.com
Role MANAGER	Status ACTIVE
Department GENERAL MANAGEMENT	

Change Password

Current Password

New Password

8-10 digits, numbers only.

Confirm New Password

Change password

Navigation Sidebar:

- OVERVIEW
 - Dashboard
 - Workload
 - Reports
- OPERATIONS
 - Job Tickets
 - Tasks
- PERFORMANCE
 - Evaluations
 - Criteria
- ADMINISTRATION
 - Employees
 - Departments

User Profile: XN XAVIER NATHANI... MANAGER **LOG OUT**

Figure 3.39: Manager's Account

Description: This interface allows managers to view and update their account information, including changing their password.

My Job Orders
Viewing as John Nino Suson

JOB ORDER	JOB TYPE	DATE	STATUS	ASSIGNED BY	YOUR ROLE	ACTION
JOB-2026-0008	Repair	8/31/2026	PENDING	Xavier Nathaniel Umbaligan	Leader	View

Navigation Sidebar:

- Job Orders
- My Performance Records

User Profile: JN JOHN NINO SUSON FIELD TECHNICIAN **LOG OUT**

Figure 3.40: Technician's Job Orders

Description: This interface allows technicians to view their assigned job orders, including job details, schedules, status, and completion progress.

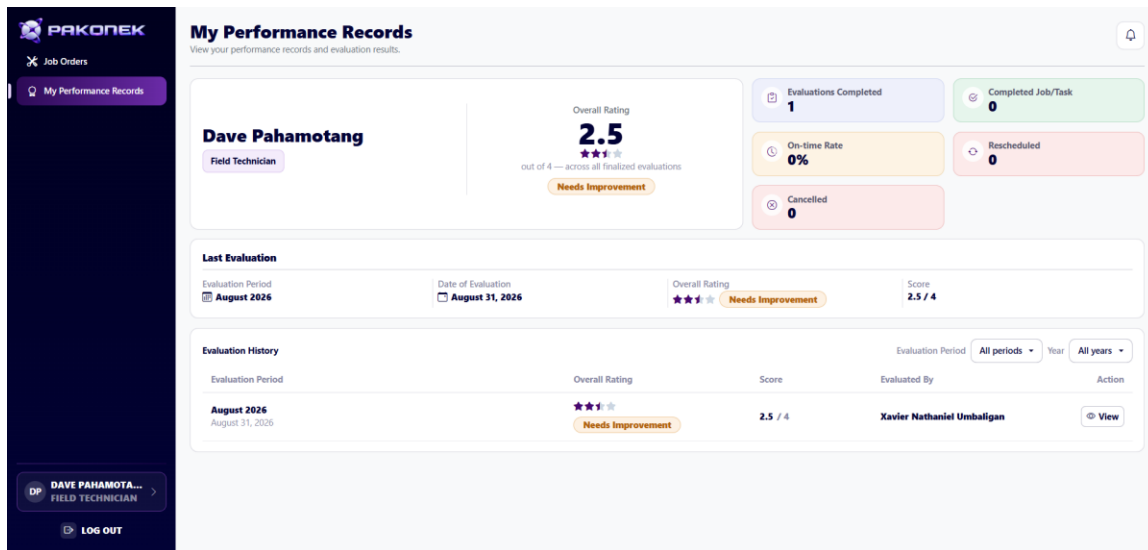


Figure 3.41: Technician & Office Staff Performance Records

Description: This interface allows technicians to view their performance records, evaluation results, feedback, and overall rating.

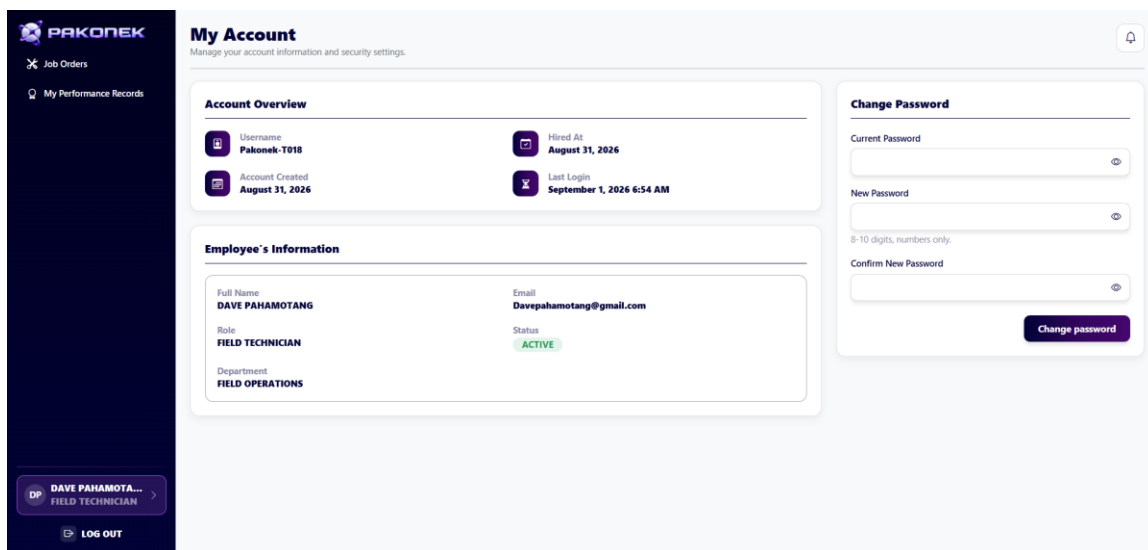


Figure 3.42: Technician's Account

Description: This interface allows technicians to view their account details and change their password.

TASK	DUE DATE	STATUS	ASSIGNED BY	OVERALL PROGRESS	YOUR ACTIVITIES	ACTION
Customer Account Verification TSK-00003	9/3/2026	COMPLETED	Xavier Nathaniel Umbaligan	100.00%	4/4 approved	View

Figure 3.43: Office Staff Assigned Tasks

Description: This interface allows office staff to view their assigned tasks, monitor deadlines, track progress, and update task completion status.

Activity	Status	Submission	History	Feedback
1 Confirm Customer Information	Approved	Umbaligan, Xa...	History	—
2 Validate Account Details	Approved	pakonek, use...	History	—
3 Contact Customer for Clarification	Approved	REAHIS-MOD...	History	—
4 Submit Verification Report	Approved	Macabuhay, L...	History	—

Figure 3.44: View Assigned Tasks

Description: This interface allows office staff to view their assigned tasks, including task details, deadlines, and current status.

PAKONEK

Employee Tasks

My Performance Records

My Account
Manage your account information and security settings.

Account Overview

Username
Pakonek-0021

Hired At
August 31, 2026

Account Created
August 31, 2026

Last Login
September 1, 2026 6:57 AM

Employee's Information

Full Name
JOSHUA PANTUAN

Email
joshuapantuan123@gmail.com

Role
OFFICE STAFF

Status
ACTIVE

Department
CUSTOMER SERVICE

Change Password

Current Password

New Password

8-10 digits, numbers only.

Confirm New Password

Change password

JOSHUA PANTUAN
OFFICE STAFF

LOG OUT

Figure 3.44: Office Staff Account

Description: This interface allows office staff to view their account details and change their password.

C. Ethical Considerations

In conducting the study on the Web-based Employee Performance Tracking System, the researchers strictly adhere to ethical standards to ensure the integrity, privacy, and safety of all participants and data. The research process emphasized transparency, confidentiality, and responsible handling of information.

Informed Consent — The management and employees of Pakonek Solutions Inc. were fully informed about the purpose and scope of the study and voluntarily participated. They understood that their responses and operational data would be used solely for system development and research purposes.

Privacy and Confidentiality — All personal data, including personal details, job assignments, performance records, evaluation results, and feedback, were treated with strict confidentiality. Only authorized researchers and system administrators had access to sensitive information.

Responsible Use of AI — AI tools were employed exclusively to improve sentence structure, grammar, and clarity in written documentation. The use of AI did not influence the ideas, findings, or decisions in the research. The original contributions, data collection, and system design remained entirely the work of the researchers.

Data Protection and Security — Collected data were securely stored and protected to prevent unauthorized access or misuse. The system ensures that employee information is encrypted and accessible only to authorized personnel.

Chapter IV

RESULT AND ANALYSIS

The System Usability Scale (SUS) is a standardized questionnaire administered after usability testing, consisting of 10 items rated on a 5-point Likert scale, where participants indicate their level of agreement or disagreement with each statement [15]. SUS is designed to measure the perceived ease of use of a wide range of systems, including both physical and digital products such as software, websites, and applications [15]. In this study, the researchers employed the System Usability Scale (SUS) questionnaire to evaluate the system with ten respondents consisting of administrators, managers, field technicians, and office staff of Pakonek Solutions Inc. The evaluation provided data on the respondents perceived usability and user acceptance of the Web-Based Employee Performance Tracking System.

1	Strongly Disagree
2	Disagree
3	Neutral
4	Agree
5	Strongly Agree

Table 2.2: 5-Point Likert Agreement Scale

1. I think that I would like to use this product frequently.

10 responses

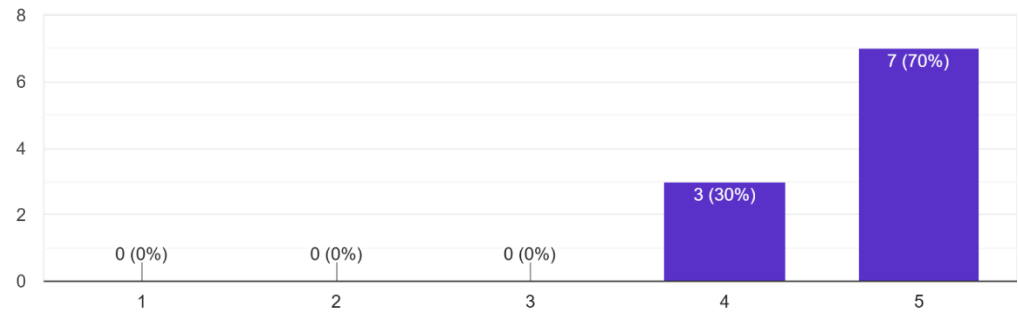


Chart 4.1: System Usability Scale for Frequently Using the Product

The survey findings indicate a strongly favorable acceptance toward the Web-Based Employee Performance Tracking System among the respondents. Of the ten respondents, 70% selected “Strongly Agree” and 30% selected “Agree” regarding their willingness to use the product frequently. No respondent selected “Neutral,” “Disagree,” or “Strongly Disagree.” The system received an average rating of 4.70, indicating a very positive response from the users. This result suggests that the system was well received by the respondents and that they expressed a strong willingness to use it regularly.

2. I found the product unnecessarily complex.

10 responses

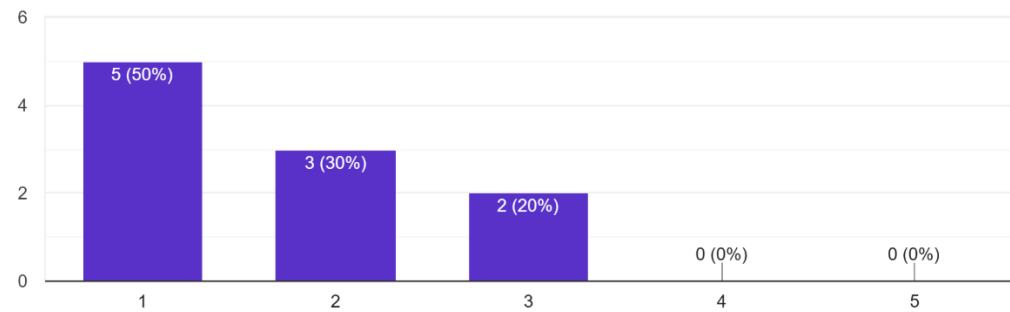


Chart 4.2: System Usability Scale for Complexity of the Product

The survey findings indicate that respondents did not perceive the system as unnecessarily complex. Of the ten respondents, 50% selected “Strongly Disagree,” 30% selected “Disagree,” and 20% selected “Neutral.” No respondent selected “Agree” or “Strongly Agree.” The average rating of 1.70 further indicates a low level of perceived complexity. These results suggest that most users found the system relatively simple and manageable to use, indicating that the system’s overall design and functionality were generally understandable to the respondents.

3. I thought the product was easy to use.

10 responses

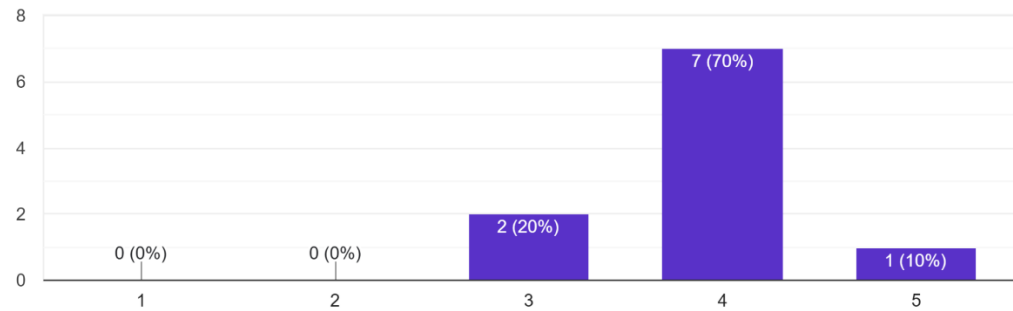


Chart 4.3: System Usability Scale for Ease of Use

The survey findings indicate a favorable perception of the system's ease of use. Of the ten respondents, 70% selected "Agree," 20% selected "Neutral," and 10% selected "Strongly Agree" that the product was easy to use. No respondent selected "Disagree" or "Strongly Disagree." The average rating of 3.90 further supports the positive result. This response suggests that the majority of users found the system straightforward and easy to operate, indicating that the system provides a generally user-friendly experience.

4. I think that I would need the support of a technical person to be able to use this product.

10 responses

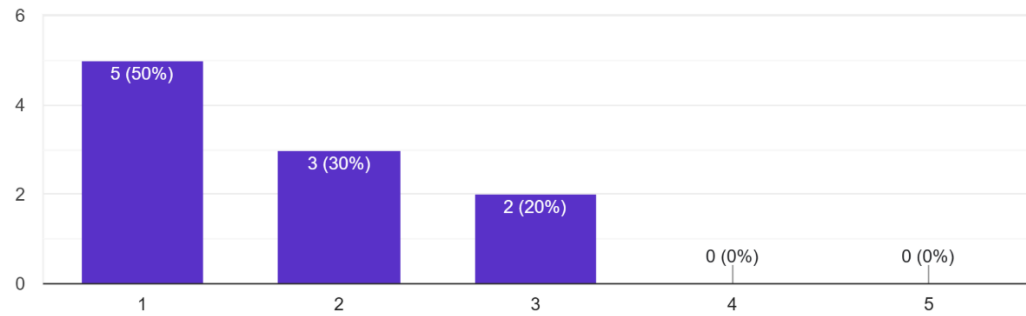


Chart 4.4: System Usability Scale for Need for Technical Support

The survey findings indicate a low perceived need for technical support when using the system. Of the ten respondents, 50% selected “Strongly Disagree,” 30% selected “Disagree,” and 20% selected “Neutral” regarding the need for support from a technical person. No respondent selected “Agree” or “Strongly Agree.” The average rating of 1.70 further indicates that most respondents felt they could use the product without requiring technical assistance. This response suggests that the system is easy to operate independently and does not require significant outside technical support.

5. I found the various functions in the product were well integrated.

10 responses

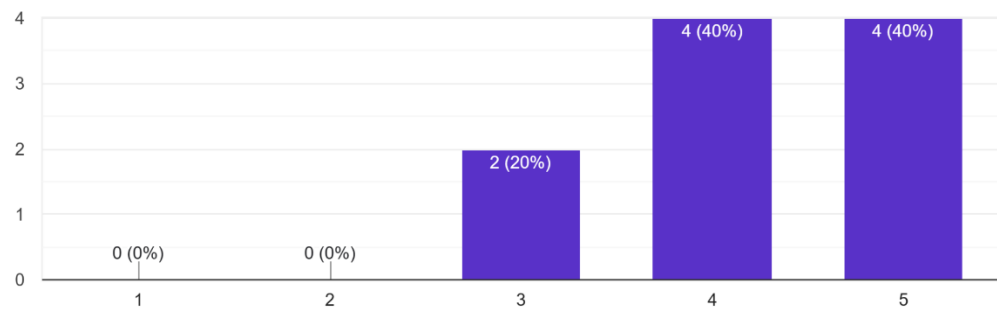


Chart 4.5: System Usability Scale for Integration of Functions

The survey findings indicate a favorable perception of the integration of the system's functions. Of the ten respondents, 40% selected "Agree," 40% selected "Strongly Agree," and 20% selected "Neutral" that the various functions in the product were well integrated. No respondent selected "Disagree" or "Strongly Disagree." The average rating of 4.20 further supports the positive result. This response suggests that the majority of users found the system's functions well integrated and working effectively together, contributing to a smooth and cohesive user experience.

6. I thought there was too much inconsistency in this product.

10 responses

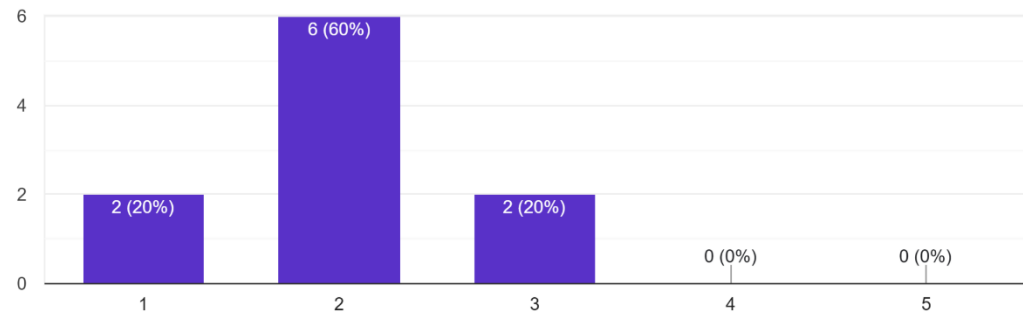


Chart 4.6: System Usability Scale for Consistency of the Product

The survey findings indicate a generally favorable perception of the system's consistency. Of the ten respondents, 20% selected "Strongly Disagree," 60% selected "Disagree," and 20% selected "Neutral" that there was too much inconsistency in the product. No respondent selected "Agree" or "Strongly Agree." The average rating of 2.00 further supports the positive result. This response suggests that the majority of users did not perceive significant inconsistencies within the system, indicating that its features and overall interface were generally consistent and reliable.

7. I imagine that most people would learn to use this product very quickly.

10 responses

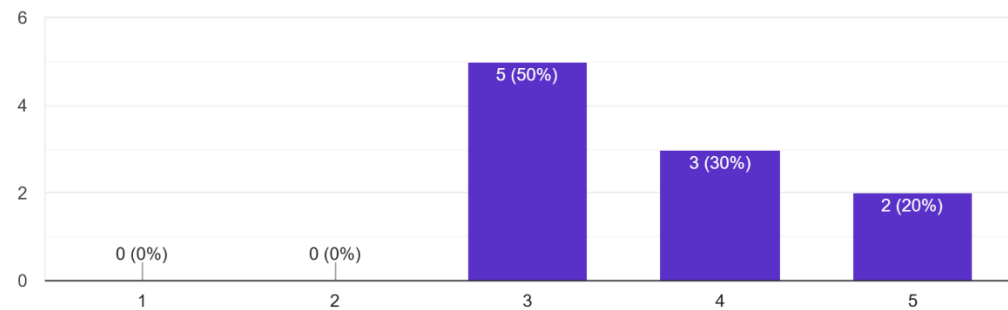


Chart 4.7: System Usability Scale for Quick Learning Curve

The survey findings indicate a moderately favorable perception of the system's learnability. Of the ten respondents, 50% selected "Neutral," 30% selected "Agree," and 20% selected "Strongly Agree" that most people would learn to use the product very quickly. No respondent selected "Disagree" or "Strongly Disagree." The average rating of 3.70 indicates a generally positive perception of the system's learning curve, although the substantial number of neutral responses suggests that some respondents were uncertain about how quickly an entirely new user would adapt to the system.

8. I found the product very awkward to use.

10 responses

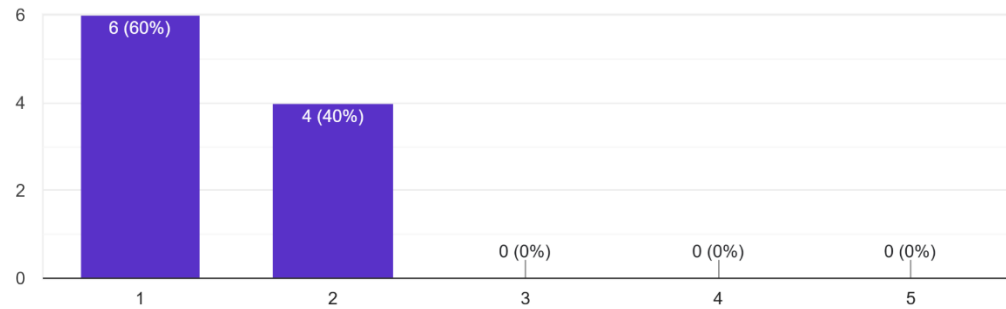


Chart 4.8: System Usability Scale for Awkwardness of Use

The survey findings indicate a highly favorable perception regarding the system's ease of handling. Of the ten respondents, 60% selected "Strongly Disagree" and 40% selected "Disagree" that the product was very awkward to use. No respondent selected "Neutral," "Agree," or "Strongly Agree." The average rating of 1.40 further supports the positive result. This response suggests that all respondents found the system comfortable and easy to use, with no significant concerns regarding awkwardness or difficulty in navigating the product.

9. I felt very confident using the product.

10 responses

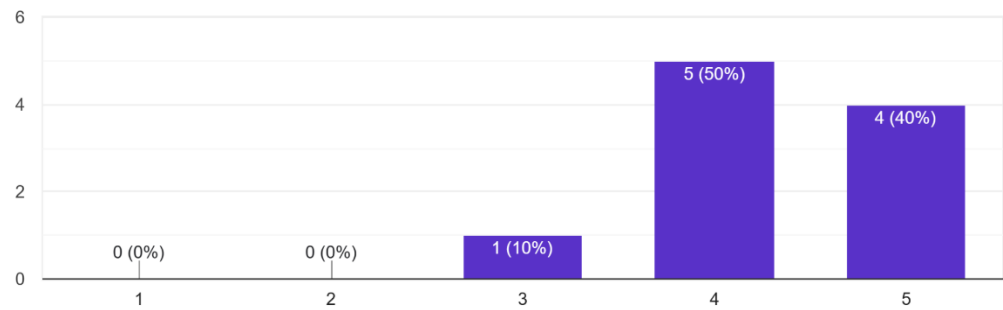


Chart 4.9: System Usability Scale for Confidence in Using the Product

The survey findings indicate a strong level of confidence among the respondents. Of the ten respondents, 50% selected “Agree,” 40% selected “Strongly Agree,” and 10% selected “Neutral” that they felt very confident using the product. No respondent selected “Disagree” or “Strongly Disagree.” The average rating of 4.30 further supports the positive result. This response suggests that the majority of users felt confident and assured when operating the system, indicating that the product is generally easy to understand and use.

10. I needed to learn a lot of things before I could get going with this product.

10 responses

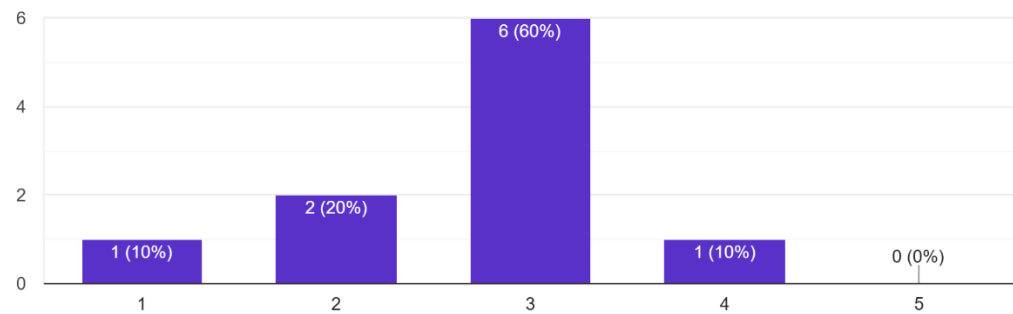


Chart 4.10: System Usability Scale for Learning Curve

The survey findings indicate a generally neutral perception regarding the amount of learning required to use the system. Of the ten respondents, 60% selected “Neutral,” 20% selected “Disagree,” 10% selected “Strongly Disagree,” and 10% selected “Agree” that they needed to learn a lot of things before they could get going with the product. No respondent selected “Strongly Agree.” The average rating of 2.70 indicates that respondents did not strongly feel that extensive learning was required, although the concentration of neutral responses suggests that a brief orientation period may still be beneficial for new users.

Chapter V

Discussion of Results

The results of the user acceptance survey demonstrate that the respondents had a positive perception of the system's usability. The overall SUS score of 78.25 is higher than the standard benchmark of 68, indicating that the system provides a satisfactory experience for its intended users. Respondents were generally able to operate the system, understand its features, and complete their work with minimal difficulty.

Most respondents gave positive ratings on the usability statements, indicating that the system was easy to operate. They expressed a strong willingness to use the system frequently (4.70) and reported a high level of confidence while using it (4.30). Respondents likewise agreed that the various functions of the system were well integrated (4.20), suggesting that its features work together rather than operating as separate, disconnected components. These results imply that the system can be used effectively even by individuals with limited technical knowledge.

Low ratings on the negatively worded statements reinforce this impression. Respondents did not find the system awkward to use (1.40), unnecessarily complex (1.70), or dependent on technical assistance (1.70), and they did not perceive significant inconsistency in its design (2.00). Taken together, these responses indicate that the interface and workflow were regarded as simple, reliable, and straightforward to operate independently.

While the overall results were positive, some respondents gave neutral ratings, particularly on the statements concerning how quickly the system could be learned (3.70)

and the amount of information needed before getting started (2.70). This suggests a distinction between operating the system and becoming familiar with it. Respondents found the system easy to use once they understood it, but were less certain how readily an entirely new user would adapt to it without guidance. Such differences may be attributed to individual experiences, familiarity with technology, or varying levels of exposure to the system, and they did not significantly affect the overall usability rating.

Certain limitations should nonetheless be acknowledged. The evaluation involved only ten respondents from a single company, which limits the generalizability of the results to other organizations. The System Usability Scale also measures perceived usability rather than actual task performance, so the findings reflect user opinion rather than measured efficiency or task completion times. Future studies may therefore involve a larger and more diverse group of respondents, and the system may benefit from onboarding features such as guided tutorials or an in-system help section to address the neutral responses concerning the initial learning process.

Overall, the findings indicate that the system can be adopted with minimal training and technical support. The usability rating demonstrates the system's potential to meet the needs of its intended users and provides a strong basis for further improvement and possible implementation in other organizations with similar performance monitoring needs

Chapter VI

Summary, Conclusions and Recommendations

A. Summary

The current process at Pakonek Solutions Inc. relied on manual methods for assigning work and monitoring employee performance, which led to difficulty in tracking job tickets and office tasks, delays in consolidating performance records, and inconsistent evaluation of employees across departments. This study developed the Web-Based Employee Performance Tracking System, a system that allows administrators, managers, field technicians, and office staff to manage employee records, create and assign job tickets and office tasks, monitor workload distribution, conduct performance evaluations based on predefined criteria, and generate performance reports through a centralized digital platform with role-based access control. Using the Rapid Application Development model the system was designed, built with programming language, framework and evaluated by ten respondents through User Acceptance Testing using the System Usability Scale (SUS). The system obtained an overall SUS score of 78.25, falling under a "Good" adjective rating, indicating that the system was generally well received in terms of usability, particularly in areas such as frequent use, user confidence, functional integration, and independence from technical support, although the speed of learning the system and the amount of preparation required before use remain areas for improvement.

B. Conclusion

Based on the results of the study, the Web-Based Employee Performance Tracking System was able to address the company's core operational needs by digitizing employee record management, job ticket and task assignment, workload monitoring, performance evaluation, and report generation, replacing the manual methods previously used at Pakonek Solutions Inc. The system consolidated processes that were previously handled separately, such as tracking field technician job orders and office staff task activities, and introduced capabilities that were not available in the existing workflow, including automated workload distribution monitoring, criteria-based scoring with configurable weights, and performance reports generated directly from existing records.

The overall SUS score of 78.25, rated as "Good," indicates that the system's usability met user expectations at this stage of development. Respondents expressed strong willingness to use the system frequently, high confidence while operating it, and favorable perceptions of how well its functions were integrated. They likewise reported low levels of perceived complexity, awkwardness, and dependence on technical assistance. However, the statements concerning how quickly the system could be learned and the amount of information needed before getting started received comparatively lower ratings, with a substantial proportion of neutral responses. This suggests that while the system's core functions were well received, the initial familiarization process could still be refined.

The study was limited to a small sample of ten respondents from a single company, which may not fully represent the experiences of organizations with different

operational scales or workforce sizes. In addition, the System Usability Scale measures perceived usability rather than actual task performance, so the findings reflect user opinion rather than measured efficiency or task completion times. The evaluation was also conducted within a short testing period, which may not have allowed respondents sufficient time to become fully familiar with the system's features or to encounter issues that typically surface during long-term use.

C. Recommendations

Based on the findings of the study, the researchers recommend that the Web-Based Employee Performance Tracking System be continuously improved so that it becomes easier to learn, more dependable, and more responsive to the daily needs of Pakonek Solutions Inc. The usability results showed that respondents were comfortable using the system once they became familiar with it, but were less certain how quickly a new user could adapt to it. The researchers therefore recommend adding guided onboarding within the system, such as a first-time walkthrough for each user role, tooltips on the job ticket and evaluation forms, and a help section that users can access while working. These materials should be prepared per role, since a field technician viewing job orders needs different guidance than a manager setting performance criteria.

The administrators and managers authorized to use the system should review it regularly to identify what needs improvement in how job tickets are assigned, how tasks and activity submissions are approved, how workload is monitored, how evaluations are scored, and how performance records are stored. Regular review allows

problems to be corrected early and keeps the system aligned with the company's actual working practices as roles and departments change. The researchers also recommend deploying the system to one department first, so that issues can surface under real workloads before it is extended to the rest of the workforce.

For further development, the researchers recommend adding automated notifications for approaching deadlines, overdue job tickets, and pending evaluations, along with mobile access that would allow field technicians to view job orders and update their status while on site. The system may also be extended with analytics that track performance trends across evaluation periods, exportable reports for management review, and the ability to attach supporting documents to completed job tickets. These improvements would strengthen the system's scalability and increase its usefulness as the workforce grows.

Finally, the researchers recommend that future studies evaluate the system under actual operating conditions over a longer period and with a larger group of respondents, including staff from other companies with similar operations. Such studies would show how the system performs as data accumulates and would allow its perceived usability to be compared with measured outcomes such as task completion times. Future researchers may also examine the system's longer-term effect on workforce productivity and on the consistency of performance evaluation across departments.

REFERENCES

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APPENDICES

Appendix A. System Specifications

Hardware Specification

System Manufacturer	Acer
System Model	Aspire A315-35
Processor	Intel(R) Celeron(R) N4500 @ 1.10GHz, 1114 MHz, 2 Core(s), 2 Logical Processor(s)
Memory	8.00 GB
Hard Disk	119 GB

Software Specification

Operating System	Microsoft Windows 11 Pro
Development Environment	Visual Studio 2022
Framework	ASP.NET Core Razor Pages
Programming Language	C#
Database	Microsoft SQL Server
Design Tool	Figma
Version Control	Git and GitHub

Appendix B. Flow of the System

Current Flow of the System

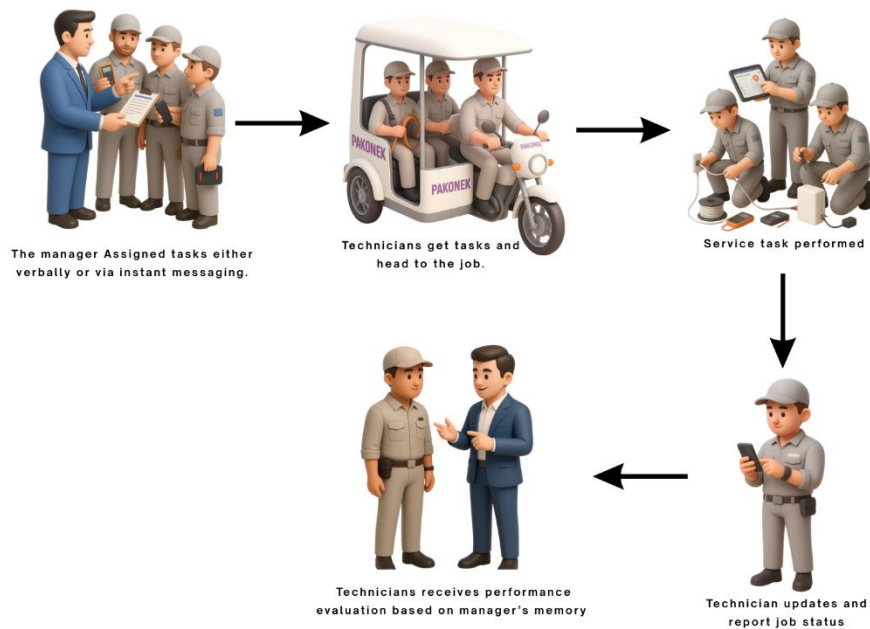
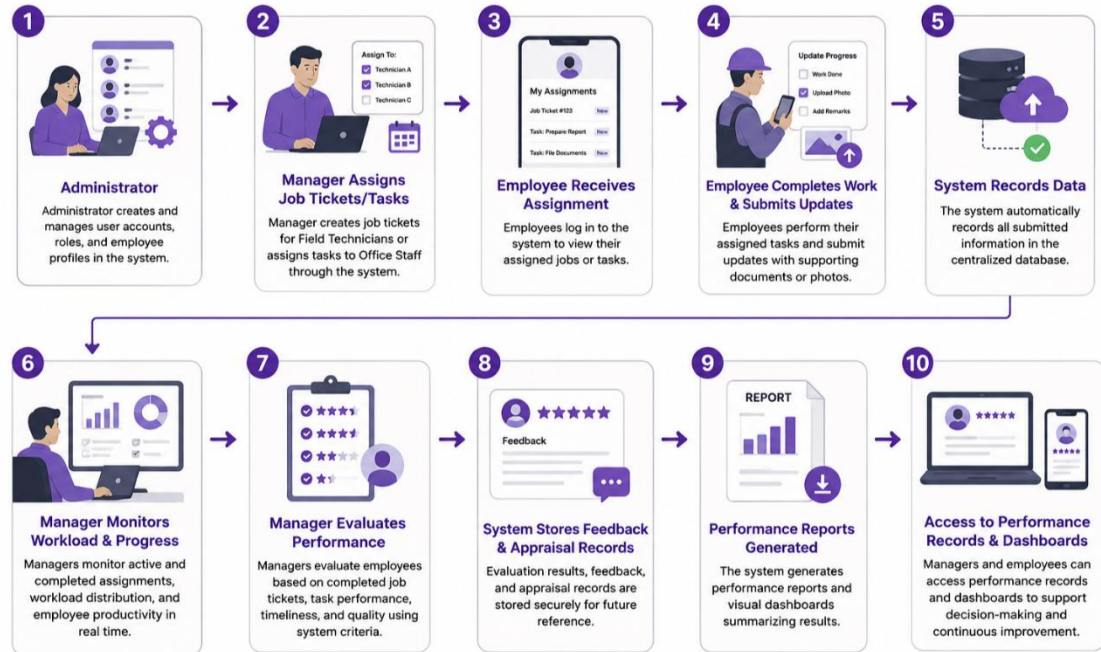


Figure B.1: Current Flow of the System

Currently, job assignments and performance monitoring at Pakonek Solutions Inc. are managed using manual coordination and basic communication tools. When a service request is received, the manager prepares a job order and assigns it to a field technician through verbal instruction or messaging. The technician proceeds to the assigned location and performs the installation, repair, or maintenance task. After completing the work, the technician informs the manager through a message or verbal update. There is no real-time system for tracking job status, and completion records are maintained only through job order forms or informal notes. Performance evaluation is conducted periodically based on the manager's observation, memory, and overall impression of the technician's work. Since job assignments, completion records, and performance information are handled separately and mostly manually, it is difficult to monitor workload distribution, track job timeliness, and generate accurate performance summaries.

Proposed Flow of the System



The proposed Web-Based Employee Performance Tracking System streamlines the process of assigning tasks, monitoring employee activities, evaluating performance, and generating reports through a centralized platform. The process begins when the Administrator creates and manages user accounts and employee records. Once employees are registered, the Manager creates job tickets for Field Technicians or assigns tasks to Office Staff through the system.

After receiving their assignments, employees log in to the system to view their assigned jobs or tasks. Field Technicians perform installation, repair, or maintenance activities and submit completion updates and supporting documentation, while Office Staff complete assigned tasks and update their progress within the system. The submitted information is automatically recorded in the centralized database.

The system continuously tracks task completion status and workload distribution. Managers can monitor active and completed assignments through dashboards and workload monitoring tools, allowing them to balance workloads and track employee productivity in real time.

Once tasks or job tickets are completed, the Manager conducts performance evaluations. Field Technicians are evaluated based on completed job tickets, timeliness, and job outcomes, while Office Staff are assessed using standardized evaluation criteria managed within the system. Evaluation results, feedback, and appraisal records are then stored for future reference.

Finally, the system generates performance reports and visual dashboards that summarize employee accomplishments, workload distribution, and evaluation results. These reports provide management with accurate information for decision-making while allowing employees to access their own performance records and feedback through the system

Appendix C. Application Prototype

The wireframe shows a login page with a two-column layout. The left column contains a placeholder for a logo or image, represented by a square box, and a label 'Employee performance tracking system' below it. The right column contains the login form. At the top of the right column is the heading 'Welcome back' followed by the subheading 'Log in to your account'. Below this are two input fields: 'Username' with a placeholder 'Enter your username' and 'Password' with a placeholder 'Enter your password'. A 'Show' button is located to the right of the password input field. Below the password field is a link 'Forgot password?'. At the bottom of the form is a 'Log in' button. Below the button is a note: 'Inline alert region for success / error messages sits above the form.'

Figure C.1: Log In

Description: This wireframe shows the initial design of the log in page, which allows managers, administrators, technicians, and office staff to log in to the system.

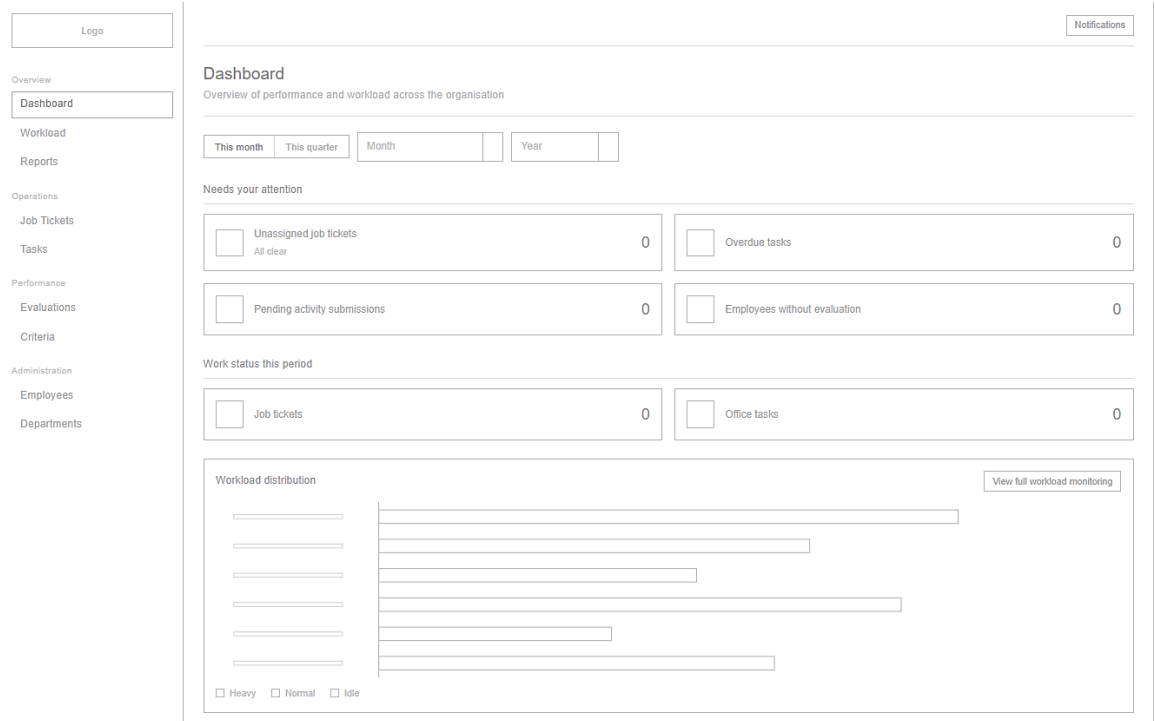


Figure C.2: Administrator and Manager Dashboard

Description: This wireframe shows the initial design of the dashboard, which provides administrators and managers with an overview of employee performance, workload distribution, evaluations, and system activities.

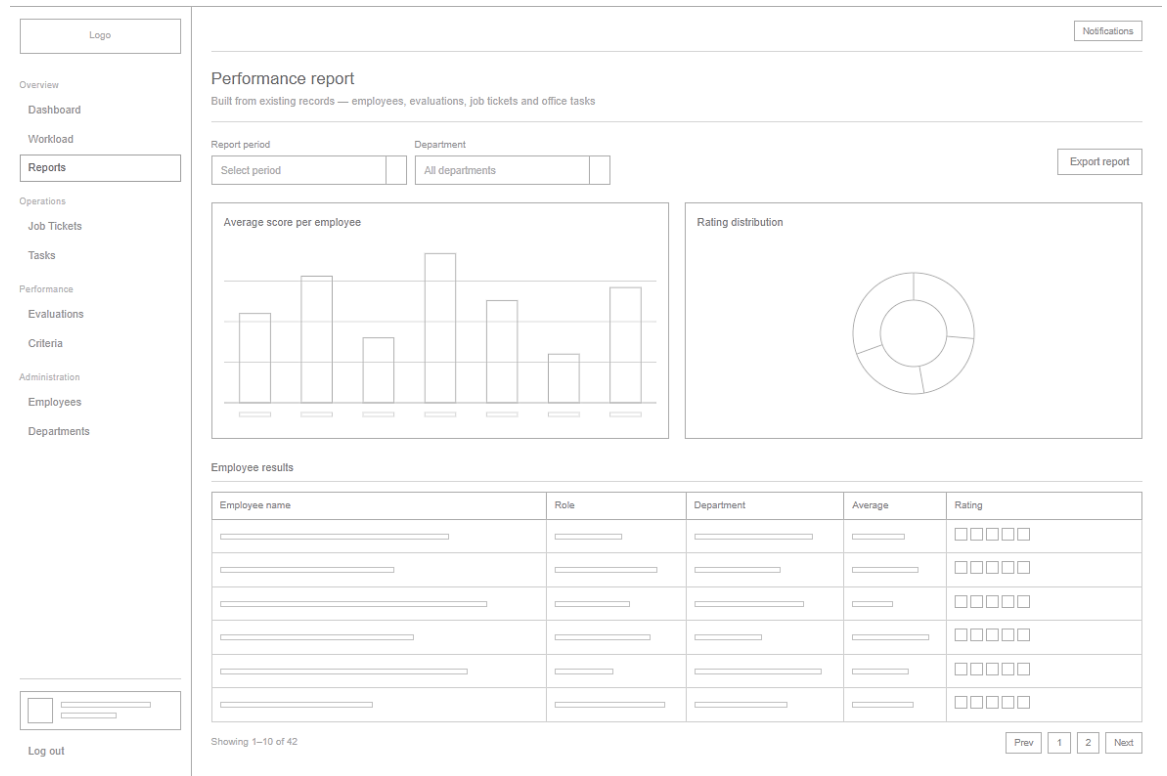


Figure C.3: Administrator and Manager Performance Reports

Description: This wireframe shows the initial design of the performance reports page, which displays the report filters, summary metrics, the average score and rating distribution charts, and the employee evaluation results table.

Logo

Overview

Dashboard

Workload

Reports

Operations

Job Tickets

Tasks

Performance

Evaluations

Criteria

Administration

Employees

Departments

Log out

Notifications

Employees

Workforce profiles and deployment status

Search by name or email...

All roles

Office staff

Field technician

Name	Department	Role	Date hired	Status
				Status
				Status
				Status
				Status
				Status
				Status

Showing 1–10 of 42

Prev

1

2

Next

Figure C.4: Employees

Description: This wireframe shows the initial design of the employee's page, which allows administrators to search, filter, view, and manage employee records, roles, departments, and employment status.

Logo

Overview

Dashboard

Performance

Performance Reports

Administration

Employees

Users

Departments

Criteria

Log out

Notifications

User management

Login accounts for every employee

Search by username or employee...

All roles

Admin

Manager

Office staff

Field technician

New user account

Username	Employee	Role	Created	Last login	Status	Actions
					Status	View Disable
					Status	View Disable
					Status	View Disable
					Status	View Disable
					Status	View Disable
					Status	View Disable

Showing 1–10 of 42

Prev

1

2

Next

Figure C.5: User Management

Description: This wireframe shows the initial design of the user management page, which allows administrators to create, search, view, and manage user accounts, roles, account status, and login activity.

Logo

Overview

Dashboard

Workload

Reports

Operations

Job Tickets

Tasks

Performance

Evaluations

Criteria

Administration

Employees

Departments

Log out

Notifications

Departments

Organisational units across the company

Search by name or description...

Department name

Role type

View employees

Department name

Role type

View employees

Department name

Role type

View employees

Department name

Role type

View employees

Department name

Role type

View employees

Department name

Role type

View employees

Department employees

Close

Name	Email	Role	Status
			Status
			Status
			Status
			Status

Modal — opens from a department card.

Figure C.6: Department

Description: This wireframe shows the initial design of the Departments page, which allows administrators to create, search, view, and manage company departments and organizational units.

Logo

Overview

Dashboard

Performance

Performance Reports

Administration

Employees

Users

Departments

Criteria

Log out

Notifications

Performance criteria

Evaluation weighting and requirements by role

Search criteria...

All rolesOffice staffField technician

New criteria

Criteria	Description	Role type	Weight	Status	Actions
				Status	ViewEdit
				Status	ViewEdit
				Status	ViewEdit
				Status	ViewEdit
				Status	ViewEdit
				Status	ViewEdit

Total active weight per role must add up to 100%.

Figure C.7: Criteria

Description: This wireframe shows the initial design of the Performance Criteria page, which allows administrators to search, view, create, and manage employee evaluation criteria.

Appendix D. Sample Program Code

```

using Microsoft.AspNetCore.Mvc;

using Microsoft.AspNetCore.Mvc.RazorPages;

using Microsoft.EntityFrameworkCore;

using TM_PE.Data;

using TM_PE.Model;


namespace TM_PE.Pages
{
    // The single front door for every role. Replaces the old "pick your name
    // from a dropdown" Select pages for Field Technician / Office Staff now
    // that real accounts exist (see Admin > User Management).

    public class LoginModel : PageModel
    {
        private readonly AppDbContext _context;

        public LoginModel(AppDbContext context) => _context = context;

        [BindProperty]

        public string Username { get; set; } = string.Empty;

        [BindProperty]

        public string Password { get; set; } = string.Empty;
    }
}

```

```

public string? ErrorMessage { get; set; }

// Set by ResetPasswordModel after a successful Forgot Password ->
// Verify Code -> Reset Password round trip, so the person lands back
// here knowing their new password actually took.

[TempData]

public string? SuccessMessage { get; set; }

public void OnGet()
{
    // Logging in again always starts clean.
    HttpContext.Session.Clear();
}

public async Task<IActionResult> OnPostAsync()
{
    var username = Username?.Trim();
    var password = Password ?? string.Empty;

    if (string.IsNullOrEmpty(username) ||
        string.IsNullOrEmpty(password))
    {
        ErrorMessage = "Please enter both a username and password.";
    }
}

```

```

        return Page();
    }

    var account = await _context.UserAccounts
        .Include(a => a.Employee)
        .FirstOrDefaultAsync(a => a.Username.ToLower() ==
username.ToLower());

    if (account == null || account.Employee == null ||
        !account.IsActive || !account.Employee.IsActive ||
        !PasswordHasher.Verify(password, account.PasswordHash,
account.PasswordSalt))
    {
        ErrorMessage = "Invalid username or password.";
        return Page();
    }

    var employee = account.Employee;

    HttpContext.Session.Clear();

    HttpContext.Session.SetInt32("AuthEmployeeId",
employee.EmployeeId);

```



```

        HttpContext.Session.SetString("AuthEmployeeName",
employee.FullName);

        HttpContext.Session.SetString("AuthRoleType",
employee.RoleType.ToString());

        HttpContext.Session.SetString("AuthUsername", account.Username);

        if (employee.RoleType == RoleType.FieldTechnician)

            HttpContext.Session.SetInt32("CurrentFieldTechnicianId",
employee.EmployeeId);

        else if (employee.RoleType == RoleType.OfficeStaff)

            HttpContext.Session.SetInt32("CurrentEmployeeId",
employee.EmployeeId);

        account.LastLogin = DateTime.Now;

        await _context.SaveChangesAsync();

        return employee.RoleType switch
        {

            RoleType.Admin => RedirectToPage("/Index"),

            RoleType.Manager => RedirectToPage("/Index"),

            RoleType.FieldTechnician
=> RedirectToPage("/FieldTechnician/Index"),

            RoleType.OfficeStaff => RedirectToPage("/OfficeStaff/Index"),

```

```
        _=> RedirectToPage("/Login")
    };
}
}
```

Appendix E. System Usability Scale

USER ACCEPTANCE FORM**SYSTEM USABILITY SCALE**

TITLE OF THE CAPSTONE PROJECT

DATE

NAME OF THE CLIENT/USER

SIGNATURE

Instruction: Please check the box that reflects your response to each statement.

	Strongly Strongly Disagree Agree
	1 2 3 4 5
1. I think that I would like to use this product frequently.	☐ ☐ ☐ ☐ ☐
2. I found the product unnecessarily complex.	☐ ☐ ☐ ☐ ☐
3. I thought the product was easy to use.	☐ ☐ ☐ ☐ ☐
4. I think that I would need the support of a technical person to be able to use this product.	☐ ☐ ☐ ☐ ☐
5. I found the various functions in the product were well integrated.	☐ ☐ ☐ ☐ ☐
6. I thought there was too much inconsistency in this product.	☐ ☐ ☐ ☐ ☐
7. I imagine that most people would learn to use this product very quickly.	☐ ☐ ☐ ☐ ☐

8. I found the product very awkward to use.	<table border="1"> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>					
9. I felt very confident using the product.	<table border="1"> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>					
10. I needed to learn a lot of things before I could get going with this product.	<table border="1"> <tr> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> </table>					

Comments/Suggestions

--

Appendix F. Plagiarism Result

This section includes the certificate or report showing the results of the plagiarism check conducted on the study. The document provides evidence that the work has been reviewed for originality and that the findings are free from unintentional plagiarism. It typically includes details such as the percentage of similarity found, the sources of any matched content, and confirmation of compliance with academic integrity standards. (NOTE: This serves as a guide on what to write in this section. **REMOVE** this paragraph when writing the paper.)

Curriculum Vitae

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EDUCATIONAL BACKGROUND

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Name of School : Asian College of Technology
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Course : Bachelor of Science in Information Technology

Senior High School Level

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Year Graduated : S.Y. 2021

